

Intelligent Mobility Management with Drive Test Data: A Hybrid Framework for Handover Prediction and Optimization

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THESIS TITLE

Intelligent Mobility Management with Drive Test Data: A Hybrid Framework for Handover Prediction and Optimization

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List of Acronym

Acronym	Full form
A1–A5	3GPP measurement event identifiers
AE	Autoencoder
AdaBoost / ADA	Adaptive Boosting
AEs	Autoencoders
ANN	Artificial Neural Network
API	Application Programming Interface
CAE	Contrastive Autoencoder
CatBoost	CatBoost
CINR	Carrier-to-Interference-plus-Noise Ratio
CO / COs	Course Outcome
CSV	Comma-Separated Values
CV	Cross-Validation
DQN	Deep Q-Network
DNN	Deep Neural Network
DRM	DRM (dataset/folder name)
eNodeB	Evolved Node B
F1	F1 score
G-NetTrack	G-NetTrack
GridSearchCV	Grid search with cross-validation
HOM	Handover Margin

HO	Handover
HOP	Handover performance metric (HOP)
IE	Information Element
K-Means	K-Means clustering
KNN	K-Nearest Neighbors
KDE	Kernel Density Estimation
KPI	Key Performance Indicator
MDP	Markov Decision Process
MSE	Mean Squared Error
ML	Machine Learning
MNO	Mobile Network Operator
Neo-7M	Neo-7M (GPS module)
OBE	Outcome Based Education
PCI	Physical Cell ID
PCA	Principal Component Analysis
Ping-pong	Ping-pong (rapid HO back-and-forth)
PO / POs	Program Outcome
P1–P7 / A1–A5 / K3–K8	Program attribute / knowledge profile ranges
Q-Learning	Q-Learning
Qweighted	Q_weighted (feature)
QSI	Quality Signal Indicator
RF	Random Forest
ReLU	Rectified Linear Unit

RL	Reinforcement Learning
ROC-AUC	Receiver Operating Characteristic — Area Under Curve
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
RLF	Radio Link Failure
S-Measure	S-Measure
SNR	Signal-to-Noise Ratio
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
t-SNE	t-distributed Stochastic Neighbor Embedding
TTT	Time-to-Trigger
TC	Time-to-Trigger (TC reference)
UE	User Equipment
XCAL-M	XCAL-M
XGBoost	eXtreme Gradient Boosting

Abstract

Efficient handover (HO) management remains a fundamental challenge in modern LTE networks, directly influencing user Quality of Service (QoS) and overall mobility performance. This thesis presents a comprehensive machine learning and reinforcement learning (RL)-driven framework for intelligent handover prediction and optimization using real-world LTE drive-test data collected in Dhaka, Bangladesh. The study integrates supervised learning, contrastive representation learning, and adaptive RL techniques to address both proactive HO event prediction and reactive HO parameter optimization. The predictive component employs supervised classifiers and neural architectures across two experimental cases: (i) mobility-agnostic datasets excluding user speed and (ii) mobility-aware datasets incorporating speed as an additional feature. Logistic Regression and Artificial Neural Network (ANN) models provided strong baselines with accuracies above 0.93 and F1-scores exceeding 0.83. The proposed Contrastive Autoencoder (CAE) framework further improved predictive balance, achieving an F1-score of 0.8471 and an accuracy of 0.9474 in Case 1, and 0.75 and 0.9243 in Case 2. These results confirm that contrastive learning effectively enhances latent-space separability and generalization under varying feature conditions. For optimization, Q-Learning and Contextual Bandit agents were trained using drive-test-based mobility traces to minimize unnecessary and ping-pong handovers. Evaluation across unseen datasets verified the generalization of learned policies, achieving up to a 13% reduction in total handovers and a 31% decrease in ping-pong events, with inference latency as low as 0.016 milliseconds. Furthermore, a hybrid Q-Learning with Subtractive Clustering approach demonstrated parameter adaptability by optimizing Handover Margin (HOM) and Time-to-Trigger (TTT) configurations over 50,000 simulations. Comparative experiments with fuzzy logic and DQN-based adaptive control validated the superiority of the proposed RL framework in both stability and responsiveness. Overall, this thesis introduces a unified and data-driven mobility management framework that combines predictive intelligence with adaptive control to improve LTE handover performance. The results demonstrate significant potential for deployment in next-generation self-organizing networks (SONs) and pave the way for scalable, learning-based handover strategies in 5G and beyond.

Chapter One

Introduction

With the rapid evolution of cellular communication technologies, efficient mobility management has become a critical challenge for maintaining service quality in dense and dynamic network environments. In Long-Term Evolution (LTE) and beyond, the handover (HO) mechanism plays a pivotal role in ensuring seamless connectivity as users move across cells. However, frequent or inaccurate handovers often lead to network inefficiencies such as increased ping-pong events, call drops, and degraded user experience. To address these challenges, the integration of machine learning (ML) and data-driven intelligence into traditional handover mechanisms has emerged as a promising research direction.

This thesis presents an intelligent framework for mobility management in LTE networks, built upon real-world drive test data. The study begins with drive test data acquisition and preprocessing, ensuring realistic network conditions and accurate reflection of field dynamics. Following that, a systematic data stratification approach is employed to enhance data quality, remove inconsistencies, and segment mobility patterns based on signal and geographical characteristics. To predict potential handover events proactively, a contrastive autoencoder-based model is developed, leveraging deep representation learning to extract robust latent features from Quality of Service (QoS) indicators such as RSRP, RSRQ, and CINR.

Building upon the predictive insights, this work further explores handover optimization using four distinct intelligent strategies: (i) Contextual Bandit with Q-Learning, (ii) Subtractive Clustering with Q-Learning, (iii) Fuzzy Logic, and (iv) Deep Q-Network (DQN) Learning. Each method is designed to balance adaptability and stability while minimizing handover failures and unnecessary transitions. The integration of reinforcement learning and fuzzy inference enables dynamic adjustment of handover parameters based on real-time context, thereby improving network robustness and user satisfaction.

In summary, this research aims to bridge the gap between conventional handover algorithms and intelligent, data-driven mobility management systems. By combining field-collected data, feature-rich prediction models, and adaptive optimization techniques, the proposed framework contributes to the development of next-generation self-organizing networks capable of autonomous decision-making and performance enhancement.

Chapter Two

Literature Review

The dataset was compiled to improve LTE handover mechanisms in Bangladesh's mobile environments. It supports machine learning models for handover prediction, QoS evaluation, and coverage optimization, addressing the lack of public datasets from the region. It is the first public dataset from a Bangladeshi LTE network, providing timestamped, field-collected measurements of serving and neighbor cell parameters with handover events. Data were collected via drive tests, capturing radio conditions during mobility across geographical zones. Prior studies, like one from Nigeria's Sabon Gari Market, offered LTE insights but lacked handover structuring [1]. An Austrian highway dataset focused on data rates, omitting raw RRC measurements or UE speed [2]. A Nigerian LTE assessment lacked timestamped granularity or machine learning formats [3]. A 2G dataset captured RF parameters but excluded newer metrics [4]. As Bangladesh's first LTE dataset, it includes UE velocity for handover analysis, raw logs with RSRP, RSRQ, and timestamps [5], and a preprocessed 64 machine learning-ready version with interpolated features and TTT handling [6]. It provides high granularity and a broad feature set for LTE mobility research in South Asia.

Building upon this data-centric foundation, to address the challenges of maintaining seamless connectivity and ensuring robust system performance, researchers have explored a variety of predictive and analytical approaches. These efforts include methods for anticipating handover events, optimizing mobility management, and mitigating the negative effects of rapid transitions, as well as techniques for extracting meaningful patterns from complex data to improve decision-making. Studies have leveraged field measurements, simulations, and advanced feature representation strategies to better understand network dynamics and system behavior. Such approaches provide a foundation for designing models that enhance both operational efficiency and overall service quality across diverse network and data-driven contexts. An ensemble learning-based handover prediction framework was developed in [12][8] for underwater wireless sensor networks (UWSNs), using simulated mobility data derived from real marine current measurements. Multiple machine learning models were evaluated on the collected handover events, and the ensemble learning approach was found to outperform traditional classifiers such as Gaussian Naïve Bayes and K-Nearest Neighbors. In terrestrial cellular networks, drive tests were conducted to evaluate RF parameters and assess handover performance using field-collected log data [15][9]. Dahouda et al. in [19][10] proposed a machine learning-based framework for non-terrestrial networks to reduce signaling storms during satellite handovers. An LTE handover mechanism combining direction prediction and adaptive time-to-trigger (TTT) reduced unnecessary handovers and handover trigger failures in [22][11]. Supervised learning approaches were applied in next-generation and O-RAN networks to predict handover events using real network measurements, enabling anticipatory service adaptations and operational cost reduction [12], [13]. The authors in [14] developed a

Q-learning-based handover optimization framework that integrates subtractive clustering and real-world LTE drive test data to reduce unnecessary and ping-pong handovers. In [15], adaptive handover optimization in LTE was evaluated using fuzzy logic and Deep Q-Network (DQN) models. Furthermore, Shafi et al. in [16] introduced a contextual bandit reinforcement learning framework for LTE handover optimization using real-world data to improve link stability.

In parallel with these algorithmic advances, various communication and networking contexts have explored Artificial Neural Networks (ANNs) for tasks such as signal equalization, mobility prediction, and handover decision-making. For instance, ANN has been applied in underwater wireless optical communication to mitigate nonlinear impairments and enhance data rates over long distances in [17]. In mobile wireless sensor networks, researchers have used ANNs to predict node mobility and manage adaptive handovers, improving network stability and efficiency [18]. Additionally, ANNs and hybrid neural models have been employed to forecast handover success rates in GSM networks and to optimize handover decisions in hybrid LiFi–WiFi systems, demonstrating their potential to improve throughput, reduce handover frequency, and maintain connectivity under varying network conditions [19], [20].

Extending beyond traditional neural architectures, deep autoencoder-based methods, including hybrid and domain-adaptation approaches, were applied for anomaly detection and medical image classification, leveraging reconstructed feature representations for improved pattern recognition in [21], [22], [23]. Authors in [24], [25] proposed multiple-input and graph-based variational autoencoders for unsupervised anomaly and outlier detection in heterogeneous and graph-structured data, improving the identification of deviations from learned normal distributions. Lu et al. in [26] developed an unsupervised locality-preserved autoencoder using complexity-invariant distance for machinery fault diagnosis, enabling discriminative feature learning from complex signals. Deep autoencoder-based methods, including deep and convolutional architectures, were proposed in [27], [28] for error mitigation and channel estimation in 5G and intelligent wireless communication systems. An autoencoder-driven semantic communication framework and a capacity-oriented coding scheme were proposed in [29], [30], where robustness against channel impairments and improved information transfer in machine-to-machine and general communication systems were demonstrated. Authors in [31], [32] proposed hybrid quantum-classical autoencoder frameworks for end-to-end communication, demonstrating feasibility, robustness against fading and noise, and reduced trainable parameters compared to classical baselines.

More recently, a contrastive predictive autoencoder was designed in [33] to learn self-supervised spatiotemporal representations from dynamic point cloud sequences by jointly optimizing contrastive prediction and reconstruction tasks. In [34], a contrastive autoencoder with multi-resolution segment-consistency discrimination was proposed for multivariate time series anomaly detection, enabling robust contextual representation learning through segment-wise contrastive objectives. In [35], Luo et al. combined contrastive learning with an autoencoder in an unsupervised framework for out-of-distribution detection, where contrastive learning enhanced feature separability and the autoencoder refined data representations. The authors proposed a contrastive variational autoencoder integrated with deep metric learning in [36] for attack detection in blockchain-cloud-IoT systems, effectively distinguishing normal and

malicious traffic through shared and private feature generation. A contrastive loss-enhanced autoencoder framework was developed for unsupervised outlier detection in IoT networks by Nikougoftar [37], integrating reconstruction and contrastive objectives to distinguish normal and anomalous data. The authors of [38] proposed a contrastive autoencoder for phoneme recognition that leverages class labels to learn task-specific invariant features, achieving superior performance on the TIMIT dataset compared to conventional and deep autoencoder-based approaches. An unsupervised cross-modal contrastive hashing framework was developed in [39], which combines autoencoder-based semantic similarity and a multi-objective contrastive loss to enhance retrieval accuracy between remote sensing images and text. Zhang et al. [40] introduced C-MELT, a contrastive-enhanced masked autoencoder framework for ECG-language pretraining, which effectively integrates physiological and textual modalities to achieve state-of-the-art cross-modal representation and diagnostic performance. A contrastive autoencoder-based framework was proposed in [7] for weather-aware RLF prediction, combining multi-modal feature fusion and survival analysis.

Although such advances have demonstrated the power of autoencoder-based learning, their application to handover prediction remains unexplored. In this work, we address this gap by developing a novel contrastive autoencoder framework with triplet loss that learns structured latent representations to accurately identify event-triggering conditions in dynamic LTE networks. By uncovering hidden patterns indicative of inefficient handover events, the framework enables early detection of unnecessary and ping-pong handovers, supporting proactive handover management and ultimately improving QoS by reducing failures, minimizing delays and lowering signaling overhead.

Complementing this learning-driven exploration, recent efforts in LTE network performance evaluation have extensively leveraged drive test data to characterize wireless connectivity in complex urban environments. A comprehensive urban LTE dataset was developed capturing detailed mobility and connectivity patterns within a busy market environment, enabling deeper analysis of wireless performance [1]. Another study focused on 2G handover evaluation through RF drive test log analysis, highlighting practical coverage assessment techniques [9]. Regional LTE performance studies in Nigeria revealed operator disparities in speed, latency, coverage, and packet loss, providing recommendations to improve infrastructure in suburban and rural areas [3]. Extensive long-term drive test data was used to apply machine learning models for predicting network metrics and understanding signal interactions [4]. KPI focused drive tests in Iraq analyzed LTE performance parameters such as RSRP, RSRQ, SINR, and throughput to guide network management [41]. A critical review of cellular network monitoring applications highlighted the importance of reliable, scalable tools for effective data-driven network optimization [42]. Handover optimization using an improved genetic algorithm dynamically adapts hysteresis tolerance for high-speed users, improving handover trigger and success rates while ensuring communication quality [43]. Load balancing self-optimization in 5G efficiently redistributes traffic from overloaded to low load cells, significantly reducing ping-pong handover, radio link failures, and improving spectral efficiency across mobility scenarios [44]. Metaheuristic algorithms like Grey Wolf and Mayfly optimizations improve vertical handover performance in heterogeneous wireless networks by reducing energy consumption, delay, call drops, and increasing throughput [45]. A fuzzy logic-based handover scheme dynamically adjusts handover

margin and time-to-trigger based on signal quality and user speed, reducing handover frequency and ping-pong effects in ultra-dense 5G networks [46]. Scalable mobility management using proximity-based clustering and aerial access networks addresses high handover rates in dense 5G environments, resolving security and privacy challenges overlooked by previous distributed schemes [47]. In [48], Alnabhana et al. propose an RSSI, mobility-direction and BS-capacity-aware handover scheme for dense femtocells, cutting unnecessary HOs by $\approx 30\%$ and lowering delay by ~ 10 ms. The work in [49] introduces a GA-based vertical-handover selection (GAfVH) for multi RAT 5G, reducing HO frequency and enabling application specific network choice. In [50], Shaye et al. present IDHPO-AWF, which fuses SINR, load, and speed metrics via an automatic-weight function to optimize per-UE HCPs, improving RSRP, HOP and RLF. The RHOT-FLC in [51] adjusts TTT and HOM using fuzzy logic, yielding significant mobility robustness gains. A self-optimization framework in [52] tunes Threshold, Hysteresis and TTT based on channel quality, mobility and latency, lowering HO rates and ping pong. The study in [53] highlights HCP trade-offs across user speeds, motivating adaptive settings, while [54] introduces CSVAG, a hybrid CS/GA method that boosts throughput ($\sim 17\%$ over CS) and reduces latency and failures. Baud Haryo Prananto et al. [55] extend RIC-based mobility prediction by incorporating UE-motion features into vector autoregression and replacing it with a neural network, improving target-cell selection in high-mobility coverage hole scenarios. In [56], supervised ML is applied to map HOM/TTT parameters to HOPP and HOF metrics, where nested cross-validation yields lower MAE and MSE than standard validation. The ANN-fuzzy logic handover protocol in [57] jointly addresses authentication and QoS, achieving $\approx 27\%$ higher success rates with notable reductions in HOF and ping-pong events. A DRL-based satellite handover scheme in [58] leverages visible time, RSS, distance and load information, reducing total handovers by over 21% under zero-failure conditions. Multi-agent RL in [59] enables distributed satellite handover coordination that balances load while lowering average handovers and user blocking probability compared to localized criteria. Q-learning in [60] dynamically selects LTE HOM/TTT pairs to minimize handover rate and latency while maximizing throughput, outperforming fuzzy-logic benchmarks. The approach in [61] integrates subtractive clustering with Q-learning to self-generate fuzzy membership functions and rules, reducing handover, ping-pong and failure rates while improving throughput and latency. Most existing HO optimization studies rely on simulations, while real drive-test datasets are typically limited to signal quality assessment; in contrast, our work integrates real-world LTE drive-test data with a novel tabular Q-learning framework for reinforcement learning-based HO optimization.

Finally, addressing the algorithmic perspective, the challenge of handover (HO) optimization lies in ensuring seamless connectivity as user equipment (UE) moves between different base stations (BS). Frequent and unnecessary handovers—especially ping-pong events—can significantly degrade network performance and user experience. The ping-pong effect, in particular, results in excessive signaling and increased power consumption, ultimately leading to Quality of Service (QoS) deterioration [66]. Traditional HO algorithms rely on static thresholds for parameters such as Reference Signal Received Power (RSRP) and Time-to-Trigger (TTT). However, these fixed heuristics often fail to adapt to real-time, dynamic radio environments, resulting in suboptimal decisions and frequent handovers [53], [67]. In response to these

limitations, Reinforcement Learning (RL) has emerged as a promising approach to dynamically adjust handover parameters such as Handover Margin (HOM) and TTT [68], [69]. Among various RL techniques, Q learning has gained popularity for its ability to optimize handover decisions without prior knowledge of the environment. It enables agents to learn optimal policies based on feedback, taking into account parameters like signal quality, mobility, and interference [70]. Several studies have extended Q-learning frameworks by integrating additional techniques. For instance, hybrid models combining Q-learning with subtractive clustering or fuzzy logic have been proposed to improve decision granularity and HO prediction accuracy [71]. These models enhance adaptability to heterogeneous radio environments by allowing finer control over threshold adjustments. Furthermore, in heterogeneous networks (HetNets), Q learning has been applied to dynamically adjust handover thresholds, thereby reducing ping-pong events and HO failures [61]. Other efforts integrate RL with optimization theory to jointly optimize base station selection and beamforming, improving throughput while maintaining low latency [72]. The use of Deep Q-Networks (DQN) has also been explored to manage real-time HO decisions in more complex and denser 5G environments. These networks incorporate a broad set of input features, including SINR, cell load, and UE speed, enabling more robust and adaptive handover mechanisms [73]. More recent proposals introduce intelligent mobility management mechanisms, where reinforcement learning is augmented with Kalman filters to predict signal trends from both serving and neighboring cells. By applying ϵ -greedy policies [74], these systems dynamically adjust TTT and hysteresis margins according to mobility patterns and signal quality [69]. To evaluate such algorithms, simulation environments like NS-3 have been widely used in prior works. NS-3-based studies simulate realistic 5G and LTE scenarios to benchmark RL-driven handover optimization under various mobility and load conditions [68]. While these simulations consistently show improved throughput, lower handover failures, and enhanced QoS, many of these studies remain theoretical. The transition from simulation to real-world deployment—where unpredictable behaviors, hardware limitations, and real radio propagation models come into play—still presents open challenges.

Extending from these insights, conventional HO schemes set fixed thresholds of RSRP, TTT and HOM but these static rules do not track rapidly changing radio conditions, which leads to poor decisions and excessive handovers [66, 53]. To overcome this rigidity, Reinforcement Learning has been used to adapt parameters such as HOM and TTT on the go [67, 68]. In particular, Q learning is attractive because it learns effective policies from experience without prior environment models, leveraging feedback tied to signal quality, user mobility, and interference levels [69]. Extensions of this approach combine Q-learning with techniques like subtractive clustering or fuzzy logic to refine decision granularity and improve HO prediction [70], offering finer control in heterogeneous settings. In HetNets, Q-learning has been employed to tune handover thresholds dynamically, curbing ping-pong events and HO failures [61]. Related work also couples RL with optimization methods to co-optimize base-station selection and beamforming, boosting throughput while keeping latency low [71]. Also, there are approaches using fuzzy logic. A fuzzy Type 1 controller to tune LTE HO parameters against multiple classic algorithms resulting in large reductions in average HOs and ping-pong events [78]. Another fuzzy-logic LTE HO parameter optimizer explicitly positions fuzzy rules to adapt TTT/HOM against speed and radio balance [79]. Fuzzy HO in heterogeneous network shows

that fuzzification mitigates ping-pong in multi-criteria HO decisions [80, 81]. An attention-enhanced rainbow-DQN-based joint traffic prediction to outperform existing methods in terms of the handover efficiency was used in [82]. An approach formalizes ping-pong HOs and shows how fuzzy decision making suppresses them better than hard thresholds by smoothing borderline scenarios [83]. The use of Deep Q-Networks (DQN) has also been explored too. An MDP is defined over LTE SON fault actions and compares heuristic queues to a DQN agent across metrics using the Vienna LTE simulator [84]. A double deep Q network (DDQN) deep reinforcement learning (DRL) framework was used to make blockage predictions based on past received signal data such that RLFs can be actively avoided [85]. DQN-ALrM formulates multi-metric HO and introduces an adaptive learning rate with momentum using DQN [86]. An ultra-dense 5G study uses three Deep-Q models to decide when to hand over, which cell to choose, and how to tune A2/A4 thresholds [87].

Chapter Three

Drive Test

This section encompasses details about the drive test location, time period, required equipment, software set up and data extraction methods.

3.1 Equipment Used in the Drive Test:

- XCAL-M software
- Samsung device with Exynos SoC chipset
- Laptop
- Android-based mobile phone
- Arduino Uno
- Neo-7M GPS module
- Vehicle

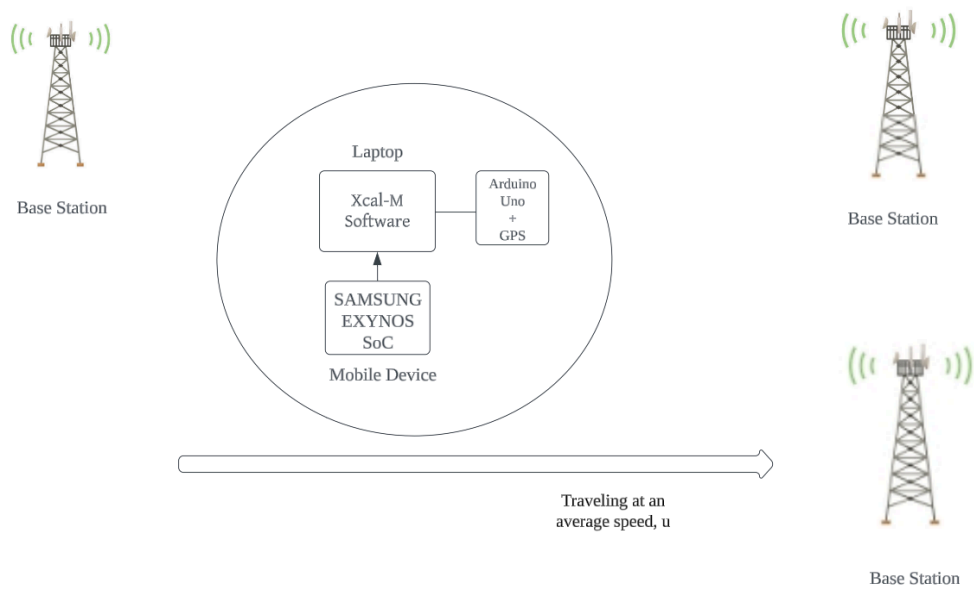


Figure 1: Drive Test Setup

3.2 Location of the Drive Test:

To select an appropriate location for the drive test, we utilized the publicly available cell ID database from the OpenCellID website, which provides approximate geolocations of cellular base stations. This database includes information for multiple Radio Access Technologies (RATs), namely GSM, UMTS, and LTE. For the purpose of this study, which focuses on LTE handover behavior, we specifically filtered and targeted LTE eNodeBs. We selected a region within Dhaka, Bangladesh characterized by a high density of LTE base stations to ensure frequent inter-cell handover events, which are critical for evaluating handover decision algorithms and mobility management performance.

The drive test route spanned approximately 13 kilometers, beginning at Le Meridien Dhaka (Uttara) and concluding at the BRAC University campus. This route includes variations in user mobility patterns such as stationary periods, pedestrian speeds, and vehicular movement under mixed traffic conditions. The measurements were collected using the Grameenphone LTE network, one of the largest mobile network operators in Bangladesh.

It is important to note that, due to the specific selection of the test area the use of a single Mobile Network Operator (MNO), the results obtained from this drive test may not be fully representative of broader network conditions across the city. Additionally, the findings may not generalize to other network operators with differing infrastructure deployments, frequency bands, or mobility management strategies.

Figure 2 and Figure 3 shows Open Cell ID website view of drive test location and Drive Test Route Map respectively.

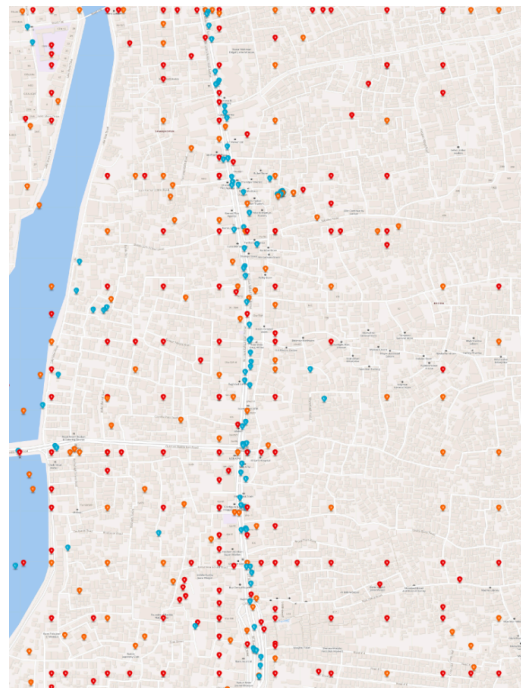


Figure 2: Open Cell ID website view of drive test location

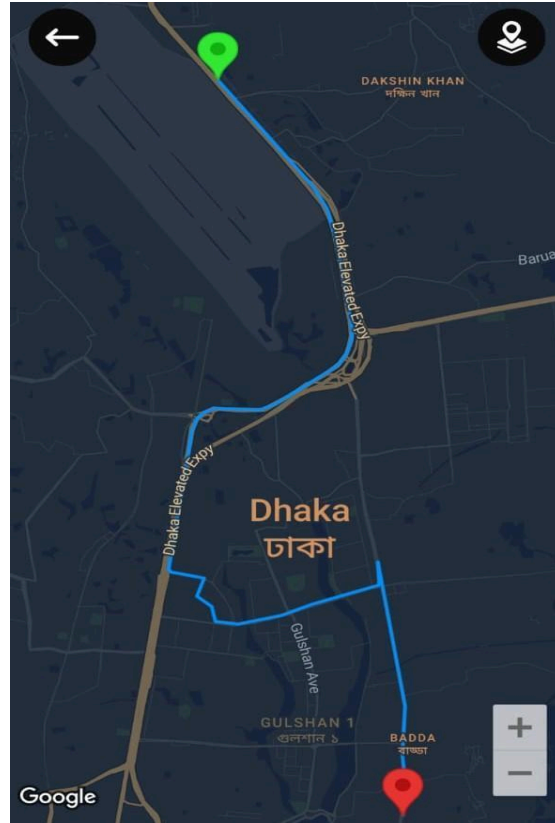


Figure 3: Drive Test Real Map

3.3 Time Period

The drive test was conducted over three separate days: October 15, November 6, and November 16, with measurements recorded during the evening hours between 16:00 and 22:00 (local time). On each day, the test route—spanning from Le Meridien Dhaka (Uttara) to BRAC University and back—was traversed in two complete rounds, with a 30-minute interval between each outbound and return trip. The time window was deliberately chosen to coincide with evening peak hours, a period characterized by high user density and network load, which can significantly influence radio resource management and handover dynamics. The selected route lies in one of the busiest corridors of Dhaka city, commonly subject to vehicular congestion and traffic delays. Such conditions introduced non-uniform mobility patterns (e.g., stop-and-go movement), which in turn affect radio measurements, particularly those related to handover triggering, RSRP fluctuation, and SINR degradation due to prolonged exposure to edge-of-cell conditions.

Traffic-induced delays allowed for denser spatial sampling of radio conditions, especially near cell edges. The short, repeatedly traversed route led to multiple encounters with the same LTE cell IDs under varying conditions, enhancing the dataset's temporal diversity. This redundancy improves the robustness and generalizability of machine learning models for handover prediction.

The network infrastructure in the area remained operationally static during the measurement period; that is, no observable commissioning or decommissioning of LTE eNodeBs took place. Therefore, temporal variations in collected Key Performance Indicators (KPIs) are attributed primarily to changes in user mobility and traffic conditions, rather than alterations in network topology or configuration. The average ambient temperature and relative humidity during the test periods were approximately 29°C and 72%, respectively—values typical of Dhaka's tropical climate in the post-monsoon season.

3.4 Software Set Up

For the execution of the drive test, we employed XCAL-M, a professional-grade drive test software developed by Accuver, installed on a laptop and interfaced with a Samsung Galaxy S10 device powered by the Exynos chipset. This setup enabled the collection of detailed Layer 1 to Layer 3 signaling information, radio frequency (RF) metrics, and protocol-level data essential for in-depth analysis of mobility and handover performance in LTE networks. The mobile device was connected via diagnostic mode to ensure direct access to chipset-level KPIs. In addition, we integrated an Arduino Uno microcontroller with a Neo-7M GPS module to continuously log the user equipment's geolocation (Latitude and Longitude) and UE velocity. This auxiliary setup ensured accurate tracking of mobility patterns, enabling correlation between user movement and observed network behavior.

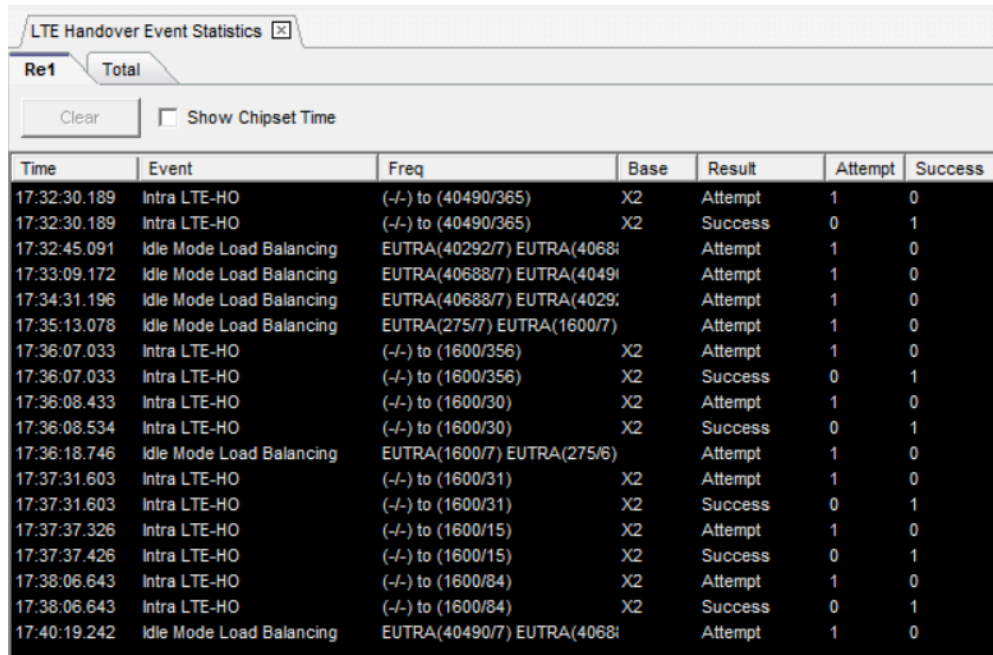
Compared to Android-based applications such as G-NetTrack Pro, which typically record measurements at one-second intervals due to OS-level constraints, XCAL-M offers millisecond-level granularity, making it substantially more suitable for fine-grained temporal and spatial analysis of handover events and transient radio conditions.

The Samsung S10 device, serving as the user equipment (UE), was placed inside a private test vehicle, and the entire system was configured for real-time data acquisition as the vehicle traversed the defined route. Throughout the drive, XCAL-M continuously captured network performance metrics and handover signaling messages, synchronized with precise GPS coordinates and time stamps.

3.5 Data Extraction from XCAL-M

Following the completion of the drive test, network performance data was extracted using the built-in post-processing capabilities of XCAL-M. To analyze mobility and handover-related events, we navigated to the "Layer3 KPI" module, selected "LTE", and then accessed the "LTE Handover Event Statistics" window. This interface provided detailed handover metrics, including the number of handover attempts, successful handovers, and corresponding event types along

with precise timestamps, enabling temporal correlation with user movement and signal conditions. Figure 4 shows the Event Statistics for Handover Information Log.



Time	Event	Freq	Base	Result	Attempt	Success
17:32:30.189	Intra LTE-HO	(-/) to (40490/365)	X2	Attempt	1	0
17:32:30.189	Intra LTE-HO	(-/) to (40490/365)	X2	Success	0	1
17:32:45.091	Idle Mode Load Balancing	EUTRA(40292/7) EUTRA(40688/7)		Attempt	1	0
17:33:09.172	Idle Mode Load Balancing	EUTRA(40688/7) EUTRA(40490/365)		Attempt	1	0
17:34:31.196	Idle Mode Load Balancing	EUTRA(40688/7) EUTRA(40292/7)		Attempt	1	0
17:35:13.078	Idle Mode Load Balancing	EUTRA(275/7) EUTRA(1600/7)		Attempt	1	0
17:36:07.033	Intra LTE-HO	(-/) to (1600/356)	X2	Attempt	1	0
17:36:07.033	Intra LTE-HO	(-/) to (1600/356)	X2	Success	0	1
17:36:08.433	Intra LTE-HO	(-/) to (1600/30)	X2	Attempt	1	0
17:36:08.534	Intra LTE-HO	(-/) to (1600/30)	X2	Success	0	1
17:36:18.746	Idle Mode Load Balancing	EUTRA(1600/7) EUTRA(275/6)		Attempt	1	0
17:37:31.603	Intra LTE-HO	(-/) to (1600/31)	X2	Attempt	1	0
17:37:31.603	Intra LTE-HO	(-/) to (1600/31)	X2	Success	0	1
17:37:37.326	Intra LTE-HO	(-/) to (1600/15)	X2	Attempt	1	0
17:37:37.426	Intra LTE-HO	(-/) to (1600/15)	X2	Success	0	1
17:38:06.643	Intra LTE-HO	(-/) to (1600/84)	X2	Attempt	1	0
17:38:06.643	Intra LTE-HO	(-/) to (1600/84)	X2	Success	0	1
17:40:19.242	Idle Mode Load Balancing	EUTRA(40490/7) EUTRA(40688/7)		Attempt	1	0

Figure 4: Event Statistics for Handover Information Log

For extracting information on serving cell parameters, we utilized the “Call Statistics” option, selected the “LTE Summary Table”, and further filtered the data by choosing the “Serving Cell” view. This yielded essential serving cell identifiers, including the Physical Cell ID (PCI) and Carrier-to-Interference-plus-Noise Ratio (CINR). Figure 5 shows the LTE Summary Table. Figure 5 shows LTE Summary Table Tab.

To obtain detailed radio measurement data, we accessed the “Message” tab, selected “Signaling Message”, and filtered for “Measurement Report” messages. These messages included Serving Cell RSRP, RSRQ, and corresponding Neighbor Cell RSRP, RSRQ, along with Neighbor Cell IDs, all timestamped for precise temporal alignment. Measurement reports play a crucial role in LTE networks, as they are fundamental to the handover decision-making process, enabling the eNodeB to evaluate surrounding cell conditions and initiate mobility procedures based on preconfigured event thresholds. Figure 6 shows Signalling Message Tab.

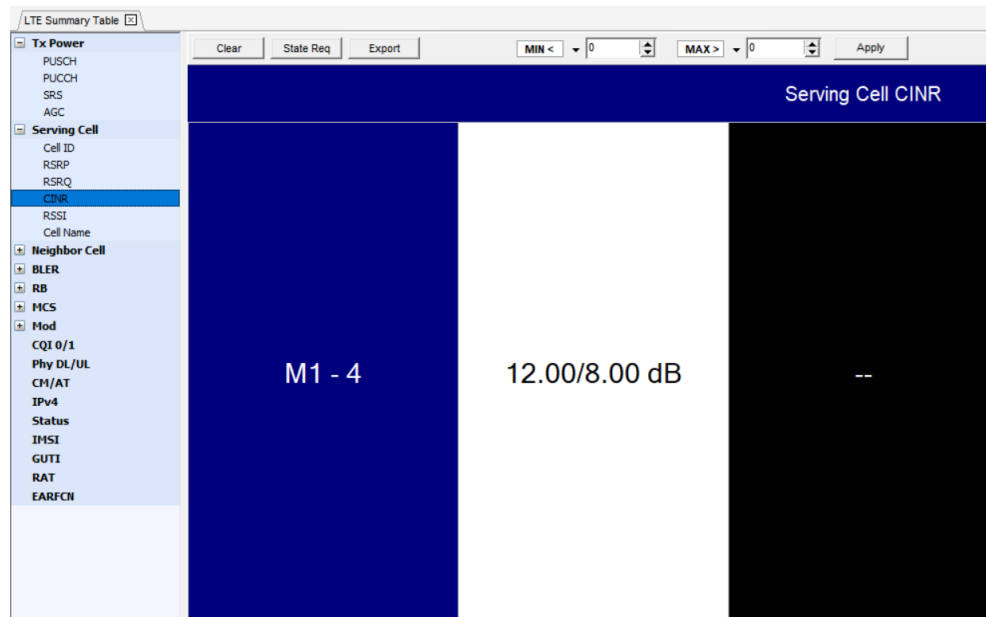


Figure 5: LTE Summary Table

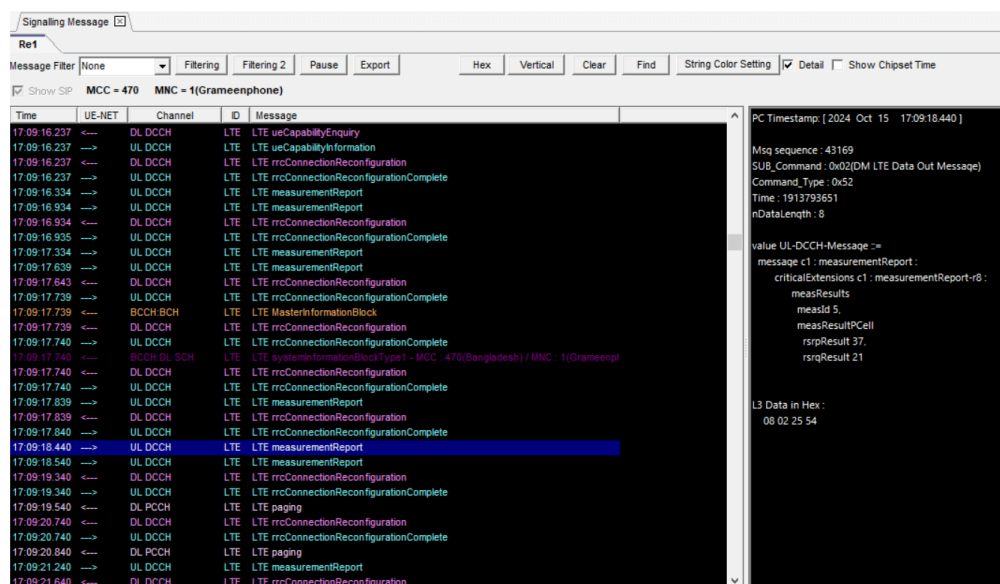


Figure 6: Signalling Message

In LTE, Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ) are critical physical layer metrics reported by the User Equipment (UE) in Layer 3 Measurement Reports, which are governed by the MeasConfig Information Element (IE). The S-Measure field within MeasConfig defines the RSRP threshold below which the UE initiates neighbor cell measurements. If S-Measure = 0, the UE is configured to perform neighbor measurements continuously, regardless of the serving cell's RSRP level.

RSRP and RSRQ thresholds are quantized using RSRP-Range and RSRQ-Range IEs, respectively. The RSRP-Range IE supports integer values from 0 to 97, each corresponding to a 1 dB interval from below -140 dBm up to and beyond -44 dBm. The RSRQ-Range IE spans 0 to 34, mapping to standard RSRQ intervals. These quantized values appear in the Measurement Report under the signaling messages tab in XCAL-M software.

3.6 Data Preprocessing:

The objective of applying machine learning (ML) to the drive test dataset is to model and predict handover decisions based on real-world LTE mobility patterns. In this context, the handover serves as the target variable, while the number of handovers is used as the primary performance metric, with a focus on minimizing unnecessary handovers and ping-pong events through reinforcement learning-based optimization to enhance network stability and efficiency. To prepare the dataset for ML models, a series of preprocessing steps were applied: missing RSRP and RSRQ values were interpolated manually based on serving cell IDs and timestamp continuity, data segments with hardware anomalies were excluded, and all non-LTE measurements were filtered out. Furthermore, the concept of Time-to-Trigger (TTT)—a standardized delay (set as 320 ms in our study) before a handover is executed—was incorporated. While raw handover labels were derived directly from Cell ID transitions in the Event Statistics log, the refined dataset assigns a handover label to those data points that contribute to a handover decision within the TTT window, thereby capturing the pre-handover context necessary for accurate model training.

Chapter Four

Overview of Handover Dataset

4.1 Dataset Structure:

The dataset is organized into five primary directories: Parent Datasets, Processed Datasets, DRM Files, Measurement Reports, and Event Statistics. Within the Parent Datasets folder, three subfolders—Raw Dataset_1, Raw Dataset_2, and Raw Dataset_3—contain 7, 2, and 3 CSV files respectively, representing the unprocessed drive test data. The Processed Datasets directory mirrors this structure with Processed Dataset_1, Processed Dataset_2, and Processed Dataset_3, each containing one CSV file with cleaned and ML-ready data. The preprocessing steps applied to these files are detailed in the following section. The Event Statistics directory comprises three subfolders—Event Statistics_Dataset_1, Event Statistics_Dataset_2, and Event Statistics_Dataset_3—containing 7, 2, and 3 CSV files respectively. These files provide detailed LTE handover attempt and success records with timestamped events. Similarly, the DRM Files folder includes DRM Files_Dataset_1, DRM Files_Dataset_2, and DRM Files_Dataset_3, which contain the original XCAL-M log files (.drm format) with 7, 2, and 3 files respectively, enabling full reconstruction of the measurement sessions. The Measurement Reports directory features Measurement Reports_Dataset_1, Measurement Reports_Dataset_2, and Measurement Reports_Dataset_3, containing 7, 2, and 3 plain text files respectively, extracted from the Layer 3 signaling messages. Including Measurement Reports, Event Statistics, and DRM log files helps researchers trace the raw signaling and verify handover behavior, ensuring reproducibility and deeper analysis of real-world LTE mobility patterns.

The dataset contains 10 features and 1 target variable. All of them are numerically valued columns. There are 647 data points and 94 handover instances, indicating imbalance in the dataset.

The dataset structure is shown in Figure 7. Table-1 shows the feature titles and the target variable.

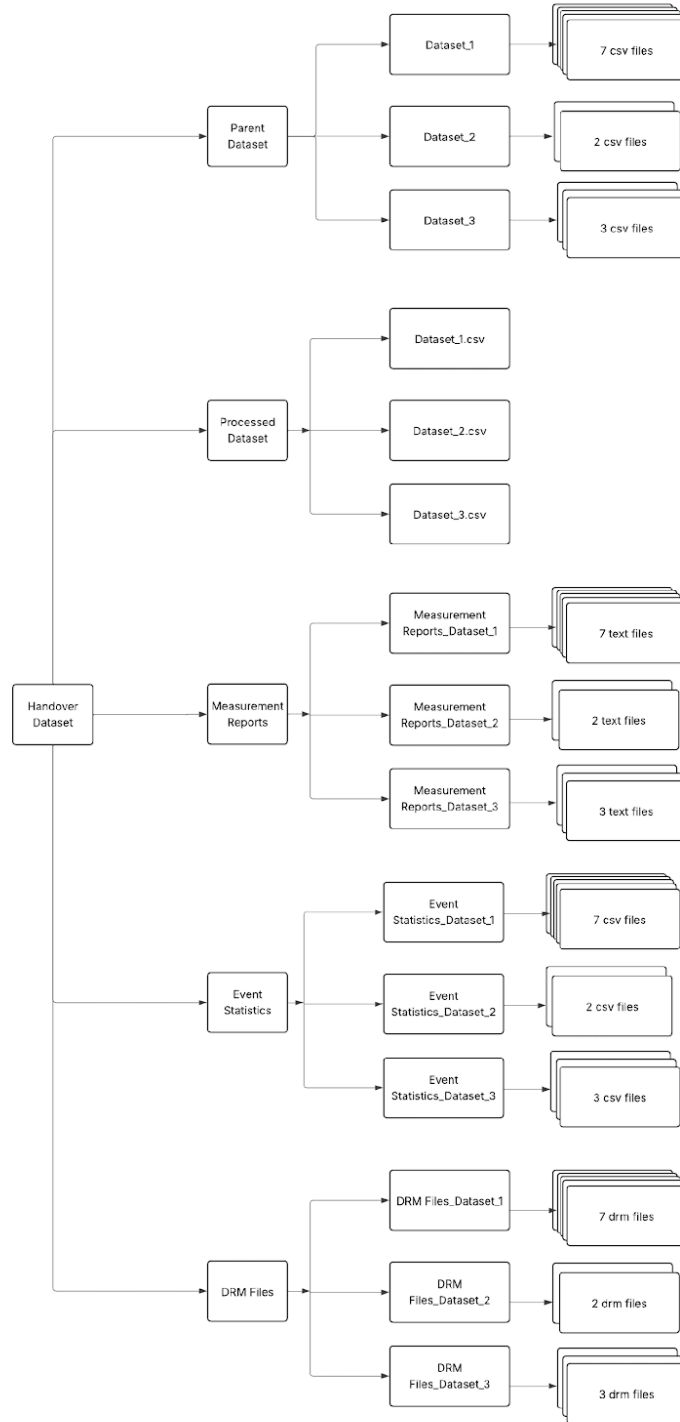


Figure 7: Dataset Structure

Table 1 Dataset Columns

Features	Serving Cell Id
	Serving Cell RSRP
	Serving Cell RSRQ
	Serving Cell CINR
	Neighbor Cell Id
	Neighbor Cell RSRP
	Neighbor Cell RSRQ
	User Speed
	Latitude
	Longitude
Target Variable	Handover

Table 2 presents the parent dataset collected from the LTE drive test, while Table 3 shows the processed version of this dataset after cleaning and feature selection. Subsequently, Table 4 illustrates the handover instances observed with respect to the traveled distance during the test route.

Table 2 Parent Dataset

Dataset Folder	Dataset Names	Number of Columns	Data Points
Dataset_1	File 1	11	132
	File 2	11	320
	File 3	11	74
	File 4	11	46
	File 5	11	243
	File 6	11	11
	File 7	11	654
Dataset_2	File 1	14	723
	File 2	14	798
Dataset_3	File 1	14	160
	File 2	14	339
	File 3	14	573

Table 3 Processed Dataset

Dataset Names	Number of Columns	Data Points
Dataset_1	11	922
Dataset_2	14	592
Dataset_3	14	641

Table 4 Handover Instances with respect to travelled distance

Time Period	Handover Count	Total Distance Travelled (km)	Distance Travelled per Handover in meter
Day 1	129	26.21	203
Day 2	94	25.54	268
Day 3	107	27.28	254

Table 2 summarizes the raw (parent) dataset collected from the LTE drive test, segmented into three folders based on distinct testing sessions or configurations. Each dataset comprises multiple files, with Dataset_1 containing seven files characterized by a consistent column count of 11, but a wide range of data points (from 11 to 654), indicating variability in recording durations or sampling intervals. Datasets 2 and 3 exhibit a slightly higher dimensionality with 14 columns, suggesting the inclusion of additional parameters latitude, longitude and user speed. The total number of data points across files provides a comprehensive foundation for further analysis but also reflects the heterogeneous nature of the raw data.

Table 3 presents the processed dataset, obtained after data cleaning, consolidation, and possibly filtering out noise or irrelevant records. The cleaned dataset shows a more balanced distribution of data points: Dataset_1 retains 922 points, while Datasets 2 and 3 hold 592 and 641 points, respectively. This reduction from the parent dataset reflects preprocessing efforts such as removing null entries, correcting inconsistencies, and ensuring uniform formatting—all critical for preparing data suitable for machine learning models.

Table 4 provides insights into handover behavior across three consecutive drive test days. Day 1 recorded 129 handovers over 26.21 km, resulting in an average of 203 meters per handover. Day 2 showed fewer handovers (95) over a similar distance, yielding 268 meters per handover, suggesting improved mobility handling or different route characteristics. Day 3 fell between the two with 107 handovers over 27.28 km, averaging 254 meters per handover. These variations reflect differences in network topology, traffic conditions, and signal propagation environments on different days. The metric "distance travelled per handover" serves as a critical performance indicator, where higher values typically imply more stable handover decisions and fewer disruptions, thus aiding in quantitative evaluation of mobility management efficiency.

4.2 Dataset Statistics:

Table 5, Table 6, and Table 7 present the descriptive statistics for Dataset_1, Dataset_2, and Dataset_3, respectively. These statistics summarize key wireless signal quality indicators such as Serving Cell RSRP, RSRQ, CINR, and their corresponding Neighbor Cell metrics. Dataset_2 and Dataset_3 also include speed as a contextual mobility parameter.

Table 5 Descriptive statistics of Dataset_1

	Serving Cell RSRP	Serving Cell RSRQ	Serving Cell CINR	NeighCell RSRP	NeighCell RSRQ
count	922	922	922	922	922
mean	47.04446855	12.40997831	-2.023318872	45.64533623	9.252711497

std	13.03553886	7.225018978	10.80873338	11.61539961	7.744573122
min	13	0	-30	1	0
25%	42	7	-6.5	41	3
50%	48	13	-2.5	46	7
75%	54	17	2	53	14
max	78	51	31.5	77	45

Table 6 Descriptive statistics of Dataset_2

	Speed (km/h)	Serving Cell RSRP	Serving Cell RSRQ	Serving Cell CINR	NeighCell RSRP	NeighCell RSRQ
count	592	592	592	592	592	592
mean	3.333044	52.12837838	11.19087838	-1.352195946	50.30152	7.787162
std	3.983686	9.307152358	6.049140385	5.856911567	9.122167	6.026897
min	0	21	0	-15	22	0
25%	0.445	48	7	-5	46	2.75
50%	1.22	53	11	-1.5	52	7
75%	5.4525	57	16	1.5	56	12
max	14.49	76	24	25	76	24

Table 7 Descriptive statistics of Dataset_3

	Speed (km/h)	Serving Cell RSRP	Serving Cell RSRQ	Serving Cell CINR	NeighCell RSRP	NeighCell RSRQ
count	641	641	641	641	641	641
mean	23.4493	52.19032761	11.82371295	-1.06318	51.04680187	7.689547582
std	14.61242	10.66835127	7.061679108	8.310685	10.07895362	6.278338252
min	0	19	0	-30	18	0
25%	11	44	6	-4.5	44	3
50%	26	53	12	-1	51	7
75%	34	60	16	4	58	11

Table 5 presents the descriptive statistics for Dataset_1, which does not include user speed but provides insight into the variability of radio signal conditions. The Serving Cell RSRP has a mean of approximately 47 dBm, with values ranging from 13 dBm to 78 dBm, indicating diverse signal strengths experienced along the test route. The CINR (Carrier-to-Interference-plus-Noise Ratio) exhibits a wide range from -30 dB to 31.5 dB, reflecting fluctuations between highly degraded and very favorable radio environments. The RSRQ values also span a broad range (0 to 51), highlighting dynamic interference and load conditions.

Table 6 shows the statistics for Dataset_2, which includes speed information, with a mean speed of around 3.33 km/h, indicating primarily stationary or low-mobility conditions. Under these circumstances, the Serving Cell RSRP and CINR metrics are generally stronger and more

stable, with average values of 52 dBm and -1.35 dB, respectively. The lower variability in signal quality metrics compared to Dataset_1 suggests improved link reliability and reduced handover frequency, aligning with the expected performance in static or pedestrian scenarios. Table 7 corresponds to Dataset_3, where the mean speed is substantially higher at 23.45 km/h, representing vehicular mobility. The Serving Cell RSRP remains strong on average (~52.2 dBm), but the CINR and RSRQ metrics show greater dispersion, with CINR ranging from -30 dB to 25 dB and RSRQ up to 31. These broader signal variations under high-speed conditions suggest more frequent signal degradation and recovery phases, likely leading to increased handover events and signaling overhead. This dataset is therefore highly relevant for training mobility-aware handover prediction models under realistic, high-mobility scenarios.

4.3 Exploratory Data Analysis (EDA):

Figure 8, Figure 9, and Figure 10 present the correlation matrices for Dataset_1, Dataset_2, and Dataset_3, respectively. These matrices reveal the pairwise linear relationships between key wireless signal parameters.

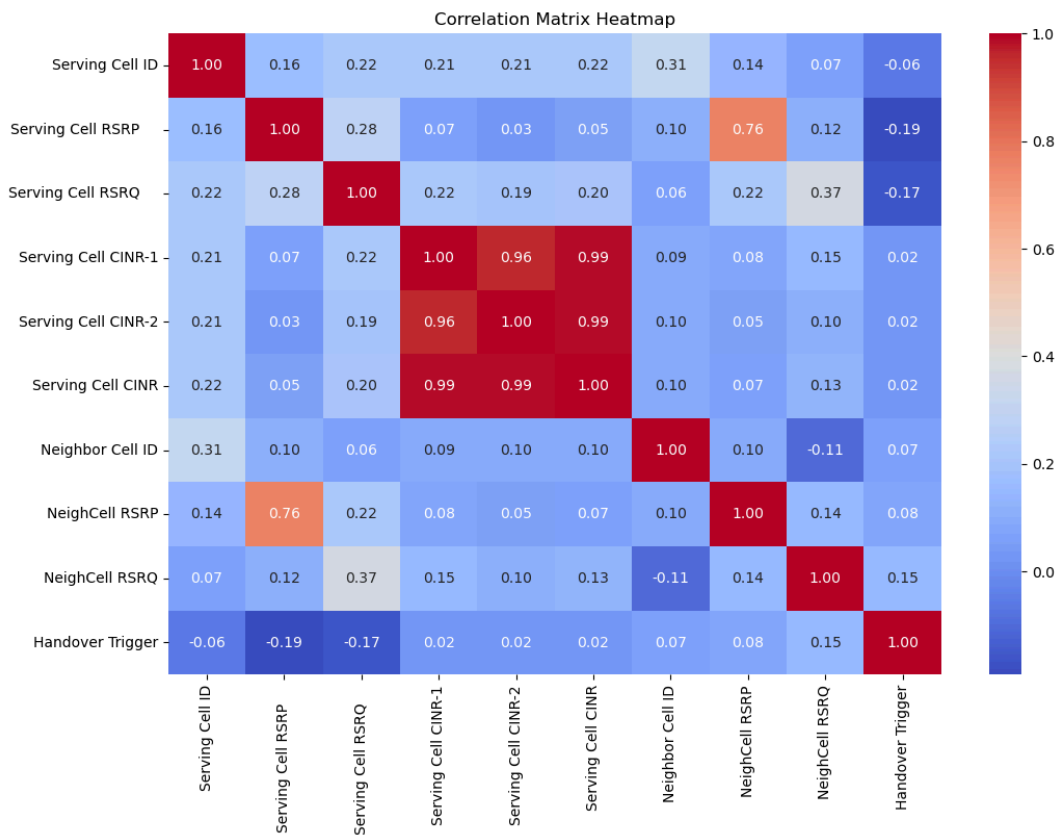


Figure 8: Correlation matrices for Dataset_1

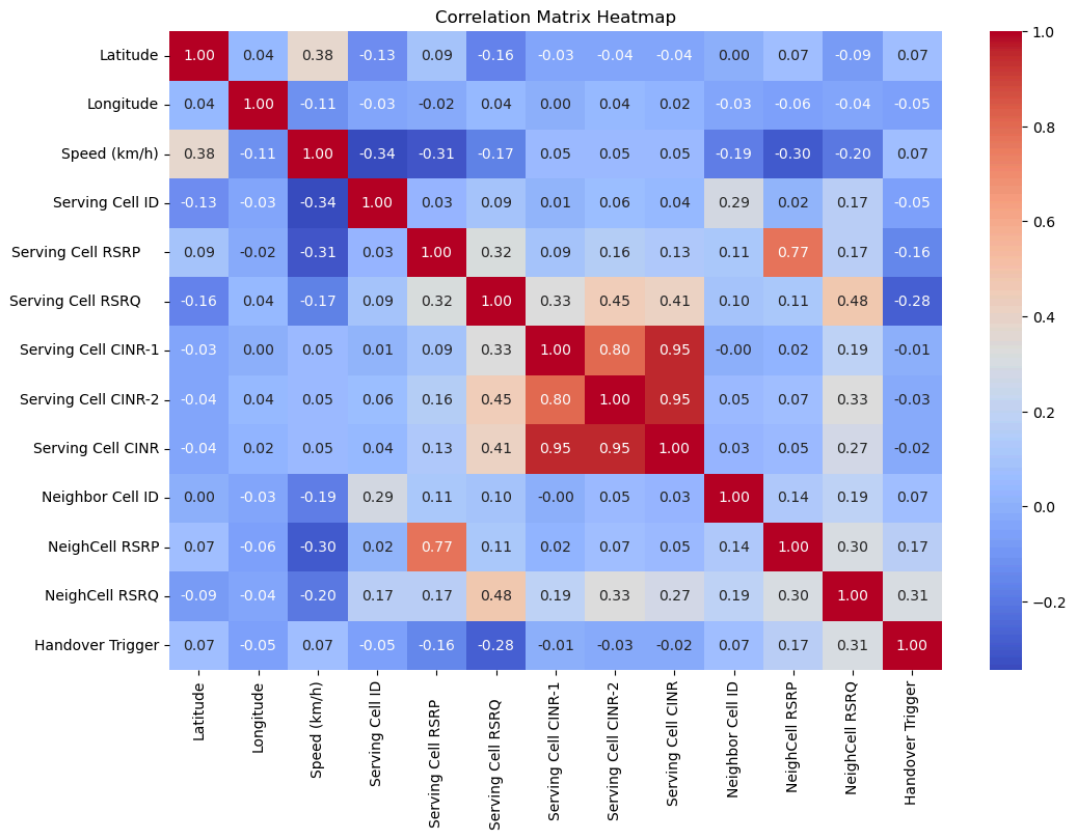


Figure 9: Correlation matrices for Dataset_2

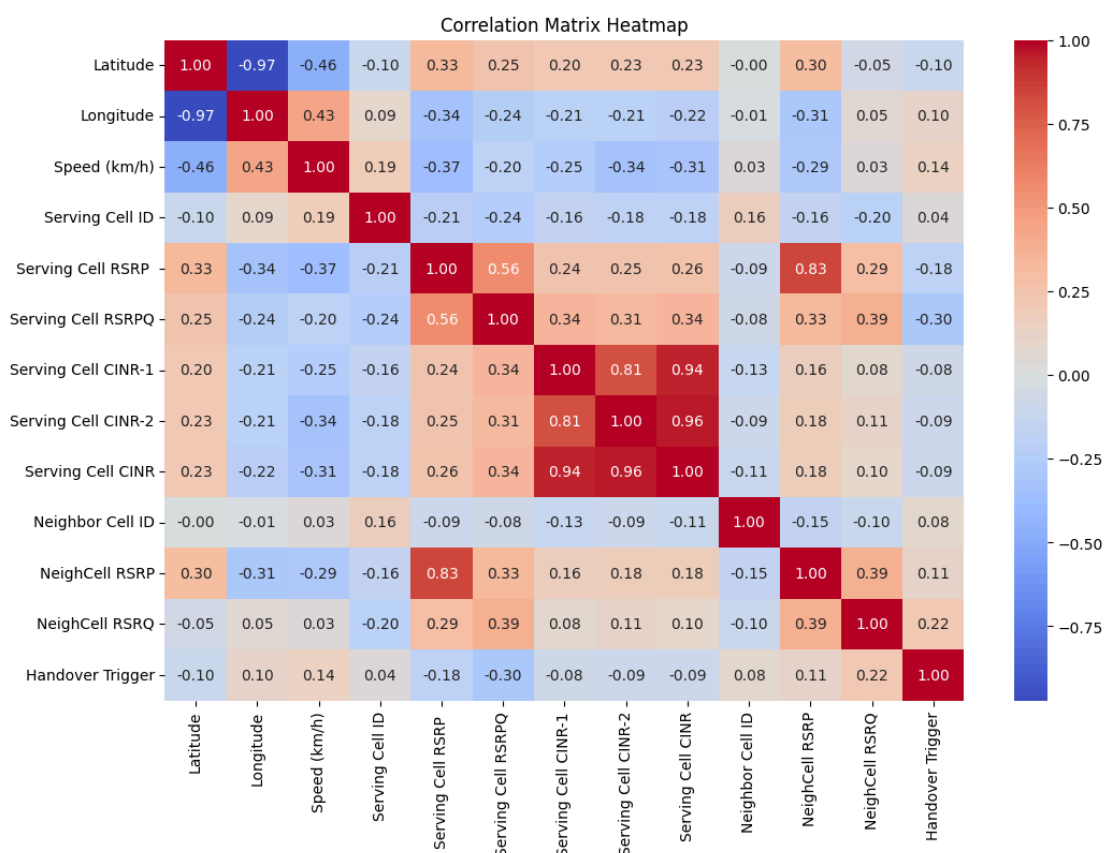


Figure 10: Correlation matrices for Dataset_3

Figure 11, Figure 12, and Figure 13 illustrate the handover frequency distribution across serving cells for Dataset_1, Dataset_2, and Dataset_3, respectively. These figures provide a visual representation of how frequently each serving cell initiated a handover during the drive test sessions.

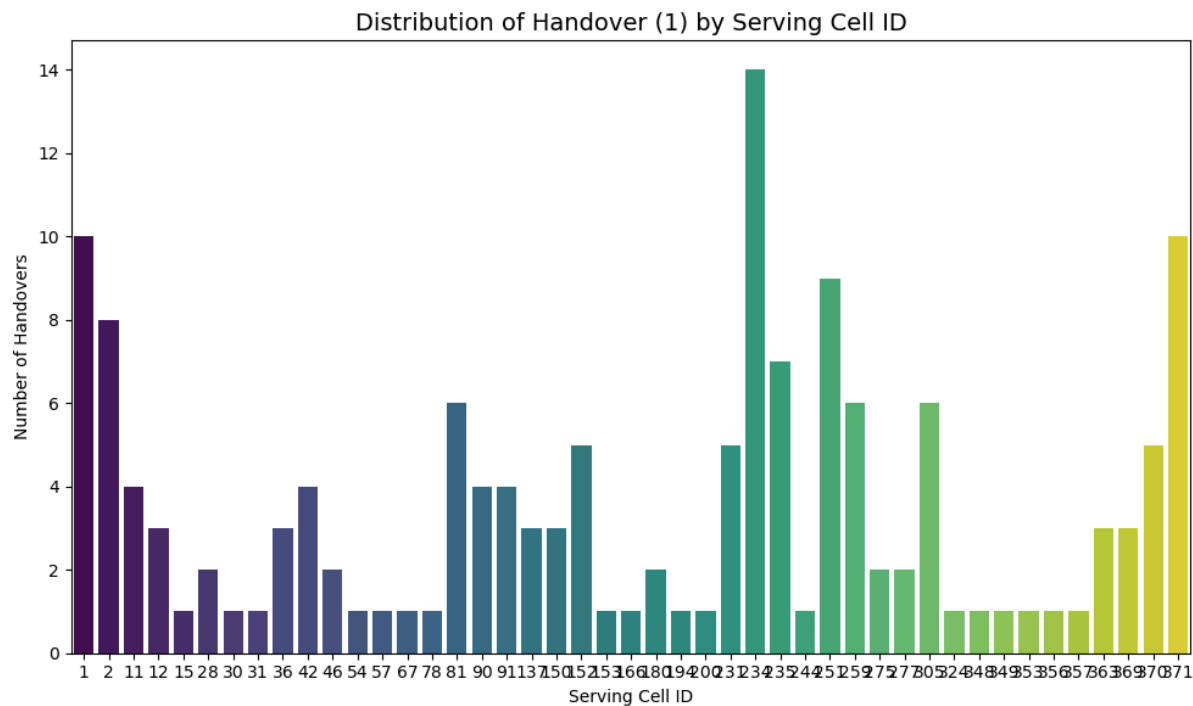


Figure 11: Handover Frequency over Serving Cells for Dataset_1

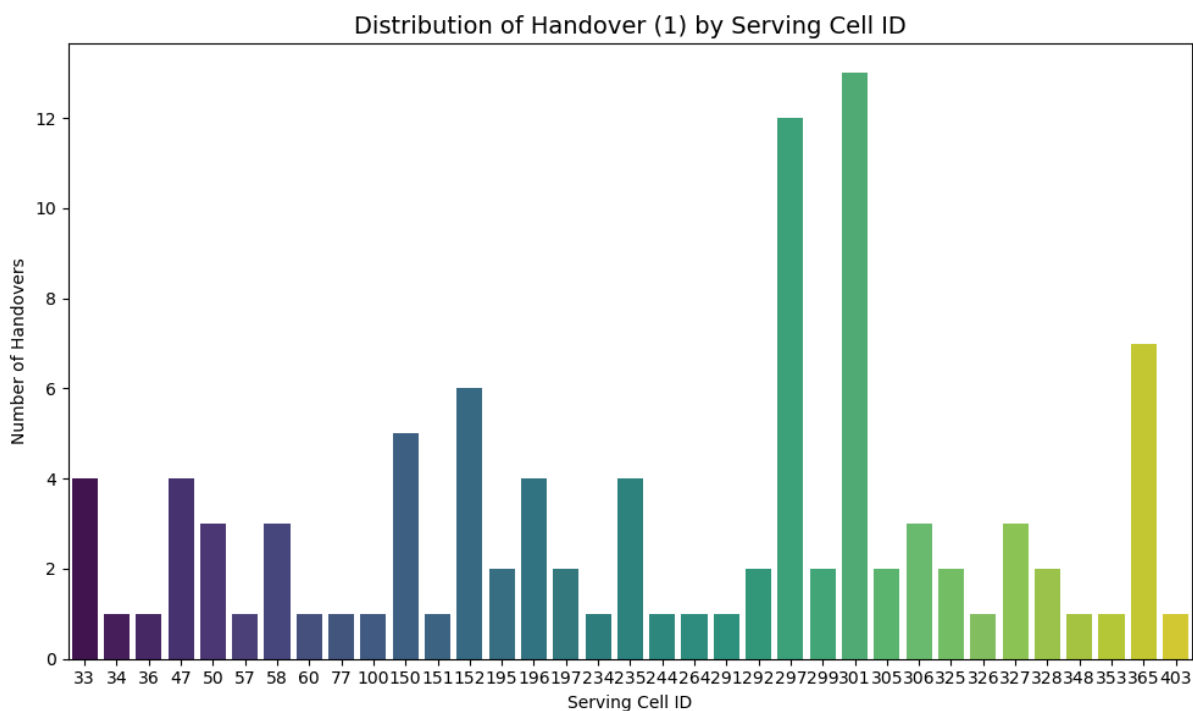


Figure 12: Handover Frequency over Serving Cells for Dataset_2

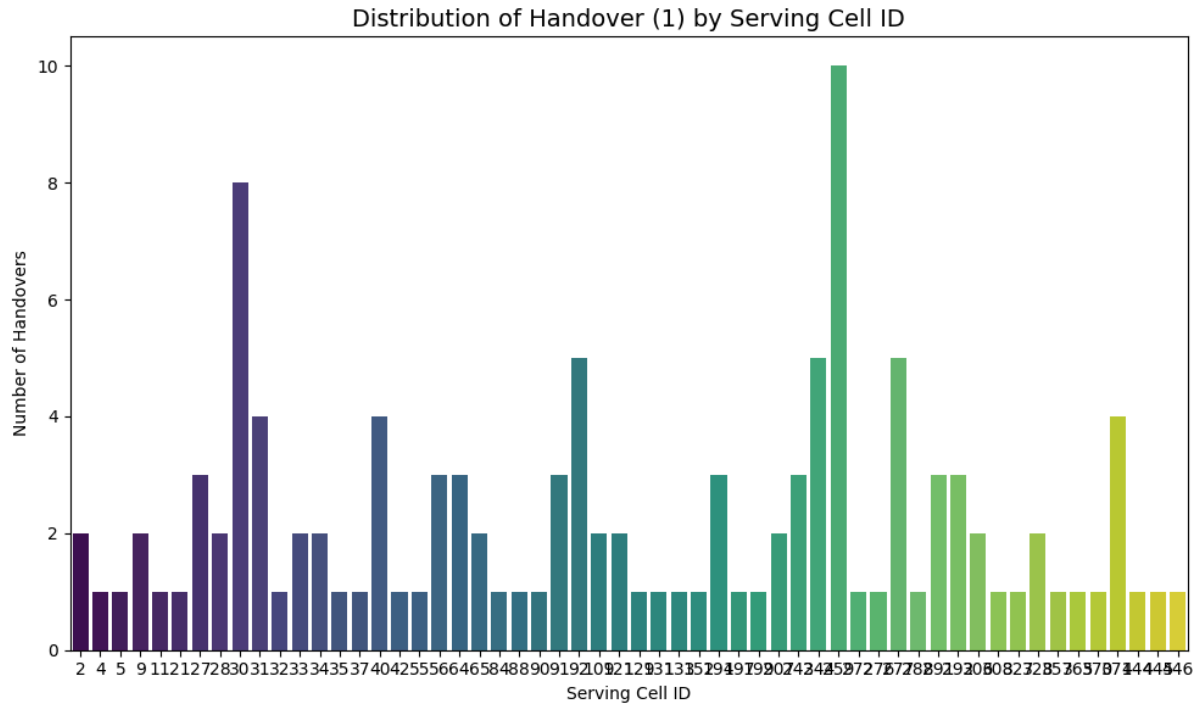


Figure 13: Handover Frequency over Serving Cells for Dataset_3

Figure 14, Figure 15, and Figure 16 illustrate the distribution of key network parameters—including UE speed, Serving Cell RSRP, RSRQ, CINR, and Neighbor Cell RSRP and RSRQ—for Dataset_1, Dataset_2, and Dataset_3, respectively. These figures provide insights into the statistical behavior of radio signal measurements and user mobility across varying conditions.

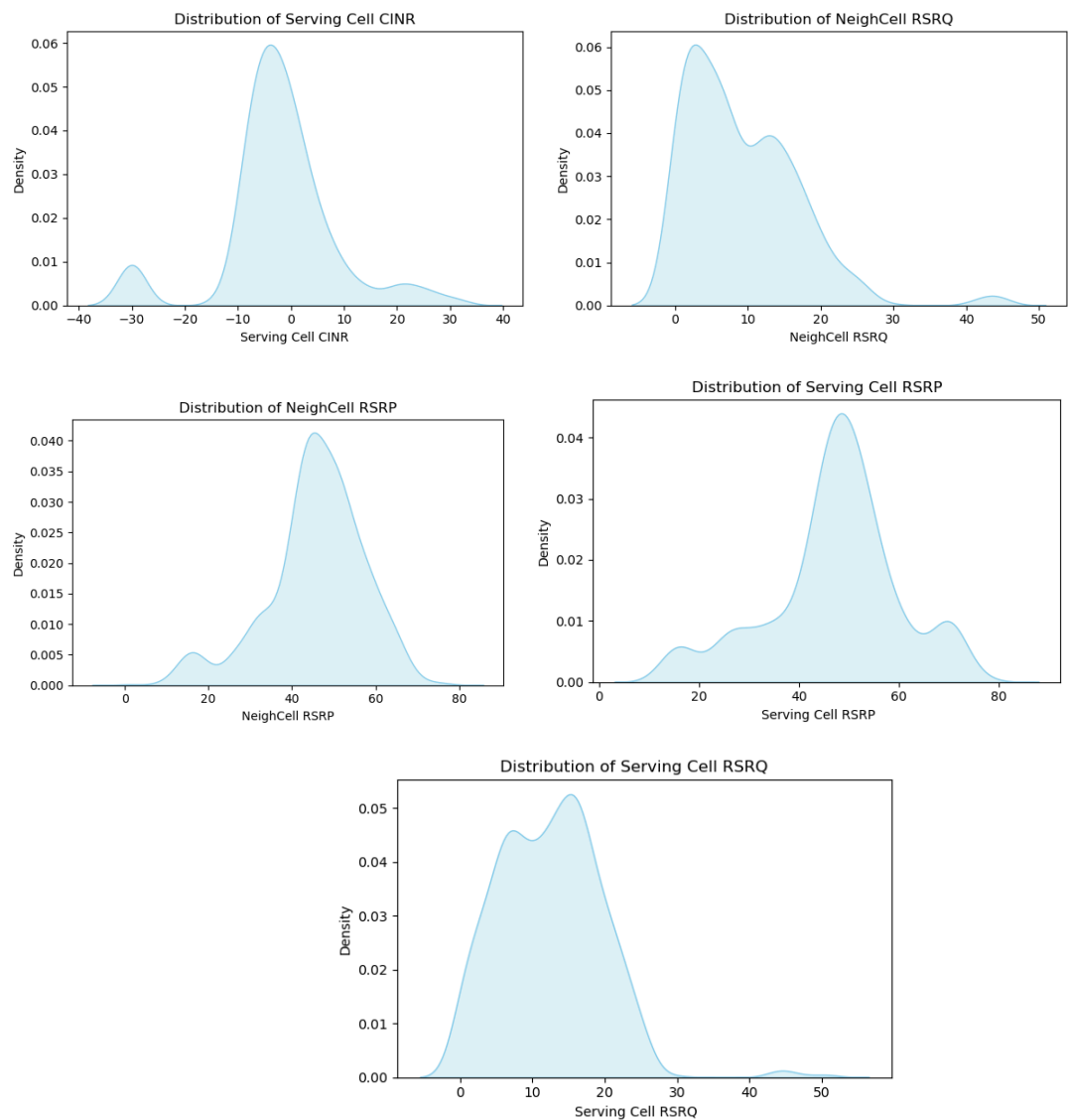


Figure 14: Distribution of Network Parameters of Dataset_1

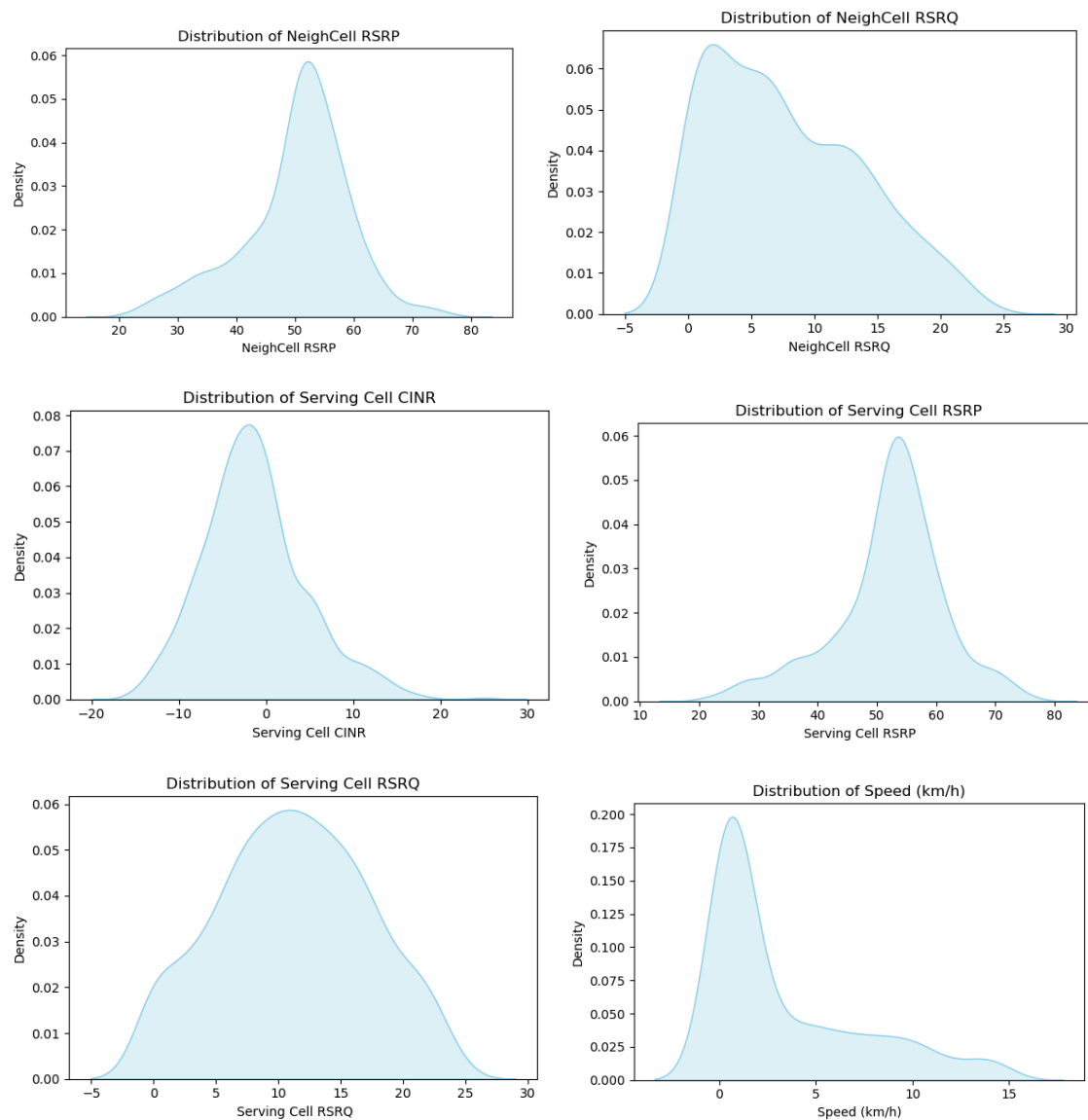


Figure 15: Distribution of Network Parameters of Dataset_2

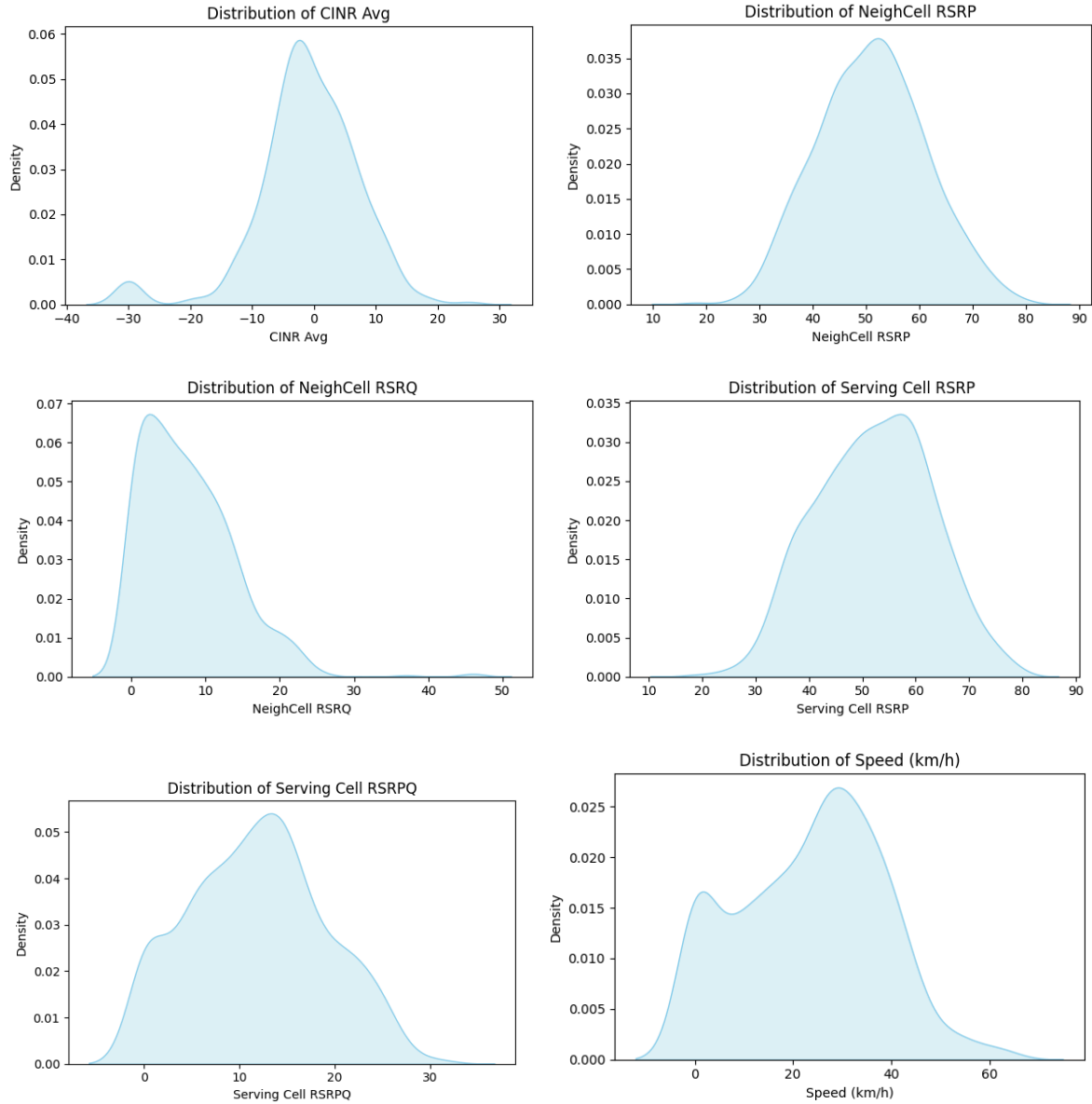


Figure 16: Distribution of Network Parameters of Dataset_3

The visual distributions indicate that the aforementioned parameters approximately follow a normal (Gaussian) distribution in each dataset. This is particularly evident in Serving Cell RSRP, RSRQ, and CINR, which display symmetric bell-shaped curves centered around their respective means. In Dataset_2 and Dataset_3, UE speed also exhibits a unimodal distribution with slight skewness due to the nature of user movement during the drive tests.

Figure 17 and Figure 18 shows the relation between RSRP and RSRQ values for both serving cell and neighbor cell in scatter plot.

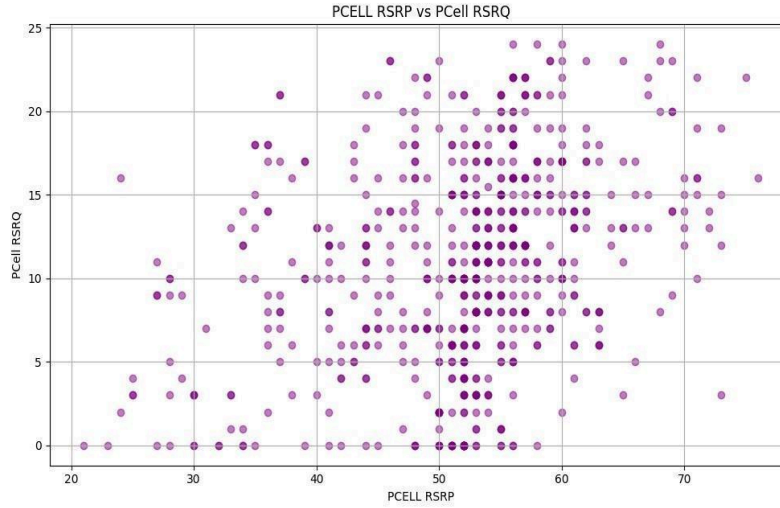


Figure 17: Scatter plot of Serving Cell RSRP and RSRQ

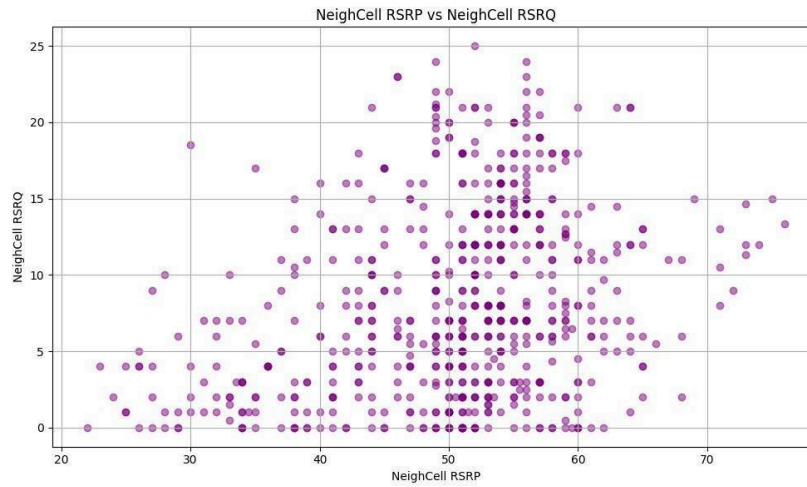


Figure 18: Scatter plot of Neighbor Cell RSRP and RSRQ

In the following chapters, we discuss the research aspects based on the dataset we generated. Our study focuses on three key aspects: Quality Signal Indicator (QSI), Handover Prediction, and Handover Optimization. We have selected Dataset_2 from the Processed Dataset folder for our experiment. Figure 19 shows the overview of our thesis.

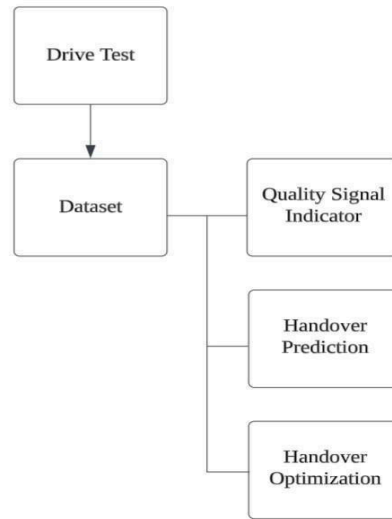


Figure 19: Thesis Overview

Chapter Five

Data Stratification

5.1 Signal Quality Labeling and Categorization

The dataset compiled for this study captures real-time radio signal quality indicators fundamental to cellular network performance assessment. These include reference signal received power (RSRP), reference signal received quality (RSRQ), and carrier-to-interference-plus-noise ratio (CINR), each measured for the serving cell. These parameters are widely recognized in 3GPP standards as primary indicators of downlink signal conditions and are instrumental in evaluating coverage, interference levels, and overall user experience within an LTE network.

To enable supervised learning-based model development, a manual threshold-based labeling scheme was employed to categorize the signal quality into four distinct classes: excellent, good, moderate, and poor. This categorization was guided by empirical domain knowledge and conventional telecom thresholds typically used in network optimization and planning processes. For instance, higher RSRP values generally correspond to stronger signal reception, while RSRQ and CINR provide insights into network load and interference environments.

This labeling approach constitutes a self-supervised learning methodology, wherein labels are derived from the input data itself using rule-based conditions rather than external annotation. Predefined thresholds were applied conditionally to each of the three signal metrics, and a composite rule was developed to assign one of the four quality labels to each data instance. For example, a sample exhibiting high RSRP, low RSRQ degradation, and a high CINR was labeled as "excellent", whereas a sample with significantly low RSRP and degraded RSRQ or CINR was labeled as "poor".

5.2 Model Training on Self Supervised Data

In this study, a comprehensive evaluation of nine machine learning classifiers was conducted to address a multiclass classification problem focused on wireless signal quality assessment. The classifiers employed include logistic regression, random forest, k-nearest neighbors (KNN), gradient boosting, XGBoost, AdaBoost, decision tree, naïve Bayes, and support vector machine (SVM). These algorithms were selected due to their proven effectiveness across a range of supervised learning tasks, as well as their complementary strengths in handling non-linearity, overfitting, interpretability, and scalability.

The input data was derived from a self-supervised signal quality labeling process using K-means clustering, ensuring that the labels represent a meaningful separation of network conditions based on data patterns rather than static thresholding. The resulting target variable, quality signal indicator (QSI), was categorized into four qualitative classes: excellent, good, moderate, and poor, based on principal wireless signal parameters.

The models were trained on key radio features that significantly influence user experience and network performance: reference signal received power (RSRP), reference signal received quality (RSRQ), and signal-to-noise ratio (SNR). These features collectively characterize both the strength and quality of the downlink radio link and are widely used in radio network planning, optimization, and handover decision-making.

Each classifier was trained and validated using stratified sampling to preserve class distribution, ensuring fair and unbiased performance evaluation. Hyperparameter tuning was performed where appropriate to optimize learning efficiency and generalization. The goal was to build a robust, scalable, and interpretable predictive model that can accurately classify the signal quality class in various network scenarios—ranging from low-mobility urban environments to high-speed mobility contexts—thus supporting intelligent decision-making in handover optimization, quality of service (QoS) assurance, and coverage analysis within LTE or other advanced cellular systems.

5.3 Dimensionality Reduction Using Principal Component Analysis (PCA)

To improve computational efficiency and reduce feature redundancy, principal component analysis (PCA) was applied as a dimensionality reduction technique prior to clustering. PCA is a powerful linear transformation method that projects high-dimensional data onto a new coordinate system, where each axis—referred to as a principal component—is a linear combination of the original features and is orthogonal to the others. These components are ordered such that the first principal component captures the maximum possible variance in the dataset, the second captures the next highest variance under the constraint of orthogonality, and so on.

In the context of this study, the original feature set comprised multiple correlated wireless signal quality metrics, including serving cell RSRP, RSRQ, CINR, and neighbor cell RSRP/RSRQ, which often exhibit interdependencies due to radio propagation behavior and network configuration. PCA was employed to transform this multi-dimensional feature space into a set of uncorrelated principal components, thereby eliminating multicollinearity and emphasizing the most informative aspects of the data.

By retaining only the top k principal components (where k is chosen based on the cumulative explained variance criterion, typically $\geq 95\%$), the dimensionality of the dataset was significantly reduced without substantial loss of information. This not only accelerates the performance of subsequent clustering algorithms such as K-means by lowering the computational burden, but also improves the stability and separability of clusters by focusing on the most discriminative directions in the feature space.

Ultimately, PCA facilitated the creation of a more compact and representative dataset, which is crucial for large-scale network performance assessments and real-time handover optimization systems, where both speed and accuracy are paramount.

5.4 Clustering and Centroid Extraction

Following PCA, we visualized the signal categories using a scatter plot to analyze the distribution of signal quality clusters. From this plot, we extracted the centroids of each category, which served as the initial cluster centroids for K-means clustering. Figure 20 shows the entire data stratification workflow.

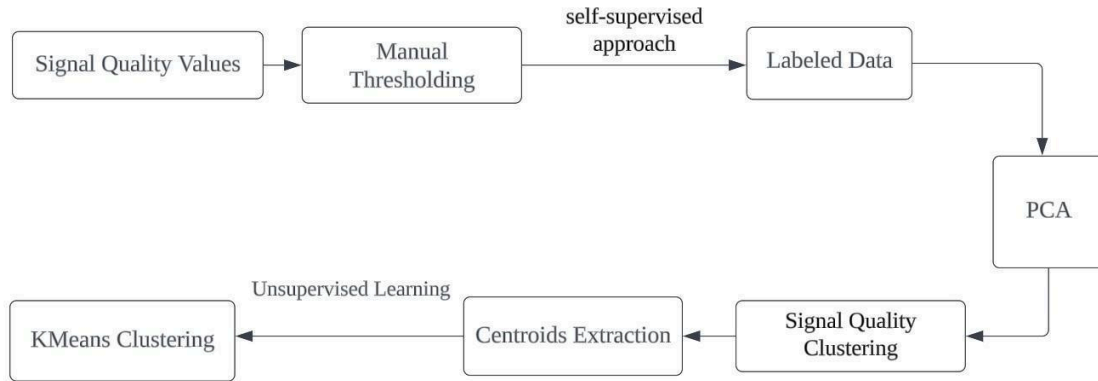


Figure 20: Workflow

5.5 Experimental result evaluation of QSI

For QSI analysis, we need only 3 features: serving cell RSRP, serving cell RSRQ and CINR. As per the methodology previously described in 4.1 section, first we performed self supervised learning and labeled the unlabeled data. We categorize them into 4 groups: Excellent, Good, Moderate and Poor. Table-9 shows the threshold mapping for the signal quality data.

Table 8: Threshold Mapping

Category	Threshold Quantile Percentage
Excellent	>90%
Good	>50%
Moderate	>15%
Poor	<15%

Figure 21 presents a comparison of the accuracy of the nine models evaluated in this study. Among the classifiers, Logistic Regression and Support Vector Machine (SVM) achieved the highest accuracy, demonstrating superior performance in predicting Quality Signal Indicator (QSI) categories. In contrast, AdaBoost showed the poorest performance, with notably lower accuracy compared to the other models. This comparison highlights the varying effectiveness of the classifiers in handling multiclass classification tasks for signal quality prediction.

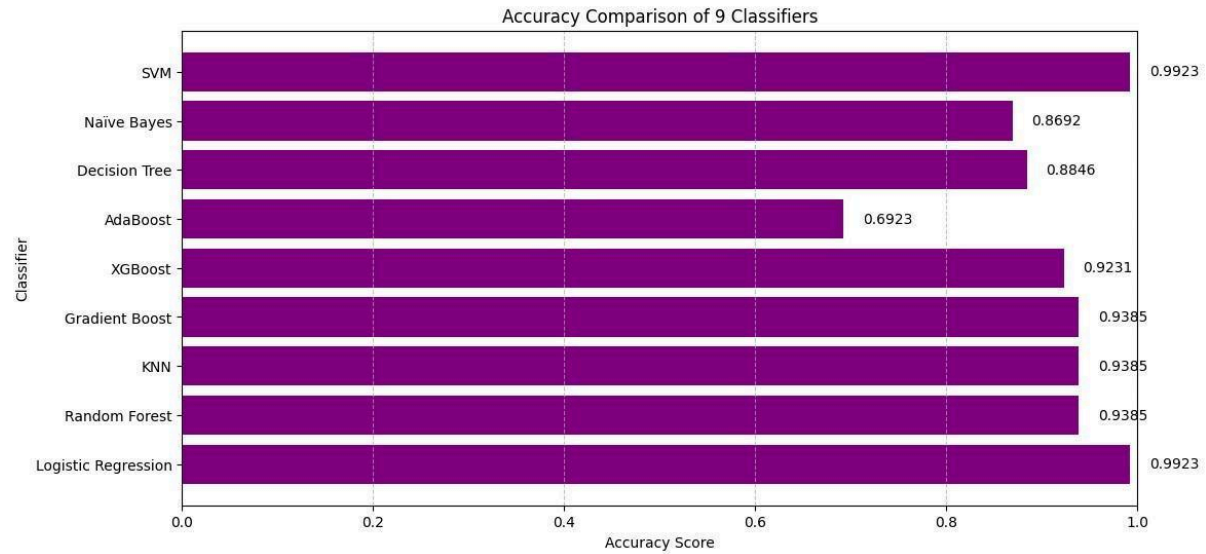


Figure 21: Model Accuracy

We applied Principal Component Analysis (PCA) to reduce the dimensionality of the dataset while preserving its key variance. Following the PCA transformation, we performed clustering to group similar data points based on their intrinsic patterns. The results were visualized using a scatter plot to illustrate the distribution and separability of the identified clusters. Figure 22 presents the scatter plot, highlighting the clustered categories and their relationships in the transformed feature space.

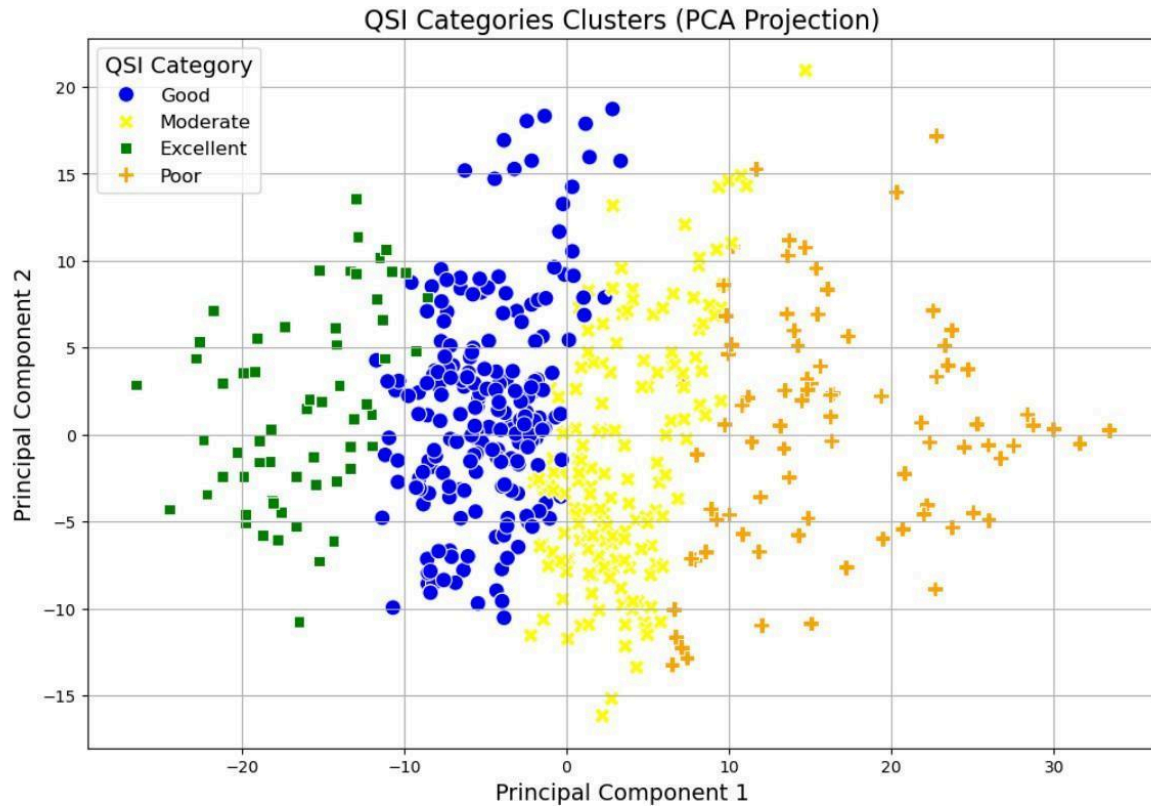


Figure 22 : QSI Clustering using Predefined Labeling

Now we extracted the centroids from this scatter plot of each category. Table-10 shows the PCA axis positions of the centroids.

Table 9: Centroid Positions

QSI_Category	PCA-1	PCA-2
Excellent	-16.042605	1.767194
Good	-4.831206	1.473416
Moderate	3.20405	-2.391944
Poor	16.350566	0.374912

Figure 23 presents the QSI clustering with centroids.

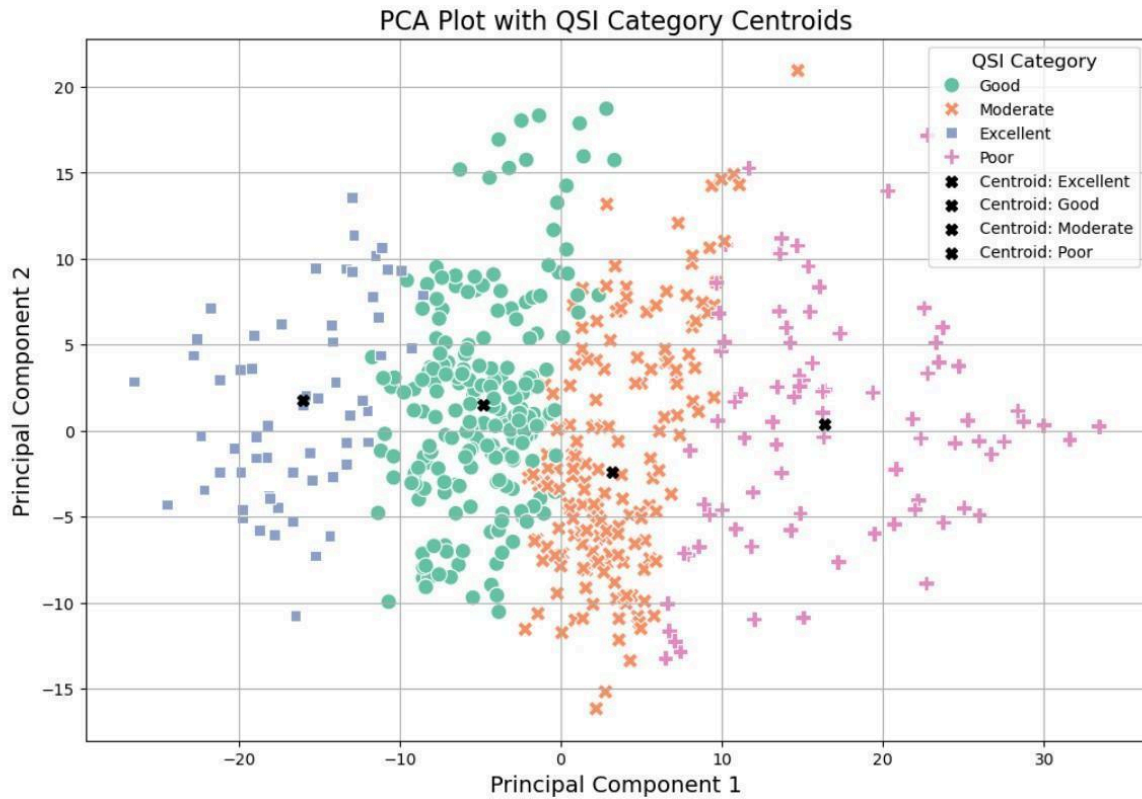


Figure 23: QSI clustering with centroids

Now, we employed K-Means clustering, an unsupervised machine learning algorithm, to classify Quality Signal Indicator (QSI) categories based on predefined centroids. The centroids were manually initialized using domain-specific knowledge, representing four distinct QSI categories: Excellent, Good, Moderate, and Poor. By specifying these centroids as the initial input, we ensured a guided clustering process that aligns with real-world network conditions. The K-Means algorithm was then executed with four clusters, a fixed initialization, and a maximum of 200 iterations, optimizing cluster assignments while maintaining consistency. Finally, the obtained cluster labels were mapped to their corresponding QSI categories, enabling a structured classification of signal quality levels in the dataset. Figure 24 presents QSI Clustering using K-Means algorithm.

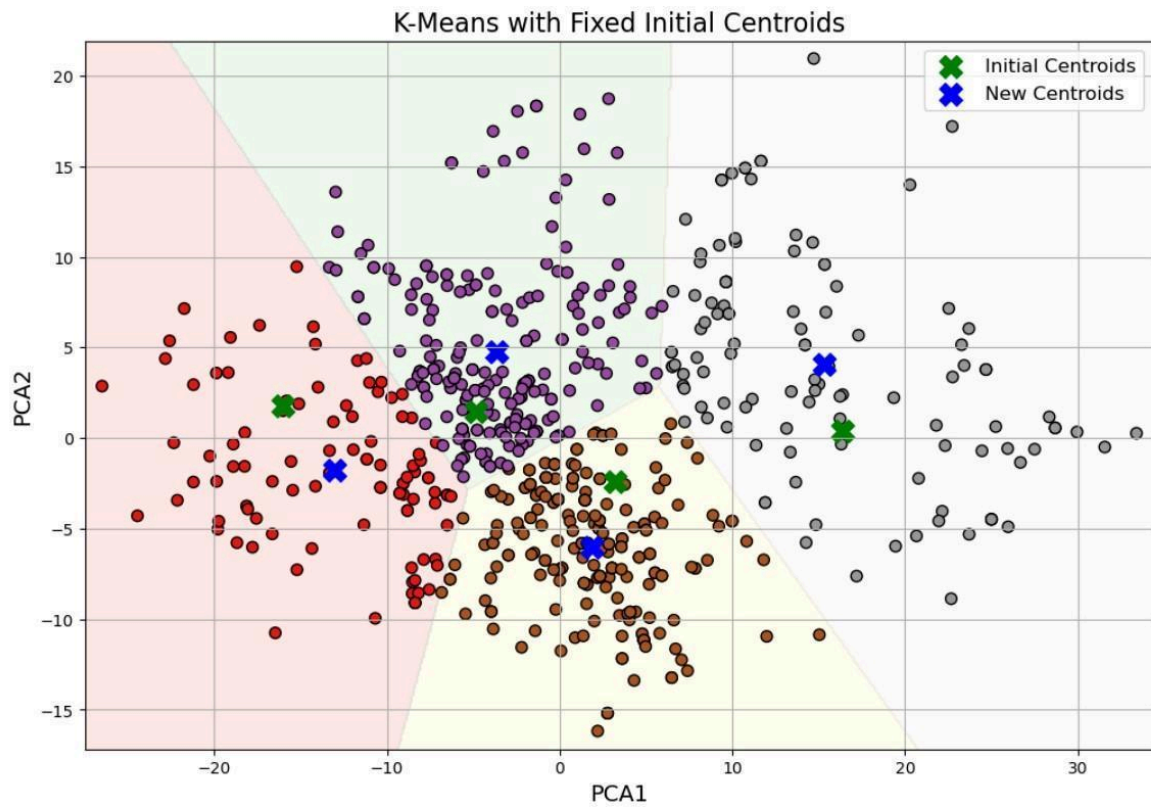


Figure 24: QSI Clustering using K-Means algorithm

Chapter Six

Handover Prediction

This section details the methodology used to develop and evaluate the proposed Contrastive Autoencoder (CAE) framework for handover prediction. It begins with an overview of the conventional Autoencoder, describing its encoder–decoder architecture, training objective, and anomaly scoring mechanism. The Autoencoder formulation for handover prediction is then presented, followed by the proposed CAE design, which incorporates contrastive representation learning with triplet loss to enhance feature discrimination. Finally, the training and testing procedures, including the encoder–decoder structure and latent space representation, are discussed.

6.1 Autoencoder Framework

Autoencoders (AEs) are a class of neural network architectures designed to learn compressed representations of input data through unsupervised learning. The model aims to capture the intrinsic structure of the data distribution by encoding the input into a latent feature space and subsequently reconstructing it with minimal loss of information. Owing to their reconstruction capability, AEs are widely employed in representation learning, noise reduction, and anomaly detection tasks, where deviations in reconstruction error serve as indicators of abnormal patterns.

An AE consists of two parametric components: an Encoder and a Decoder.

Let the input observation be denoted as $x^{(i)} \in \mathbb{R}^d$ where d represents the input dimensionality and $i=1, 2, \dots, N$ indexes the total N training samples.

Encoder

The Encoder performs a non-linear transformation from the input space to a compact latent representation

$z^{(i)} \in \mathbb{R}^{d_z}$, where $d_z < d$. This mapping is defined as:

$$z^{(i)} = f(x^{(i)}; \phi) = \sigma(W_e x^{(i)} + b_e) \quad (1)$$

where:

- is the encoding function parameterized by ,
- is the weight matrix of the Encoder,
- is the Encoder bias vector,
- is a non-linear activation function such as ReLU or tanh,
- represents the compressed latent embedding

Decoder

The Decoder reconstructs the original data from its latent representation through a parametric function g , expressed as:

$$\hat{x}^{(i)} = g(z^{(i)}; \theta) = \sigma(W_d z^{(i)} + b_d) \quad (2)$$

where:

- g is the decoding function parameterized by θ ,
- W_d denotes the weight matrix of the Decoder,
- b_d is the bias vector of the Decoder,
- $\hat{x}^{(i)}$ is the reconstructed approximation of the original input $x^{(i)}$.

Training Objective

The AE is trained to minimize the discrepancy between the input x and its reconstruction $\hat{x}$. The reconstruction loss is typically computed using the mean squared error (MSE) across all samples:

$$\mathcal{L}_{AE}(X; \phi, \theta) = \frac{1}{N} \sum_{i=1}^N \|x^{(i)} - \hat{x}^{(i)}\|_2^2 \quad (3)$$

where $\|\cdot\|_2$ denotes the Euclidean norm, and X represents the training dataset. The minimization of $\mathcal{L}_{AE}$ enforces the Encoder–Decoder pair to learn compact yet informative latent features that effectively capture the distribution of normal data.

Anomaly Scoring

Once trained, the AE evaluates a new observation x by computing its reconstruction error, which serves as the anomaly score:

$$s_{AE}(x^{(i)}) = \frac{1}{d} \sum_{t=1}^d (x_t^{(i)} - \hat{x}_t^{(i)})^2 \quad (4)$$

Here, $x_t^{(i)}$ and $\hat{x}_t^{(i)}$ denote the t -th feature of the input and reconstructed sample, respectively. Higher values of $s_{AE}(x^{(i)})$ indicate a greater deviation from the learned data manifold, implying a higher likelihood of anomaly.

In general, AE-based anomaly detection assumes that the majority of data samples are normal , enabling the model to predominantly learn the characteristics of nominal behavior. Post-training, the top samples with the largest anomaly scores are designated as anomalous instances. Figure 25 shows the general structure of an autoencoder.

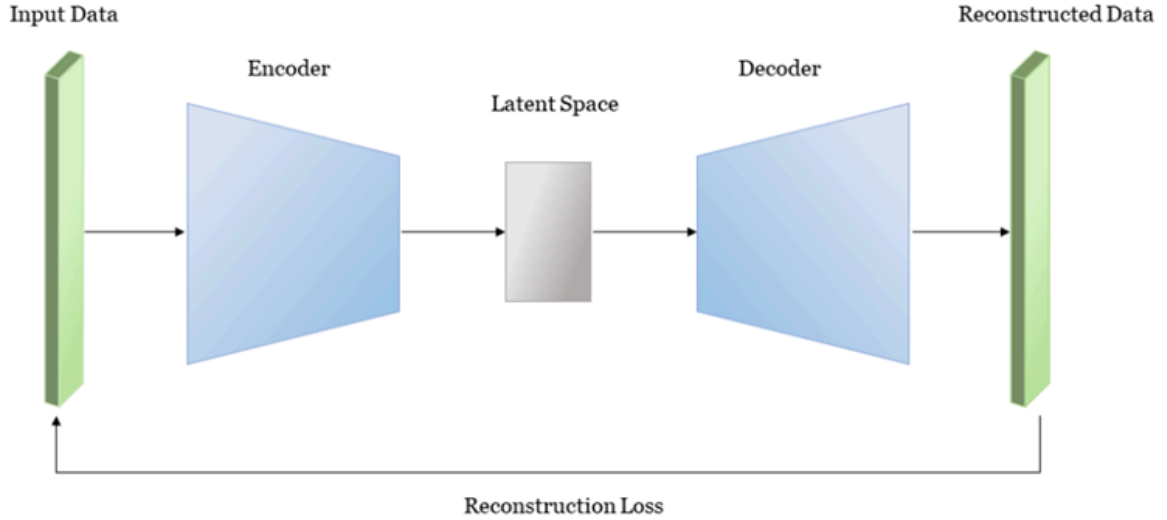


Figure 25. General structure of an autoencoder

6.2 Autoencoder Formulation for Handover Prediction

In the context of LTE networks, the handover (HO) process is initiated when the signal quality of the serving cell degrades below a predefined threshold while a neighboring cell exhibits comparatively higher quality. Such conditions are governed by event-triggering mechanisms, as standardized in 3GPP specifications. Typical events include A1–A5, where for instance, *Event A3* is triggered when the received signal strength of a neighboring cell exceeds that of the serving cell by a hysteresis margin for a specific duration. These event-triggering instances indicate potential transitions between cells and represent dynamic changes in the wireless environment.

Let the time-series data of radio measurements at a given UE location be denoted by

$$x^{(i)} = [RSRP_s, RSRQ_s, RSRP_n, RSRQ_n, CINR, \dots] \in \mathbb{R}^d \quad (5)$$

where the subscript *s* and *n* refer to serving and neighboring cells, respectively, and *d* represents the number of measurement parameters collected at each instant. Each sample corresponds to the network state at a particular time instance.

Under normal operating conditions, the measurements exhibit stable patterns reflecting nominal connectivity with the serving cell. These are denoted as normal data points, forming the distribution $\mathcal{N}$. Conversely, when event-triggering conditions (e.g., A3, A5) are met, the feature vector deviates significantly due to rapid variations in received power and quality metrics. Such samples constitute anomalous data points, represented by $\mathcal{A}$ where $x^{(i)} \in \mathcal{A}$. These anomalies correspond to potential handover events, reflecting the moment when the network must decide whether a cell transition is necessary.

An AE can be formulated to learn the underlying manifold of the normal radio conditions and identify deviations that indicate imminent handover triggers. The Encoder function f compresses each measurement vector into a latent representation:

$$z^{(i)} = f(x^{(i)}; \phi), \text{ for } x^{(i)} \in \mathcal{X} \quad (6)$$

For nominal conditions $\mathcal{N}$, the encoded representation $z^{(i)}$ lies within a compact latent subspace that captures typical channel behavior. For anomalous states $\mathcal{A}$, the latent representation tends to diverge from this manifold due to the non-stationary signal variations associated with the HO event.

The Decoder g reconstructs the input measurement from its latent embedding:

$$\hat{x}^{(i)} = g(z^{(i)}; \theta) \quad (7)$$

where the reconstruction aims to approximate the nominal feature distribution. The corresponding reconstruction error,

$$s_{HO}(x^{(i)}) = \frac{1}{d} \sum_{t=1}^d (x_t^{(i)} - \hat{x}_t^{(i)})^2 \quad (8)$$

acts as an HO anomaly score that quantifies the deviation of current radio conditions from the learned normal baseline. When $s_{HO}(x^{(i)})$ exceeds a predefined threshold τ_{HO} , i.e.;

$$s_{HO}(x^{(i)}) > \tau_{HO} \quad (9)$$

the instance $x^{(i)}$ is identified as an anomalous state, indicating an impending or ongoing event-triggered handover. The threshold τ_{HO} can be empirically chosen or derived through percentile-based or statistical estimation techniques.

This formulation establishes an equivalence between autoencoder-based reconstruction error and handover event detection, wherein:

- Normal samples correspond to stable serving-cell attachment, and
- Anomalous samples correspond to HO-triggering events (A1-A5 conditions).

Hence, the Autoencoder serves as a data-driven model that captures normal radio behavior and predicts the onset of handover events based on deviations in reconstruction error. This provides the foundational principle upon which the Contrastive Autoencoder (CAE) is proposed to enhance discriminative representation and improve HO prediction robustness under rapidly varying network conditions.

6.3 Proposed Contrastive Autoencoder (CAE) Architecture

The proposed Contrastive Autoencoder (CAE) is designed with a quasi-symmetric, fully connected architecture that compresses high-dimensional handover (HO) feature vectors into a compact latent representation and subsequently reconstructs them back to the original dimension. The encoder progressively reduces the feature space, allowing the network to learn low-dimensional embeddings that capture the intrinsic characteristics of the input data. Conversely, the decoder expands this representation to approximate the original feature distribution. Figure 26 shows the structure of our proposed CAE architecture.

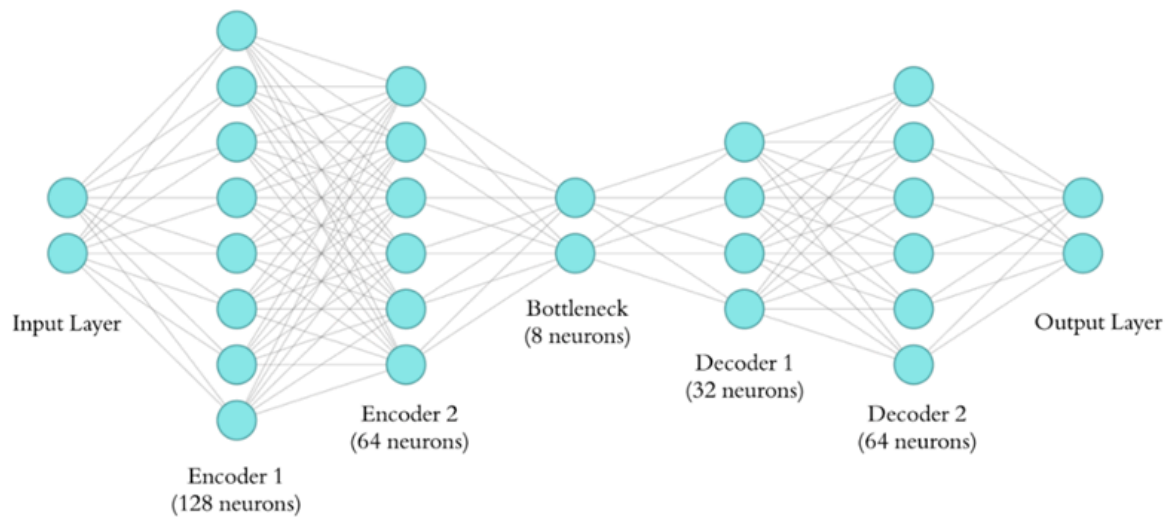


Figure 26. Proposed CAE architecture

The architecture is lightweight yet sufficiently expressive to capture nonlinear feature dependencies commonly present in radio signal measurements such as RSRP, RSRQ, and

CINR. The latent dimension is deliberately kept small (8 neurons) to enforce strong compression, which facilitates discriminative representation learning when integrated with the contrastive objective in later stages. While the general structure of the proposed CAE follows a symmetric fully connected design, the number of hidden layers and neurons was not fixed throughout all experiments. During ablation studies, these configurations were adaptively tuned to identify the most effective architecture under varying radio conditions. Therefore, the Table 11 presents a representative configuration used in the main experiments, while acknowledging that the hidden layer dimensions were flexibly adjusted in specific cases for performance optimization.

Table 10 Representative Structural Configuration of the Proposed CAE

Path	Layer	No. of Neurons	Activation Function	Functional Role
Encoder	Input Layer	Input dimension	-	Accepts normalized handover feature vectors as input
	Dense-1	128	ReLU	Performs initial nonlinear projection to capture broad feature interactions
	Dense-2	64	ReLU	Further compresses and abstracts key radio signal features
	Dense-3	8	ReLU	Generates the latent embedding representing the compact HO state space
Decoder	Dense-4	32	ReLU	Begins reconstruction by expanding latent features toward the input space.

	Dense-5	64	ReLU	Refines intermediate reconstruction for structural recovery.
	Output Layer	Input dimension	Linear	Produces the final reconstructed feature vector corresponding to the original input.

This architectural configuration ensures a smooth information bottleneck between encoder and decoder, enabling the CAE to efficiently compress, represent and later distinguish between normal and event-triggering handover states once contrastive training is applied.

6.4 Training and Testing Methodology of the Proposed CAE

The proposed Contrastive Autoencoder (CAE) is composed of three principal modules: the encoder, the latent representation space, and the decoder. The encoder and decoder operate jointly during both training and testing phases, while the latent space acts as the structural bridge that governs the discriminative mapping between normal and event-triggering (handover) conditions.

Latent Space Representation and Triplet Loss Optimization

The latent space of the proposed Contrastive Autoencoder (CAE) plays a pivotal role in transforming the input feature space into a structured, discriminative manifold that distinctly separates *normal* and *handover-triggering* instances. Unlike traditional autoencoders that rely solely on minimizing reconstruction error, the CAE explicitly constrains the latent space using a triplet-based contrastive learning objective, ensuring that the encoded representations carry semantic separation aligned with network behavior.

(a) Role of Loss Function in CAE

The loss function defines the optimization target guiding the encoder–decoder training process. In conventional autoencoders, the reconstruction loss

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x^{(i)} - \hat{x}^{(i)}\|_2^2 \quad (10)$$

encourages the network to reproduce its inputs with minimal distortion, but it does not enforce any structural discrimination between samples of different operational states. Consequently, the embeddings of normal and handover-triggering samples may overlap in latent space, leading to ambiguous reconstruction behavior.

To address this limitation, the CAE replaces the conventional reconstruction objective with a triplet loss formulation, which introduces metric learning within the latent domain. This formulation explicitly shapes the geometry of the embedding space such that samples belonging to the same class (normal conditions) are drawn closer together, while samples corresponding to distinct classes (handover-triggering events) are repelled beyond a specified margin. The result is a well-separated latent topology where anomalies manifest as embeddings distant from nominal clusters.

(b) Triplet Loss Formulation

Formally, each triplet consists of three input samples:

- Anchor (x_A): a reference sample from the normal (non-handover) condition,
- Positive (x_P): another sample from the same class as the anchor (normal condition),
- Negative (x_N): a sample from the opposite class, representing a handover-triggering instance.

These samples are projected into the latent domain by the encoder :

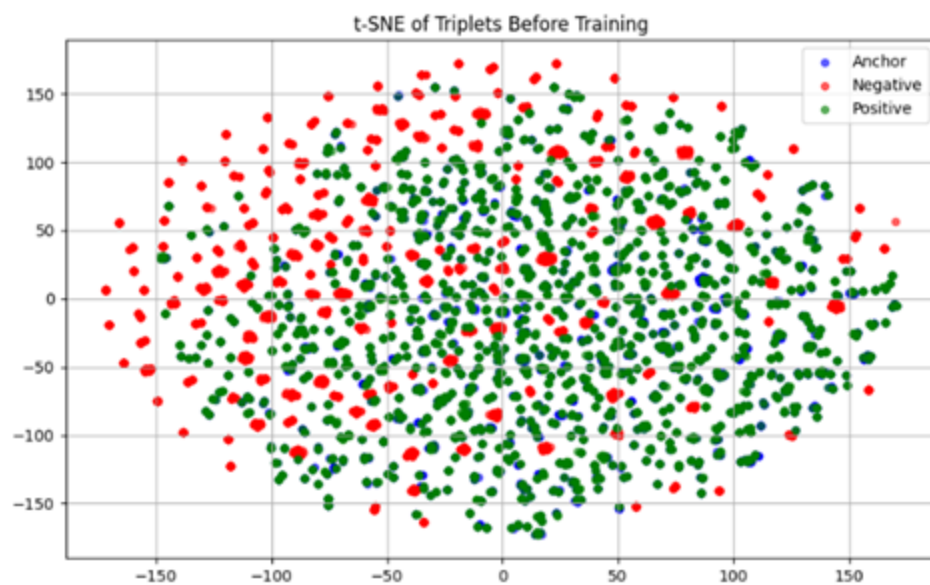
$$z_A = f_\phi(x_A), \quad z_P = f_\phi(x_P), \quad z_N = f_\phi(x_N) \quad (11)$$

where z denotes the latent representation and ϕ represents encoder parameters.

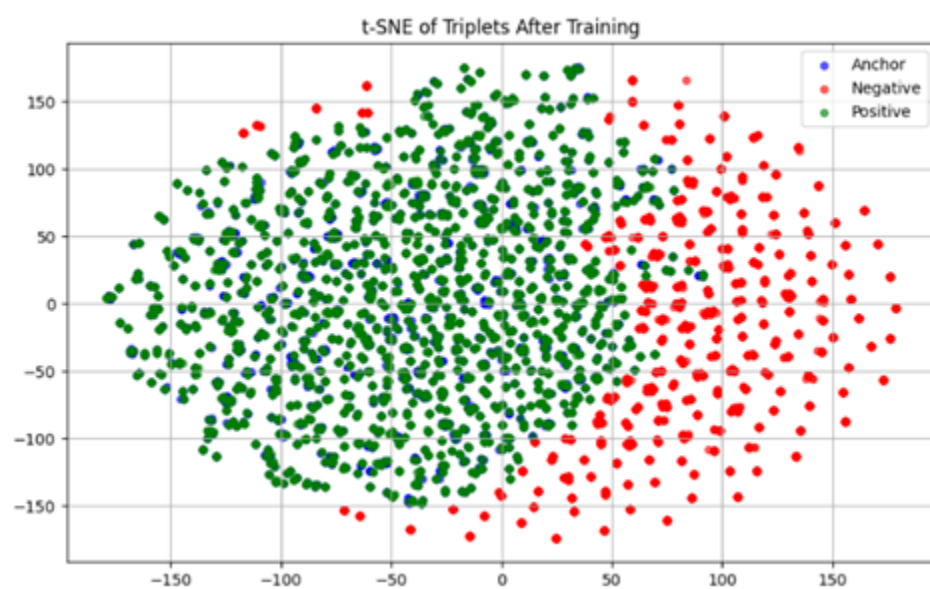
The triplet loss is defined as:

$$\mathcal{L}_{triplet} = \frac{1}{M} \sum_{i=1}^M \max (\| z_A^{(i)} - z_P^{(i)} \|_2^2 - \| z_A^{(i)} - z_N^{(i)} \|_2^2 + \alpha, 0) \quad (12)$$

Figure 27 shows the Visualization of the effect of triplet loss in the proposed CAE framework.



(a)



(b)

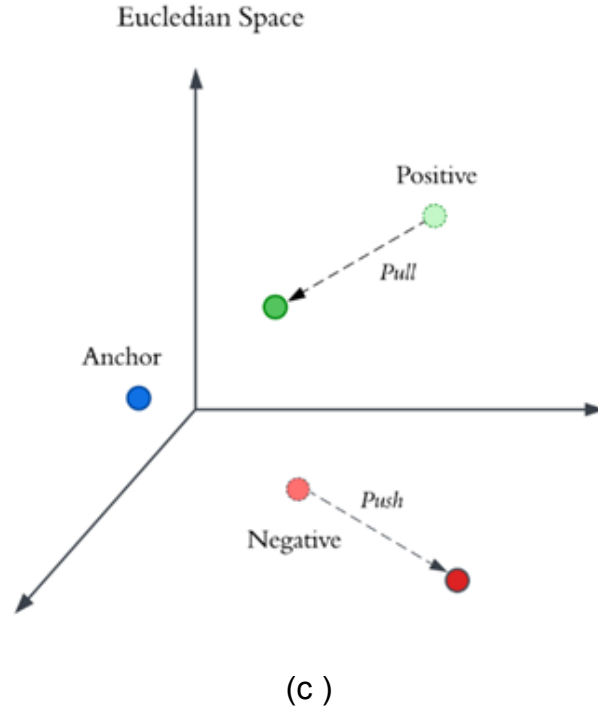


Figure 27. Visualization of the effect of triplet loss in the proposed CAE framework: (a) t-SNE representation before applying triplet loss, (b) t-SNE representation after applying triplet loss showing improved class separability and (c) schematic illustration of the triplet loss mechanism.

(c) Function of the Margin Parameter

The margin defines the minimal inter-class separation required between the anchor–positive and anchor–negative pairs. It ensures that:

$$\|z_A - z_N\|_2^2 \geq \|z_A - z_P\|_2^2 + \alpha \quad (13)$$

thereby preventing trivial convergence (e.g., collapsing all embeddings into a single point). A properly tuned margin enforces robust class separation in the latent manifold, making the embedding space resilient to small perturbations or overlapping radio measurements common in LTE environments.

(d) Effect on Latent Space Geometry for HO Prediction

During training, the encoder receives supervision through the triplet loss and gradually molds the latent space such that:

- Non-handover (stable) samples form compact, high-density clusters.
- Handover-triggering samples are projected to distinct, sparsely distributed regions separated by at least margin .

This separation provides the decoder with a structured latent domain: embeddings within dense normal regions yield low reconstruction error, whereas embeddings lying in sparse or boundary regions (near the HO cluster) produce higher reconstruction error due to insufficient decoder generalization in those areas.

Encoder Module

The encoder module performs a nonlinear projection that compresses high-dimensional radio measurements (e.g., RSRP, RSRQ, CINR) into a compact, information-rich latent representation. During training, the encoder is jointly optimized with the triplet loss objective, which enforces relative similarity constraints among triplets of samples — an anchor, a positive (same class), and a negative (opposite class). This supervision instructs the encoder to learn a topological mapping in which embeddings of samples from similar radio states (non-handover) are drawn close together, while those representing dissimilar conditions (handover-triggering events) are pushed apart by at least a fixed margin.

Through this optimization process, the encoder gradually discovers an intrinsic manifold structure that reflects the underlying dynamics of network stability and transition behavior. Normal samples form dense, smooth manifolds characterized by minimal intra-cluster variability, while handover-triggering samples occupy peripheral or sparsely populated regions of the latent domain. This structured separation is not explicitly programmed but emerges naturally as the encoder minimizes the triplet loss across the dataset.

At inference time, the encoder functions in a label-free mode. It receives an unseen input sample and directly projects it into the pre-organized latent space using the previously learned geometric mapping.

- Inputs exhibiting stable radio conditions are automatically positioned within the dense, well-trained regions corresponding to non-handover states.
- Inputs exhibiting transitional or degraded signal characteristics are projected toward the outer or boundary regions associated with handover events.

This enables the encoder to maintain class-aware latent projections without requiring explicit class labels during testing, effectively converting complex multivariate radio data into discriminative latent coordinates suitable for anomaly-based HO prediction.

Decoder Module

The decoder performs the inverse mapping, reconstructing input-like representations from the latent embeddings. Although no explicit reconstruction loss is used, the decoder still participates in training through the triplet loss, which is computed in the reconstruction (decoder output) space rather than the latent space.

In this setting, the triplet loss simultaneously updates both encoder and decoder parameters so that:

- Reconstructions of normal (anchor–positive) pairs become similar.
- Reconstructions of anomalous (anchor–negative) pairs diverge beyond the margin.

This implicit regularization causes the decoder to learn consistent mappings for normal samples — effectively mimicking the behavior of a reconstruction loss. By minimizing the distance between reconstructed normal samples, the decoder develops a structural bias to accurately reproduce normal patterns while distorting reconstructions of anomalous ones.

As a result, the reconstruction error still emerges as a meaningful indicator of handover likelihood:

- For samples encoded within the dense normal manifold, reconstructions remain accurate, producing low errors.
- For samples lying near the sparse or boundary HO manifold, reconstructions deviate significantly, yielding higher errors.

During testing, this encoder–decoder interaction acts as a self-organized, label-free predictor. The encoder maps new inputs into their corresponding latent regions, while the decoder’s reconstruction fidelity reflects how closely these inputs align with learned normal behavior. Consequently, even without an explicit reconstruction objective, the triplet-induced mapping ensures that reconstruction error remains a reliable and discriminative metric for identifying handover-prone conditions.

Training Procedure

The model was implemented in TensorFlow/Keras using a quasi-symmetric fully connected architecture. LTE drive-test datasets were split 80–20, then combined for training. Features such as RSRP, RSRQ, CINR and speed were standardized using z-score normalization.

Triplet samples were created where anchors and positives represented non-handover states, and negatives represented handover-triggering states. Around 10,000 triplets were used for training. The model was trained for 50 epochs using the Adam optimizer with a margin-based triplet loss, batch size of 128. The training encouraged the decoder to consistently reconstruct normal radio conditions while separating the latent representations of handover states, enabling

effective handover inference. Table 12 presents the Training Configuration of the Proposed CAE.

Table 11 Training Configuration of the Proposed CAE

Parameter	Value
Optimizer	Adam
Batch size	128
Epochs	50
Margin (α)	1
Number of triplets	10000

6.5 Reconstruction Error Thresholding for Handover Prediction

In the context of handover prediction, thresholding on reconstruction error enables the CAE to distinguish between stable link conditions and those likely to trigger a handover event. After training, the CAE reconstructs non-handover (normal) patterns accurately while producing higher reconstruction errors for feature combinations indicative of handover-triggering states. Applying a suitable threshold to this error distribution converts continuous deviations into binary decisions — identifying when radio link dynamics shift from stability to a potential handover condition. To ensure robustness and adaptability across varying scenarios, five thresholding techniques have been adopted in this study.

I. Percentile-Based Dynamic Thresholding

This method establishes an adaptive cutoff on the reconstruction error distribution using a percentile of the non-handover (normal) class. Given the absolute reconstruction error

$$e_i = \frac{1}{d} \sum_{j=1}^d |x_{ij} - \hat{x}_{ij}|, \text{ where } d \text{ is the feature dimension, the threshold } \tau \text{ is selected as the 97th percentile of } e_i \text{ among normal samples. Any test instance with } e_i > \tau \text{ is identified as anomalous (handover). This non-parametric approach assumes that normal errors follow a stable distribution and higher deviations signify degraded radio conditions, enabling efficient detection without requiring probabilistic modeling or classifier training.}$$

II. Isolation Forest–Based Anomaly Segmentation

The Isolation Forest (IF) algorithm treats anomaly detection as a process of isolating samples through recursive random partitioning. In this approach, reconstruction errors from

non-handover samples are used to train the isolation model. The algorithm builds multiple random trees and computes an average path length $h(x)$ for each point, where anomalous samples typically have shorter path lengths due to their isolation in fewer splits. The anomaly score is derived as $s(x) = 2^{-\frac{E(h(x))}{c(n)}}$, with $c(n)$ as a normalization term. Samples yielding high anomaly scores are labeled as potential handover-triggering instances. This technique effectively models nonlinear error distributions and is robust to skewed or multimodal data.

III. Kernel Density Estimation (KDE)–Driven Likelihood Thresholding

Kernel Density Estimation provides a smooth, non-parametric estimate of the probability density function of reconstruction errors from normal conditions. The density $p(e)$ is approximated as

$$p(e) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{e-e_i}{h}\right) \quad (14)$$

where $K(\cdot)$ is a Gaussian kernel and h is the bandwidth. For each test sample, the log-likelihood $\log(p(e))$ is computed, and anomalies are identified when the likelihood falls below the 5th percentile of the non-handover log-likelihood distribution. This approach captures the underlying probability landscape of normal reconstruction patterns and adaptively rejects samples that reside in low-density regions, indicating unusual network behavior.

IV. Youden’s J–Optimized ROC Thresholding

This supervised thresholding method derives an optimal cutoff from the Receiver Operating Characteristic (ROC) curve, balancing true positive rate (TPR) and false positive rate (FPR). The Youden’s J statistic is defined as

$J = TPR - FPR$, and the threshold that maximizes J corresponds to the best trade-off between sensitivity and specificity. Using this criterion, the optimal reconstruction error boundary is data-driven and tuned according to the discriminative behavior of the CAE’s reconstruction quality between normal and handover samples. This ensures performance-optimized decision boundaries under varying network dynamics.

V. Latent–Reconstruction Fusion with XGBoost Classification

In this final approach, a hybrid thresholding mechanism was implemented using a supervised XGBoost classifier trained on features derived from the CAE. The dataset was partitioned through a stratified multi-stage split: 35% of the data was used for triplet-based CAE training, 45% for training the XGBoost classifier, and the remaining 20% was reserved as an unseen evaluation set, ensuring class balance and preventing information leakage. All input features were standardized based on the CAE training subset. The CAE generated latent embeddings representing compressed radio-state information, along with reconstruction errors capturing deviations from normal conditions. These complementary representations were concatenated to form a fused feature vector, combining structural and anomaly-sensitive cues. The XGBoost classifier learned a discriminative boundary between normal and handover-triggering states, while evaluation on the held-out set assessed generalization. By integrating latent geometry

with reconstruction-based deviations, this method provides an adaptive, data-driven thresholding mechanism that enhances handover prediction across heterogeneous radio conditions.

6.6 Vanilla Autoencoder with MSE Loss

The baseline architecture in this study employed a *vanilla autoencoder* trained to reconstruct radio condition features of non-handover (non-HO) samples using a mean squared error (MSE) objective. The AE acts as an unsupervised anomaly detector by learning the underlying distribution of normal handover-free network states. During inference, deviations from this learned manifold are quantified through reconstruction error, which serves as an implicit anomaly score for identifying potential handover triggers.

Prior to training, all features were normalized using z-score scaling to eliminate scale dependency and improve gradient stability. Only samples labeled as non-HO were used for training to ensure that the model captured the intrinsic radio environment characteristics under stable connectivity conditions. The network comprised a symmetric, fully connected encoder–decoder configuration: four encoding layers with progressively decreasing units (64-32-16-8-4) compressed the high-dimensional input into a 4-dimensional latent representation, while the decoder mirrored this structure to reconstruct the input feature space. ReLU activations were employed throughout the hidden layers to promote non-linear expressiveness, and a lightweight L1 activity regularization was introduced in the first encoding layer to enhance sparsity and mitigate overfitting. The model was trained using the Adam optimizer for 100 epochs with mini-batches of 32 samples, using 10% of the data for validation.

Once trained, the reconstruction errors were computed for all non-HO training samples, and their distribution was analyzed to establish an anomaly threshold. The 80th percentile of the training reconstruction error distribution was selected as the decision boundary, representing the upper tolerance limit of normal reconstruction behavior. During evaluation, both HO and non-HO test samples were passed through the AE; samples exhibiting reconstruction errors exceeding this threshold were classified as potential handover events.

Unlike the proposed *Triplet-based CAE*, which leverages both HO and non-HO samples to learn a discriminative latent structure, the vanilla AE relies solely on reconstruction fidelity. Consequently, it focuses on minimizing overall feature-level deviations rather than explicitly modeling inter-class separability. The absence of a contrastive learning mechanism causes the latent space to lack structural discrimination between normal and abnormal states—leading to substantial overlap between the reconstruction errors of HO and non-HO samples. While the AE effectively captures stationary radio patterns, it struggles to generalize when channel dynamics vary abruptly during imminent handover conditions.

Limitations of MSE-Based Reconstruction Loss

The MSE loss function optimizes for global reconstruction accuracy without considering the relational geometry among samples in the latent space. This objective inherently encourages the network to approximate an average representation of the input distribution, resulting in *latent*

representation collapse when the input features of normal and HO conditions are numerically similar. In dynamic cellular environments, where signal degradation and interference evolve gradually, the resulting reconstruction errors from MSE training often remain small even for HO-triggering samples, yielding poor anomaly separability.

In contrast, the Triplet-based loss explicitly enforces *relative distance constraints* between embeddings of normal and HO samples, guiding the encoder to cluster homogeneous (non-HO) representations tightly while pushing dissimilar (HO) representations apart. This structured latent space enables the decoder to inherently assign higher reconstruction errors to unseen or anomalous channel states, yielding superior precision and recall in HO detection. Thus, while the vanilla AE serves as a robust baseline for modeling normal connectivity conditions, its reliance on MSE reconstruction loss fundamentally limits its discriminative capability compared to the contrastive triplet learning paradigm.

6.7 Artificial Neural Network Based Handover Prediction

Artificial Neural Networks (ANNs) were employed as a supervised learning baseline to model the nonlinear dependencies between radio parameters and handover (HO) trigger events. The ANN learns a discriminative mapping from multi-dimensional radio features—including serving and neighbor cell RSRP/RSRQ, CINR, and user speed—to a binary decision space representing handover and non-handover states. Through iterative backpropagation, the network optimizes its weights to minimize the binary cross-entropy loss, effectively capturing the underlying decision boundary associated with HO-triggering conditions.

The designed ANN architecture consists of multiple fully connected (Dense) layers with rectified linear unit (ReLU) activations, interleaved with batch normalization and dropout layers to improve convergence stability and prevent overfitting. The final output layer employs a sigmoid activation to produce probabilistic handover predictions. The configuration of hidden layers and neuron counts was empirically tuned across different simulation runs to identify the most effective topology under varying channel conditions. Table 13 presents the Parameter Configuration of the ANN Model.

Table 12 Parameter Configuration of the ANN Model

Component	Description	Description
Input Layer	6 features	Radio parameters (RSRP, RSRQ, CINR, speed, etc.)
Hidden Layers	[128, 64, 64, 16] neurons (tuned)	Fully connected layers for nonlinear feature extraction
Activation Function	ReLU	Enables sparse and efficient gradient propagation
Normalization	Batch Normalization after each hidden layer	Stabilizes learning and mitigates internal covariate shift
Regularization	Dropout (0.1–0.2)	Reduces overfitting by random neuron deactivation

Output Layer	1 neuron, Sigmoid activation	Generates binary probability of HO trigger
Optimizer	Adam	Adaptive learning rate for stable convergence
Loss Function	Binary Cross-Entropy	Measures classification error between predicted and actual HO labels
Epochs / Batch Size	100 / 32	Tuned for optimal trade-off between accuracy and convergence speed

The ANN operates as a supervised classifier, learning explicit decision boundaries based on labeled handover data. By mapping radio-condition features to a probabilistic output, it identifies patterns that precede HO events. This structure enables the network to generalize across varying radio conditions, offering an interpretable baseline for performance comparison against the proposed contrastive autoencoder framework.

6.8 Ablation and Benchmarking with Conventional Supervised Models

To provide a rigorous ablation against the proposed contrastive autoencoder, a set of conventional supervised classifiers was implemented and systematically optimized. The supervised pipeline comprised three disciplined steps: (i) training-time imbalance correction using SMOTE (Synthetic Minority Over-Sampling Technique) applied only within training folds, (ii) feature preprocessing (z-score scaling where required) encapsulated in a pipeline to avoid leakage, and (iii) hyperparameter selection by grid search with stratified k-fold cross-validation.

SMOTE addresses class imbalance by synthesizing minority-class samples in feature space, thereby reducing bias toward the majority (non-HO) class; critically, SMOTE was applied inside the cross-validation pipeline so that synthetic samples never contaminate validation folds. Grid search (scikit-learn GridSearchCV) used stratified 5-fold CV, `n_jobs = -1` and deterministic seeds; the primary optimization metric was the F1 score for the positive (HO) class, with ROC-AUC retained as a secondary metric. For algorithms sensitive to feature scale (logistic regression, SVM, KNN, Naïve Bayes), a standard scaler was included in the pipeline; tree-based and boosting classifiers received raw features. Early-stopping or `n_estimators` tuning was used for gradient-boosting methods to control overfitting and reduce training cost.

Below (Table 14) is a compact technical description and the hyperparameter search focus for each implemented classifier. The entries highlight the algorithmic principle, the principal hyperparameters tuned, and the specific rationale for its inclusion as an HO-prediction baseline.

Table 13 Supervised Classifier Descriptions

Classifier	Technical description	Suitability
Logistic Regression	Linear discriminative model with convex optimization; L2/L1	Fast, interpretable baseline; tests whether HO is (approximately) linearly separable in feature space.

	regularization controls weight magnitude.	
Random Forest	Bagged ensemble of decision trees; reduces variance via bootstrap aggregation.	Robust to heterogeneous features and noise; provides feature-importance baseline.
Support Vector Machine (SVM)	Maximum-margin classifier; kernelized (RBF / linear) enables non-linear decision surfaces.	Effective for medium-dimensional data; requires careful scaling and C/gamma tuning.
Gaussian Naïve Bayes	Generative probabilistic model assuming feature conditional independence and Gaussian likelihoods.	Lightweight probabilistic baseline; useful to assess how well simple generative assumptions perform for HO.
K-Nearest Neighbors (KNN)	Instance-based, nonparametric classifier using distance in feature space.	Simple local decision metric; provides insight into local separability of HO patterns.
AdaBoost	Stagewise additive model that reweights misclassified samples; often uses decision stumps as weak learners.	Effective for moderately nonlinear patterns; sensitivity to noisy labels tested.
GradientBoosting	Sequential tree boosting (Friedman) minimizing differentiable loss; regularized via learning rate & subsampling.	Strong baseline for tabular data; trade-off between bias and variance controlled by depth & learning rate.
XGBoost	Optimized gradient boosting with regularization and efficient tree construction.	High performance on heterogeneous, imbalanced data; scale_pos_weight useful for class imbalance.
CatBoost	Gradient boosting optimized for categorical features and ordered boosting to reduce target leakage.	Robust default settings, strong regularization and fast convergence; included as an advanced boosting baseline.

Implementation notes (methodological rigor)

- **Pipelines & leakage prevention:** For scalable reproducibility, each model was wrapped in a Pipeline that placed StandardScaler (if needed) and SMOTE (from imblearn) before the classifier. When performing grid search, SMOTE was nested inside the CV pipeline so synthetic samples were generated only from training folds. This is essential to avoid overly optimistic validation scores.
- **Cross-validation & selection criterion:** GridSearchCV employed stratified 5-fold CV; the primary scoring objective was F1 (positive class) because early detection of HO (minority class) is prioritized; ROC-AUC acted as a secondary check for overall separability. For boosting methods, early stopping was used where possible to prevent overfitting and to shorten hyperparameter searches.

- **Reproducibility and compute:** All estimators were fit with fixed `random_state` values and `n_jobs=-1` where supported. For large parameter grids, randomized search or successive halving was considered to balance thoroughness with computational resources.
- **Evaluation:** Final model selection used the held-out test set (distinct from CV) to produce unbiased metrics (precision, recall, F1, ROC-AUC, confusion matrix) reported in the Results section.

Rationale for design choices

The chosen ensemble of classifiers spans linear, distance-based, probabilistic, bagging and boosting paradigms. This breadth permits a comprehensive ablation that isolates whether performance gains of the proposed CAE stem from superior representation learning (latent separation) or simply from downstream classifier capacity. SMOTE is critical because HO events are scarce; without adequate minority-class augmentation the supervised algorithms would be biased toward the majority class, yielding poor recall. Grid search provides a principled mechanism to tune model complexity and regularization so that comparisons are fair (each classifier evaluated at near-optimal hyperparameters).

Chapter Seven

Handover Optimization

7.1 Overview of the Handover Optimization Framework

Handover (HO) optimization in Long Term Evolution (LTE) networks is a critical enabler of seamless mobility, user-perceived quality of service (QoS), and efficient utilization of radio resources. Conventional handover mechanisms rely on static thresholds for parameters such as Reference Signal Received Power (RSRP), Handover Margin (HOM), and Time-to-Trigger (TTT). These fixed rules are inherently inflexible and fail to adapt to the dynamic nature of real-world radio environments—particularly in dense urban settings characterized by rapid signal fluctuations, shadowing, interference, and heterogeneous user mobility patterns.

To overcome these limitations, this chapter presents a unified and comprehensive methodology that integrates four adaptive learning-based frameworks for handover decision-making: (i) Q-learning with contextual bandit modeling, (ii) Q-learning enhanced with subtractive clustering for state abstraction, (iii) fuzzy logic-based adaptive control, and (iv) Deep Q-Network (DQN)-based reinforcement learning. All frameworks are trained and evaluated using a rich, real-world LTE drive-test dataset collected in Dhaka, Bangladesh—a dense urban environment with complex propagation conditions.

The core philosophy across all methods is to replace static thresholds with context-aware, data-driven policies that dynamically adjust HOM and TTT based on instantaneous and historical radio measurements. These policies are learned through interaction with real network data, guided by carefully designed reward functions that penalize unnecessary and ping-pong handovers while rewarding stable, high-quality connections.

The overall architecture begins with data collection via professional drive-test equipment, followed by feature engineering and preprocessing. The processed data is then fed into one of the adaptive frameworks, each of which defines its own state space, action space, and reward mechanism. Crucially, all methods incorporate HOM and TTT not merely as fixed parameters but as tunable, state-dependent control variables that govern both the triggering condition and temporal stability of handovers.

7.2 Feature Engineering and Preprocessing

7.2.1 Data Collection

The data used was collected from the drive test mentioned in Chapter 3.

7.2.2 Data Preprocessing Pipeline

Raw drive-test logs contain missing values, outliers, and non-LTE measurements that must be cleaned before model training. The preprocessing steps are as follows:

- 1. Timestamp Standardization:** All timestamps converted to milliseconds since epoch for precise event sequencing.
- 2. Invalid Value Removal:** Entries with RSRP < -140 dBm or RSRQ < -20 dB are discarded as physically implausible.
- 3. Interpolation of Missing Values:** Short gaps (< 500 ms) in RSRP/RSRQ are linearly interpolated using neighboring valid samples from the same serving cell.
- 4. Non-LTE Filtering:** All 2G/3G measurements are removed to ensure LTE-only analysis.
- 5. Hardware Anomaly Detection:** Segments exhibiting abrupt, non-physical jumps in RSRP (> 20 dB within 100 ms) are flagged and excluded.

6. Feature Construction:

$\Delta\text{RSRP} = \text{Neighbor RSRP} - \text{Serving RSRP}$

$\Delta\text{RSRQ} = \text{Neighbor RSRQ} - \text{Serving RSRQ}$

$\text{RSRP_trend3} = 3\text{-point moving average slope of } \Delta\text{RSRP}$

$Q_weighted = 0.6 \cdot \Delta\text{RSRP} + 0.3 \cdot \Delta\text{RSRQ} + 0.1 \cdot \text{CINR}$ (used in DQN)

7. Normalization:

Min-Max Scaling: Used in subtractive clustering (Reducing Handover...) to balance feature influence in Euclidean distance.

StandardScaler (Z-score): Used in contextual bandit Q-learning (FinalManuscript.pdf) for zero-mean, unit-variance features.

The final preprocessed dataset consists of three independent subsets:

Day 1: 922 samples, 150 HO events → used for training in most experiments

Day 2: 592 samples, 99 HO events → reserved for testing

Day 3: 641 samples, 118 HO events → used for cross-validation and robustness checks

7.3 Q-Learning and Contextual Bandit Modeling

7.3.1 Reinforcement Learning Implementation

The reinforcement learning (RL) agent is designed with a defined state space, binary action set, and a reward function that incorporates handover margin (HOM) and time-to-trigger (TTT) parameters to optimize handover decisions based on signal quality and stability.

State

Each state is a single measurement snapshot represented by a 5-dimensional feature vector (all features standardized using the *StandardScaler* fitted on the training set):

$s = [\text{Serving RSRP}, \text{Serving RSRQ}, \text{Neighbour RSRP}, \text{Neighbour RSRQ}, \text{Serving CINR}]_{\text{normalized}}$

Action

A binary action space is used:

- $a=0$: No handover (stay on serving cell)
- $a=1$: Trigger handover to neighbour cell

7.3.2 Reward Function

The reward is a conditionally structured scalar signal that (i) rewards correct retention of a good serving link, (ii) rewards handovers that improve both signal strength and quality, and (iii) heavily penalizes unnecessary or unstable handovers. The design explicitly integrates two conventional mobility-control primitives: HOM (handover margin) and TTT (time-to-trigger).

The reward function is designed to strongly discourage unstable or poor-quality handovers through large negative rewards (e.g., -20 , -8), effectively reducing ping-pong effects and unnecessary signaling. Positive rewards (e.g., $+5$, $+3$) promote maintaining strong serving links and avoiding needless handovers. Intermediate rewards ($+1$, -5) capture marginal improvements or missed handover opportunities, guiding the agent towards balanced decisions that optimize network performance and stability.

In the implemented agent, HOM and TTT are parameters that gate and stabilize handover decisions.

7.3.3 Handover Margin (HOM)

The handover margin is defined as a fixed signal strength threshold, expressed in decibels, by which the received signal power of a neighbouring cell must exceed that of the serving cell before a handover is considered. This threshold mitigates the risk of unnecessary handovers caused by short-term measurement fluctuations or minor variations in signal quality, thereby enhancing decision stability.

7.3.4 Time-to-Trigger (TTT)

The time-to-trigger parameter specifies the minimum duration for which the HOM condition must remain satisfied before a handover is executed. During this interval, signal measurements are continuously monitored to ensure that the neighbouring cell maintains its advantage over the serving cell. If the advantage is lost at any point within this duration, the handover is aborted. This persistence mechanism prevents premature or oscillatory handovers and ensures that only sustained improvements in link quality lead to a cell change.

Each dataset entry is a timestamped but independent snapshot of radio conditions, with no influence on future states. Thus, the problem is modeled as a contextual bandit, aiming to learn a mapping that maximizes instantaneous reward. This supports per-snapshot rewards without state-transition modeling, using tabular Q-learning to estimate action utilities. HOM and TTT effects are embedded in the reward to ensure compliance with quality thresholds and temporal stability.

7.3.5 Training Phase

The training process employed a Q-learning-based reinforcement learning framework, wherein the agent incrementally updated its Q-table to represent the expected cumulative rewards for each state–action pair.

At each training episode, a random measurement point was sampled from the dataset to simulate the agent’s perception of the environment. The current state was represented by normalized feature values, and an action was selected following an ϵ -greedy exploration–exploitation strategy. This ensured that the agent initially explored a broad range of state–action combinations before gradually converging toward exploitation of the learned policy as ϵ decayed.

The Q-table was initialized with zero values for both available actions (handover, no handover) for each state. The update of Q-values followed the canonical Q-learning rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (15)$$

where:

- $Q(s, a)$ is the current Q-value for state s and action a
- α is the learning rate
- r is the immediate reward
- γ is the discount factor for future rewards
- $\max_a Q(s', a')$ is the estimated value of the best action in the next state

This iterative process enabled the Q-table to gradually approximate the optimal action-value function, allowing the agent to make informed handover decisions based on accumulated experience. Table 15 presents the Q-learning parameter settings.

Table 14 Q-learning Parameter Settings

Parameter	Value
Learning rate	0.1
Discount factor	0.9
Initial exploration rate	1

Minimum exploration rate	0.01
Exploration decay	0.995
Training episodes	10,000

7.4 Q-Learning via Subtractive Clustering

This section outlines the comprehensive methodology used to model, optimize, and evaluate handover decisions in a 4G/LTE cellular network using reinforcement learning. The proposed framework begins with real-world data collection through drive tests, followed by state space construction via subtractive clustering, action-reward design using domain-specific mobility logic, and Q-learning-based policy optimization. The methodology concludes with a parameter search across handover margins (HOM), time-to-trigger (TTT), and clustering weights to identify an optimal policy that minimizes unnecessary handover.

Figure 28 illustrates a flowchart depicting the entire process of the study, starting with Drive Test Data and proceeding through Subtractive Clustering. It continues with Q-Learning Training using HOM, TTT, Weights, and concludes with Final Evaluation in the testing phase.

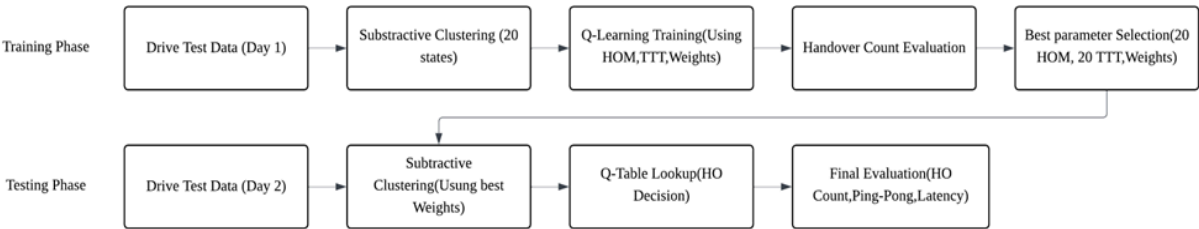


Figure 28: Methodological Pipeline

7.4.1 State Abstraction via Subtractive Clustering

To discretize the continuous radio environment into meaningful mobility contexts, subtractive clustering was employed for state space construction. This unsupervised clustering method is well-suited for sparse and non-uniform datasets, as it does not require a predefined number of clusters and dynamically selects representative data points based on local density.

Each data instance was represented as a five-dimensional feature vector comprising serving and neighboring cell RSRP (in dBm), serving cell RSRQ and CINR, and neighboring cell RSRQ. Features were normalized using Min-Max scaling to balance their influence in distance-based similarity computation. The clustering process begins with the input dataset, followed by

checking cluster numbers and adjusting parameters. Next, cluster quality is calculated, and clusters are sorted by quality. Cluster states are then assigned, followed by label assignment, resulting in clustered data with state labels as the output. On the parallel track, density computation using Euclidean distance initiates the selection of the first cluster center. This is followed by density reduction near centers, and the process repeats until a threshold is reached, as depicted in Figure 29.

The algorithm generated 20 distinct clusters, each representing a unique radio condition profile or "state". These states reflect the heterogeneous signal conditions experienced by user equipment during mobility, from strong-serving-cell dominance to interference-prone cell-edge zones.

Compared to grid-based or heuristic discretization, subtractive clustering provides a data-driven, topology-aware abstraction [15]. It captures nuanced patterns in signal interactions and enables the reinforcement learning agent to generalize over similar conditions rather than overfitting to raw measurements. Figure 29 illustrates the detailed clustering process for state discretization used in our proposed model.

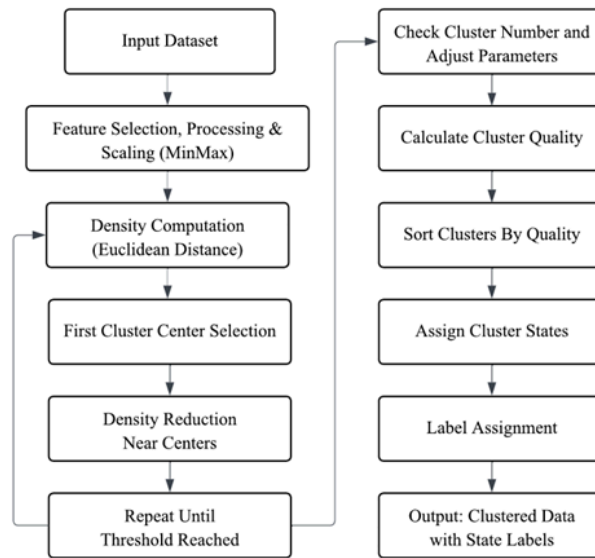


Figure 29: Clustering Process for State Discretization

7.4.2 Action and Reward Design for Q-Learning

The agent operates with two discrete actions: to trigger a handover (HO) or maintain the current serving cell connection (No_HO). This binary action space reflects practical decision-making in LTE mobility management and simplifies the policy optimization problem.

The reward function is designed to reflect signal quality trade-offs and operational constraints. A positive reward is assigned when a handover occurs and results in improved RSRP, while a penalty is applied for handovers that offer no improvement or introduce instability. Conversely, retaining the current cell yields a small positive reward if justified by signal conditions and a penalty if it results in missed opportunities.

To discourage ping-pong handovers, additional negative rewards are applied when consecutive handovers occur within the Time-to-Trigger (TTT) window. All reward values are clipped within $[-3, +3]$ to stabilize learning dynamics.

This reward structure aligns with real-world operator objectives: optimizing handover decisions to reduce signaling overhead, avoid unnecessary transitions, and improve user experience.

To illustrate the underlying logic, Table 16 summarizes the key decision cases and corresponding reward signals used in training.

Table 15: Reward Logic and Action Mapping in Q-Learning

Condition	Action Taken	Reward / Penalty	Comment
Neighbor RSRP > Serving RSRP + HOM	HO	+4	Desirable handover to a better cell
Neighbor RSRP $\leq$ Serving RSRP + HOM	HO	-0.7	Unnecessary handover with no benefit
Neighbor RSRP > Serving RSRP + HOM	No_HO	-4	Missed opportunity, leads to poor signal
Neighbor RSRP $\leq$ Serving RSRP + HOM	No_HO	+0.7	Correct decision to retain serving cell
Repeated HO within TTT duration	HO	-10	Ping-pong handover penalty
Any reward outcome	—	Clipped to $[-3, +3]$	Stabilizes training and prevents outliers

7.4.3 Q-Learning Implementation and Decision Mapping

The framework uses Q-learning to derive an optimal handover policy over a discretized state space formed from radio conditions [16]. Each state-action pair—HO or No_HO—is evaluated to maximize cumulative rewards under specific HOM and TTT settings. The four-dimensional Q-table is indexed by state (via subtractive clustering), HOM value (in dB), TTT index (in ms), and action. An ϵ -greedy strategy guides exploration, while rewards are assigned based on RSRP delta and adherence to TTT constraints. Q-values are iteratively updated using the Bellman equation, enabling policy refinement over successive episodes [17].

This structured mapping helps the agent converge toward stable and context-aware handover behavior. The use of state-specific HOM and TTT values further enhances granularity, allowing the policy to adapt to heterogeneous signal conditions across the coverage area. Figure 30 depicts the Q-learning framework, outlining the training and testing phases for decision-making in handover (HO) scenarios.

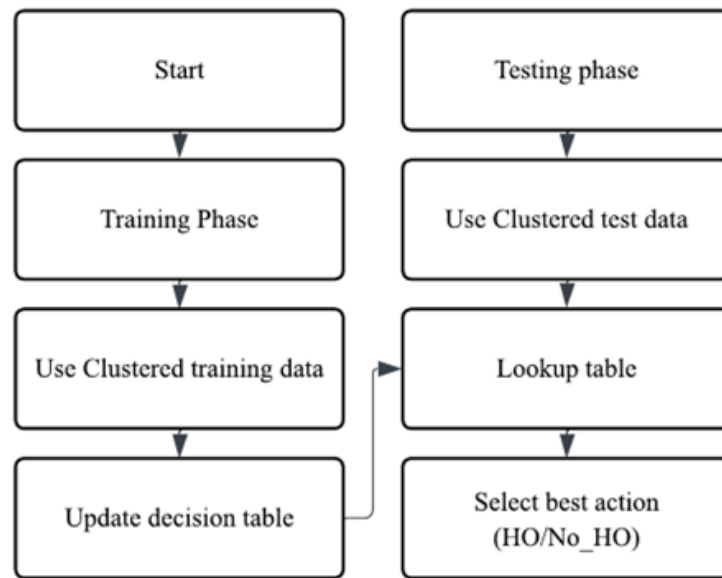


Figure 30: Q-learning Framework

7.4.4 Parameter Optimization Strategy

To enhance handover efficiency and reduce unnecessary transitions in real-world LTE environments, this work incorporates a dedicated parameter optimization routine targeting three core components: Handover Margin (HOM), Time-to-Trigger (TTT), and feature weights for subtractive clustering. These jointly shape the state abstraction, reward evaluation, and triggering logic of the Q-learning framework.

A large search space was systematically defined: HOM values ranged from 0.0 to 2.0 dB, TTT values spanned from 40 ms to 800 ms, and five clustering feature weights were varied to reflect differing priorities for signal quality metrics (e.g., RSRP, RSRQ, CINR). Parameter maps were structured for each of the 20 Q-learning states, enabling state-specific granularity in HO control.

To comprehensively explore this space, the system was run over 50,000 unique combinations of HOM, TTT, and feature weights. For each combination, the training dataset (Day 1) was clustered using the associated weights, and Q-learning was applied over 1000 episodes to learn the optimal handover policy.

The learned policy was then evaluated on an independent test dataset (Day 2), and the total number of handovers was recorded as the primary performance metric. The optimal parameter set was selected based on the lowest resulting HO count.

This full-scale optimization required approximately 56 hours and 37 minutes of simulation time, conducted under real-world mobility conditions. The feature weights shaped the clustering granularity to reflect heterogeneous signal environments, while the dynamic HOM and TTT values adapted policy aggressiveness across network states. As a result, the final learned policy significantly reduced both total and ping-pong handovers, offering a robust and data-driven alternative to conventional static handover configurations.

7.5 Fuzzy Logic–Based Adaptive Handover Control

Fuzzy logic provides a rule-based decision-making framework capable of reasoning under uncertainty and imprecision. Unlike static threshold-based methods, it maps quantitative radio measurements into qualitative linguistic categories (e.g., low, medium, high) and infers decisions through a comprehensive set of rules. In this study, fuzzy logic was used to dynamically adapt Time-to-Trigger (TTT) and Handover Margin (HOM) parameters, which are critical in determining the timing and aggressiveness of handovers in LTE networks. The main objective was to minimize unnecessary handovers while maintaining reliability under real-world mobility conditions.

The overall fuzzy-based handover optimization process is illustrated in Fig. 31.

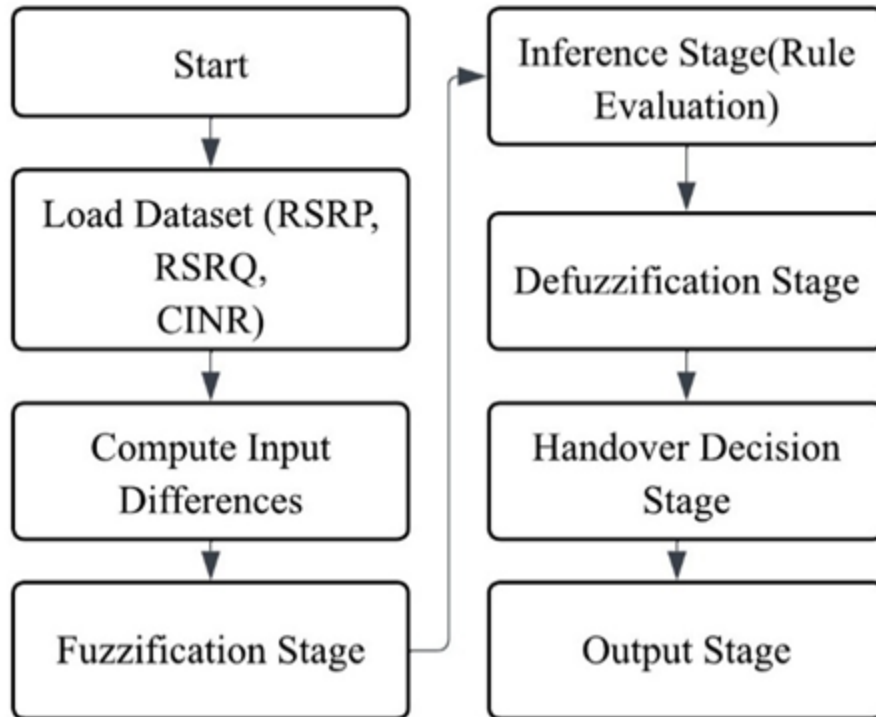


Figure 31. Fuzzy Logic Handover Decision Process

7.5.1 Input Parameters

The fuzzy inference system employed three input parameters derived from the drive test dataset:

- **RSRP difference (Neighbor – Serving):** Indicates the relative signal strength advantage of the neighboring cell.
- **RSRQ difference (Neighbor – Serving):** Represents the relative signal quality difference.
- **Serving Cell CINR:** Captures the link quality under interference and noise conditions.

Each input variable was mapped into three linguistic categories (low, medium, high) using triangular membership functions, with adjustments made to ensure adequate coverage of realistic radio conditions.

7.5.2 Output Parameters

Two adaptive outputs were defined to regulate handover behavior:

- **Adaptive TTT (25–200 ms):** Categorized as short, medium, or long.
- **Adaptive HOM (0 to 2 dB):** Categorized as low, medium, or high.

Lower TTT and HOM values encourage aggressive handovers, whereas higher values enforce conservative behavior to prevent frequent switching.

7.5.3 Fuzzy Rule Base

The adaptive fuzzy logic system employs eight rules mapping RSRP and RSRQ differences along with serving cell CINR to adaptive TTT and HOM values, targeting reductions in missed and unnecessary handovers, as summarized in Table 17.

Table 16 Fuzzy Rule Mapping for Handover Decision Parameter

Fuzzy Rule Base	Input		Output	
Rule No.	RSRP & RSRQ diff	CINR	HOM	TTT
1	High	low	Low	Short
2	High	medium	Low	Medium
3	High	high	Medium	Medium
4	Medium	low	High	Medium
5	Medium	medium	Medium	Medium
6	Medium	high	High	Medium
7	Low	low	Medium	Medium
8	Low	medium /high	High	Long

The purposes of the rules outlined in Table 17 are as follows. Rule 1 ensures rapid handover execution to mitigate missed handovers when the neighboring cell exhibits superior strength and the serving link quality is degraded. Rule 2 achieves a balance between missed and unnecessary handovers, facilitating handover without excessive aggression. Rule 3 minimizes unnecessary handovers by introducing a delay, given the sustained quality of the serving link. Rule 4 reduces the incidence of missed handovers by enabling handover when the serving cell quality is suboptimal, despite a moderately better neighboring cell. Rule 5 maintains equilibrium between missed and unnecessary handovers under moderate operational conditions. Rule 6 decreases unnecessary handovers, prioritizing retention on a sufficiently robust serving cell. Rule 7 addresses missed handovers by supporting handover when the neighboring cell offers a slight improvement over a poor serving cell. Finally, Rule 8 diminishes unnecessary handovers, promoting retention on the current cell when the neighboring cell does not provide significant improvement and the serving cell remains adequate.

7.6 Deep Q-Network (DQN)–Based Reinforcement Learning

7.6.1 Overview of the Framework

Reinforcement Learning (RL) is employed to optimize handover (HO) decision-making in LTE networks by formulating the problem as a Markov Decision Process (MDP). At each time step t , the RL agent observes the current state s_t , selects an action a_t from the action space, and receives a reward r_t from the environment based on the effectiveness of the chosen handover parameters. The objective is to maximize the expected cumulative discounted reward:

$$(\pi)^* = \underset{\pi}{\operatorname{argmax}} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r_t | \pi \right] \quad (16)$$

where π represents the handover policy and $\gamma \in [0,1]$ is the discount factor.

To approximate the optimal action-value function $Q^*(s,a)$, a Deep Q-Network (DQN) is implemented. The DQN predicts the long-term expected reward for each possible action given the current state. Two separate neural networks are used:

- **Online Network** → Continuously updated during training.
- **Target Network** → Stabilizes learning by periodically synchronizing with the online network.

This dual-network structure mitigates oscillations and improves convergence stability.

7.6.1 State Space Definition

The input to the DQN agent is a 7-dimensional continuous state vector derived from LTE measurement reports.

For each time step t , the state vector is defined as:

$$s_t = [Q_{\text{weighted}}, \Delta\text{RSRP}, \Delta\text{RSRQ}, \text{CINR}_{\text{serv}}, \Delta\text{RSRP}_{t-1}, \Delta\text{RSRP}_{t-2}, \text{RSRP}_{\text{trend3}}] \quad (17)$$

where:

- $Q_{\text{weighted}} = 0.6 \cdot \Delta\text{RSRP} + 0.3 \cdot \Delta\text{RSRQ} + 0.1 \cdot \text{CINR}_{\text{serv}}$
- $\Delta\text{RSRP} = \text{Neighbor RSRP} - \text{Serving RSRP}$
- $\Delta\text{RSRQ} = \text{Neighbor RSRQ} - \text{Serving RSRQ}$
- $\text{CINR}_{\text{serv}} = \text{Serving cell CINR}$
- $\Delta\text{RSRP}_{t-1}, \Delta\text{RSRP}_{t-2} = \text{Previous-step RSRP differences}$
- $\text{RSRP}_{\text{trend3}} = \text{3-step RSRP trend}$

Thus, the state space is: $S \subseteq \mathbb{R}^7$

7.6.2 Action Space Definition

The agent selects actions corresponding to different handover parameter configurations. The action space consists of all possible combinations of Time-to-Trigger (TTT) and Hysteresis Offset Margin (HOM) values:

$$\mathcal{A} = \{(t, h) \mid t \in \text{TTT levels}, h \in \text{HOM levels}\} \quad (18)$$

In this work, the following discrete values were defined:

- **TTT levels** = [25, 50, 100, 150, 200] ms
- **HOM levels** = [0, 0.5, 1.0, 1.5, 2.0] dB

Total number of possible actions:

$$|\mathcal{A}| = 5 \times 5 = 25$$

Each action corresponds to a unique (TTT, HOM) pair, allowing the DQN to dynamically control handover aggressiveness based on network conditions.

7.6.3 Environment and Reward Design

A custom LTE handover simulation environment was developed where the agent interacts sequentially with real LTE measurement data.

For each state-action pair, the environment simulates the next state and computes a reward r_t based on the quality of the handover decision.

The reward structure is summarized in Table 18.

Table 17 Reward Structure

Scenario	Condition	Reward
Beneficial HO	RSRP gain $\geq$ 6 dB	+6.0
Bad HO	HO executed but RSRP gain < 6 dB	-5.0
Unnecessary HO	HO triggered when not required	-2.0
Missed HO	No HO when needed (RSRP gap $\geq$ margin)	-3.0
Stable Connection	No HO needed and agent stays	+2.0
Ping-Pong HO	HO within 500 ms of previous HO	-6.0

This reward design balances handover efficiency and network stability, enabling the agent to avoid unnecessary HOs while preventing radio link failures.

7.6.4 Deep Q-Network Architecture

The Q-function $Q(s,a;\theta)$ is approximated using a deep neural network implemented in Pytorch. The architecture is illustrated in Figure 32.

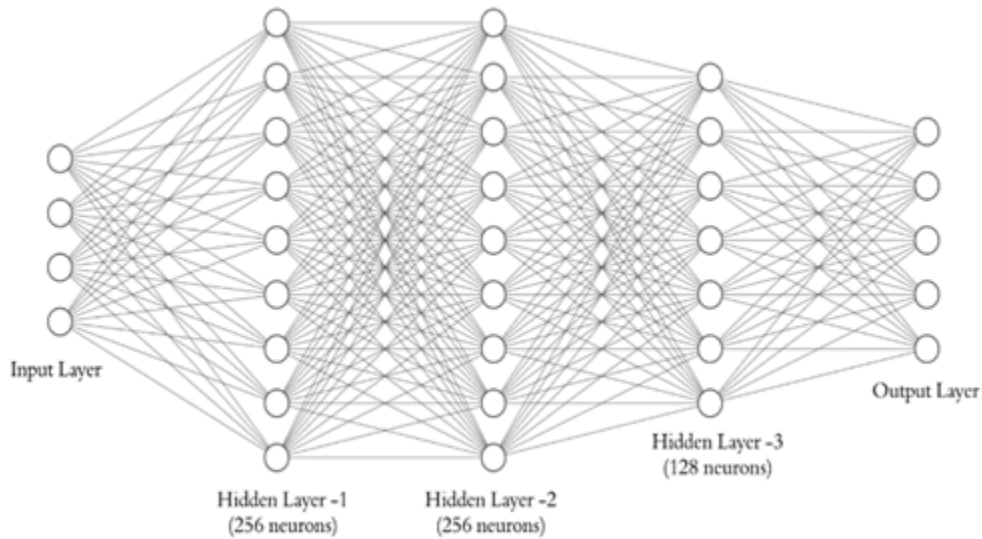


Figure. 32. Deep Q-Network Architecture

Key Components

- Input Layer: Accepts the 7-dimensional state vector.
- Hidden Layers:

- Fully connected layers: [256, 256, 128]
 - Activation: Leaky ReLU
 - Normalization: LayerNorm
 - Regularization: Dropout = 10%
- Output Layer: Produces $Q(s,a)$ values for all possible actions.

Role of Replay Buffer

A Replay Buffer of fixed capacity is used to store experience tuples (s_t, a_t, r_t, s_{t+1}) . During training, the agent samples mini-batches uniformly, which:

- Breaks temporal correlations between sequential samples.
- Stabilizes gradient updates.
- Improves overall learning efficiency.

7.6.5 Deep Q-Network Agent and Learning Policy

The DQN agent is responsible for approximating the optimal action-value function $Q^*(s,a)$ and selecting handover actions. The agent employs an epsilon-greedy policy to balance exploration and exploitation:

$$a_t = \begin{cases} \text{random action} & \text{with probability } \epsilon \\ \operatorname{argmax}_a Q_{\text{online}}(s_t, a) & \text{with probability } 1 - \epsilon \end{cases}$$

where ϵ decays linearly from an initial ϵ_{start} to a final ϵ_{end} over a fixed number of steps. This mechanism ensures sufficient exploration during early training while gradually favoring exploitation of learned policies.

7.6.6 Online and Target Q-Networks

In this work, two separate neural networks are maintained to stabilize the Deep Q-Learning process: the online Q-network and the target Q-network.

- Online Q-Network: Responsible for selecting actions during training. It is continuously updated via gradient descent to minimize the prediction error between the current Q-values and the expected future rewards.
- Target Q-Network: A delayed copy of the online network that is used to compute stable target Q-values. Unlike the online network, its parameters are updated less frequently by periodically copying the weights from the online network.

This dual-network architecture addresses the moving target problem in reinforcement learning, where the Q-value targets shift rapidly if the same network is used both for predicting and evaluating actions. By introducing a target network with delayed updates, the learning process becomes more stable, prevents oscillations, and improves convergence.

Furthermore, a replay buffer is employed to store past transitions (s, a, r, s'). During training, the agent samples mini-batches from this buffer to break temporal correlations and ensure more efficient and stable updates for the online network. The difference between the predicted Q-values (from the online network) and the target Q-values (computed using the target network) is minimized using the mean squared error (MSE) loss. The Adam optimizer with gradient clipping is used to ensure stable training and avoid exploding gradients.

7.6.7 Hyperparameters

The key hyperparameters of the DQN agent are summarized in Table 19.

Table 18 Hyperparameter Values

Parameter	Value / Range
Learning Rate	0.001
Discount Factor	0.99
Replay Buffer Size	100,000
Batch Size	64
Min Replay Size	1,000
Target Update Frequency	1,000 steps
Epsilon Start / End	1.0 $\rightarrow$ 0.05
Epsilon Decay Steps	15,000
Gradient Clipping	10.0

Chapter Eight

Evaluation

8.1 Performance Evaluation of Handover Prediction

Among the three datasets, Dataset_2 and Dataset_3 did not include user speed information, whereas Dataset_1 contained additional user speed measurements obtained from the drive test equipment. Based on this distinction, the evaluation has been divided into two separate experimental cases:

- Case 1: Drive test data from Dataset_2 and Dataset_3, which do not include user speed as a feature.
- Case 2: Drive test data from Dataset_1, which includes user speed as an additional input feature.

This dual-case evaluation enhances the robustness of the ablation study by testing model performance under both mobility-aware and mobility-agnostic conditions. Case 1 evaluates prediction solely from network parameters, while Case 2 incorporates user speed to capture mobility dynamics. This design verifies that the proposed framework remains effective and generalizable across varying feature availability, reflecting realistic handover prediction scenarios.

We have used five evaluation metrics for our study, namely Accuracy, Precision, Recall, F1 Score and ROC–AUC Score, to provide a comprehensive assessment of model performance in predicting handover events. These metrics are described in brief here.

Accuracy

Accuracy is a fundamental metric used for evaluating the performance of a classification model. It represents the proportion of correct predictions made by the model out of all predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (19)$$

Here, TP (True Positives) and TN (True Negatives) denote the correctly predicted positive and negative instances, respectively, while FP (False Positives) and FN (False Negatives) represent the incorrectly predicted instances. Accuracy provides an overall measure of how well the model performs in correctly classifying data samples.

However, accuracy alone may not always provide a complete picture of model performance, particularly when dealing with imbalanced datasets. In such cases, a model might achieve a high accuracy by primarily predicting the majority class while failing to correctly identify the minority class. Hence, while accuracy offers a useful general overview, additional metrics such as Precision, Recall, F1 Score, and ROC–AUC are necessary to obtain a more balanced and reliable evaluation of the model's predictive capability.

Precision

Precision evaluates the reliability of positive predictions by quantifying the proportion of correctly predicted positive instances among all instances classified as positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (20)$$

It indicates how accurate the model's positive classifications are and reflects its ability to minimize false alarms. A high Precision value signifies that when the model predicts a positive event, it is likely to be correct.

Recall or Sensitivity

It measures the model's ability to correctly identify actual positive instances. High recall ensures that the model captures most of the relevant events, which is particularly important for critical tasks such as handover prediction:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (21)$$

Recall is especially important in applications where missing a positive case can lead to significant consequences. A high Recall value indicates that the model successfully captures most of the relevant positive events, minimizing missed detections.

F1 Score

It is a performance metric that evaluates how well a classification model balances Precision and Recall, particularly useful when dealing with imbalanced datasets where one class occurs more frequently than the other. It is defined as the harmonic mean of Precision and Recall:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

The harmonic mean ensures that the F1 Score is high only when both Precision and Recall are high, making it a more balanced indicator of performance than a simple average. It penalizes extreme differences between the two metrics—if either Precision or Recall is low, the F1 Score will also decrease significantly. This makes the F1 Score especially suitable for evaluating models in imbalanced classification problems, where maintaining an appropriate trade-off between correctly identifying positive instances and minimizing false alarms is crucial.

ROC AUC Score

The ROC–AUC (Receiver Operating Characteristic – Area Under Curve) is a widely used performance metric for evaluating binary classification models. It measures how well a model can distinguish between positive and negative classes across different threshold levels. The ROC curve is a plot of the True Positive Rate (TPR) — also known as Recall or Sensitivity — against the False Positive Rate (FPR), defined as:

$$\text{FPR} = \frac{FP}{FP + TN} \quad (23)$$

Each point on the ROC curve represents a different decision threshold, showing the trade-off between correctly identifying positives and incorrectly classifying negatives. The AUC (Area Under the Curve) quantifies the overall ability of the model to rank a randomly chosen positive instance higher than a randomly chosen negative one:

$$\text{ROC AUC Score} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR} \quad (24)$$

Thus, a higher AUC value indicates better model performance. In essence, the ROC–AUC score provides a threshold-independent measure of model quality, reflecting how effectively the classifier separates the two classes.

These metrics collectively capture different aspects of classification effectiveness, ensuring a balanced evaluation of both overall correctness and the ability to identify critical events. This multifaceted evaluation is particularly important in wireless network scenarios, where both accurate and timely identification of handover events directly impact network quality and user experience.

8.1.1 Ablation Study on Case 1

Table 19 Comparative analysis of supervised machine learning algorithms- Baseline Model

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC AUC Score
Logistic Regression	0.9271	0.7955	0.7955	0.7955	0.9711
Random Forest	0.9231	0.8378	0.7045	0.7654	0.9672
SVM	0.9352	0.8889	0.7273	0.8000	0.9709
KNN	0.9028	0.7941	0.6136	0.6923	0.9083
Naïve Bayes	0.8421	0.7273	0.1818	0.2909	0.9037
Gradient Boost	0.9352	0.8684	0.7500	0.8049	0.9694
AdaBoost	0.8988	0.7568	0.6364	0.6914	0.9425
XGBoost	0.9312	0.8462	0.7500	0.7952	0.9620
CatBoost	0.9190	0.9000	0.6136	0.7297	0.9723

Table 20 Comparative analysis of supervised machine learning algorithms- SMOTE and Grid Search

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC AUC Score
Logistic Regression	0.9312	0.7647	0.8864	0.8211	0.9659
Random Forest	0.9393	0.8718	0.7727	0.8193	0.9680
SVM	0.9271	0.7708	0.8409	0.8043	0.9727
KNN	0.9150	0.7805	0.7273	0.7529	0.8961
Naïve Bayes	0.8704	0.7000	0.4773	0.5676	0.9240
Gradient Boost	0.9352	0.8889	0.7273	0.8000	0.9661
AdaBoost	0.9150	0.7674	0.7500	0.7586	0.9637
XGBoost	0.9352	0.8684	0.7500	0.8049	0.9689
CatBoost	0.9312	0.8140	0.7955	0.8046	0.9738

Table 20 represents the performance metrics of the baseline supervised models for Case 1. After applying SMOTE with Grid Search optimization, the results improved slightly as shown in Table 21. Among the nine classifiers, Logistic Regression showed the most balanced results with F1 score of 0.8211 and accuracy 0.9312. It also attained the best recall of 0.8864. The highest attained precision was 0.8889 and the ROC AUC Score ranged between 0.8961 and 0.9738.

Table 21 Comparative analysis of ANN and Vanilla AE

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC AUC Score
ANN	0.9433	0.8571	0.8182	0.8372	0.9735
Vanilla AE	0.6721	0.3011	0.6364	0.4088	0.7669

The performance metrics of ANN and Vanilla autoencoder for case 1 are summarized in Table 22. ANN achieved F1 score of 0.8372 and accuracy of 0.9433. But, Vanilla autoencoder could attain F1 score of only 0.4088, accuracy of 0.6721 and a poor precision of 0.3011, indicating limited capability in distinguishing handover events without contrastive learning.

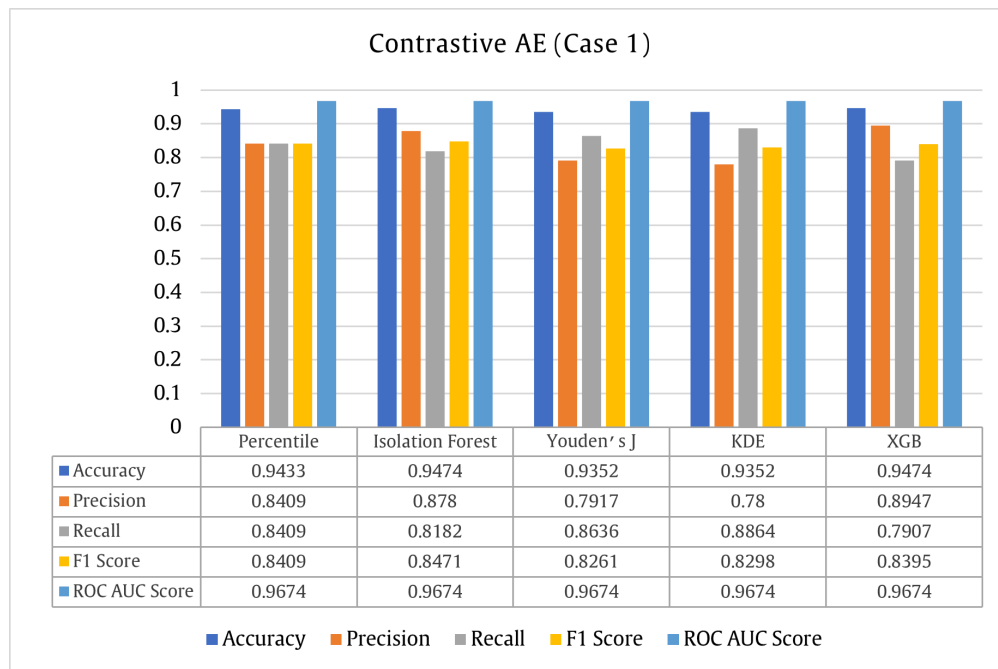
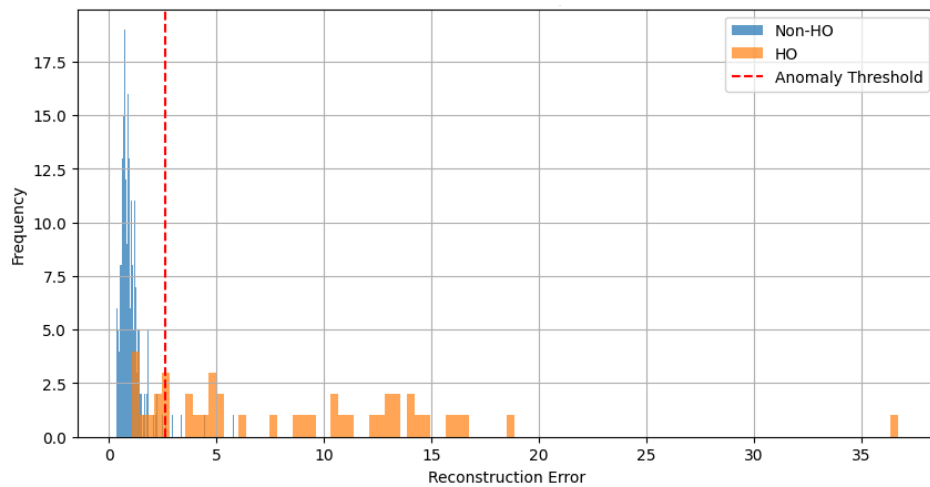


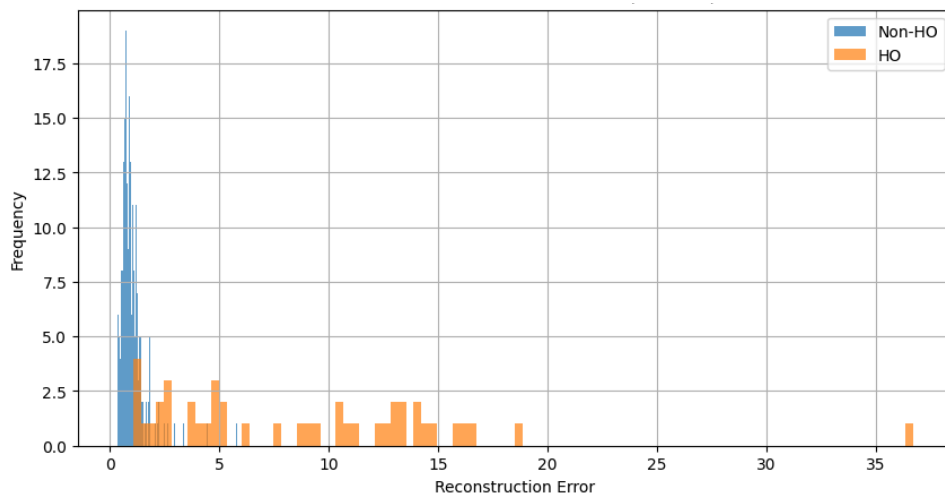
Figure 33. Evaluation Metrics of CAE for different thresholding techniques.

For case 1, Figure 33 illustrates the F1 score, accuracy, precision, recall, and ROC–AUC score of the Contrastive Autoencoder framework with five different thresholding mechanisms – Percentile, Isolation Forest, Youden's J threshold, KDE and XGBoost. The model achieved the highest F1 score of 0.8471 with the Isolation Forest threshold, indicating a strong balance between correctly predicting handovers and minimizing unnecessary predictions. Other thresholds also performed well, with KDE attaining an F1 score of 0.8298, XGBoost achieving 0.8395, Percentile reaching 0.8409, and Youden's J obtaining 0.8261, showing the robustness of the framework across different thresholding strategies. Accuracy reached 0.9474 with both

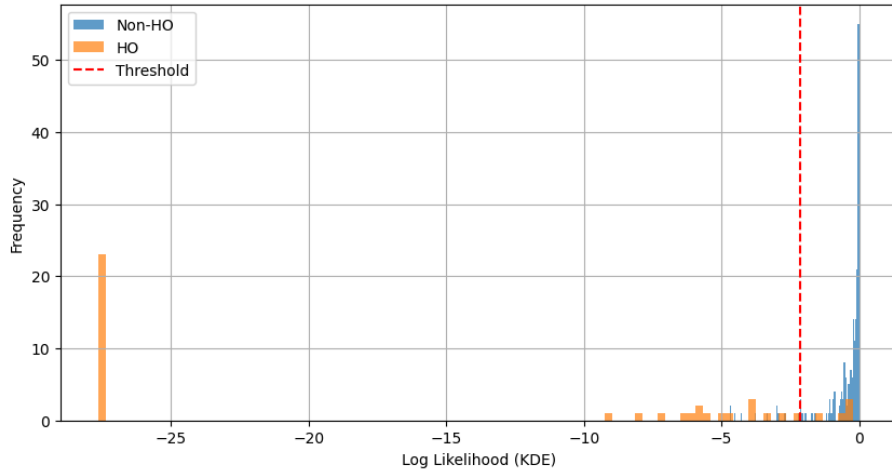
Isolation Forest and XGB thresholds, reflecting that the model reliably identified both handover and non-handover instances. Precision was highest with the XGBoost threshold at 0.8947, meaning most predicted handovers were correct, minimizing unnecessary triggers. While recall ranged from 0.7907 to 0.8864, demonstrating the model's ability to capture a large proportion of true handovers, the ROC–AUC scores remained consistently high at 0.9674 for all thresholds, indicating that the learned latent space effectively separates handover and non-handover events.



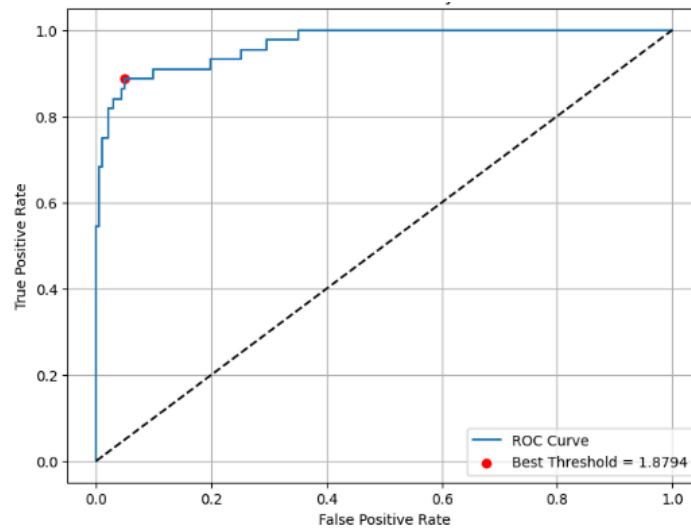
(a)



(b)



(c)



(d)

Figure 34. Thresholding of Reconstruction Error Based on (a) Isolation Forest, (b) Percentile, (c) KDE, and (d) Youden's J Threshold.

Figure 34 illustrates the thresholding mechanisms applied to the reconstruction error distribution for distinguishing Handover (HO) and Non-HO samples. Among the four methods—Isolation Forest, Percentile, Kernel Density Estimation (KDE), and Youden's J—only the latter three differ in adaptivity and data dependency. The Isolation Forest adaptively learns anomaly boundaries through recursive feature partitioning, while KDE and Youden's J determine thresholds dynamically from the empirical distribution and ROC optimization, respectively. In contrast, the percentile-based approach relies on a fixed quantile, providing a simple yet non-adaptive baseline. Together, these thresholding techniques enable a comparative assessment between static and data-driven decision boundaries for robust reconstruction error-based HO event detection.

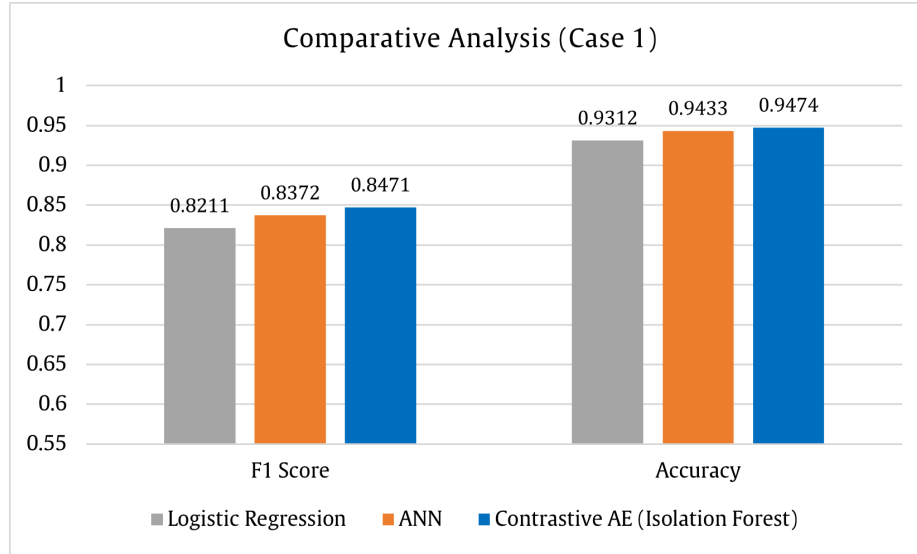


Figure 35. Comparative Analysis of Best-Performing Models

Figure 35 presents a comparative analysis of the best-performing supervised model, ANN and the proposed CAE with Isolation Forest thresholding for Case 1. The CAE framework achieved the highest F1 score of 0.8471 and accuracy of 0.9474, outperforming both the supervised model and ANN. The consistent improvement in both metrics demonstrates that the proposed contrastive learning framework enhances the model's ability to accurately identify handover events while maintaining high reliability in overall classification. These results confirm the effectiveness of the CAE framework in capturing discriminative patterns and improving handover prediction performance compared to conventional supervised and ANN-based approaches.

8.1.2 Ablation Study on Case 2

Table 22 Comparative analysis of supervised machine learning algorithms- Baseline Model

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC AUC Score
Logistic Regression	0.8270	0.4286	0.2000	0.2727	0.8903
Random Forest	0.9135	0.8889	0.5333	0.6667	0.9462
SVM	0.9027	0.9286	0.4333	0.5909	0.9275
KNN	0.9135	0.9375	0.5000	0.6522	0.8278
Naïve Bayes	0.8324	1.0000	0.0312	0.0606	0.7330
Gradient Boost	0.9189	0.9474	0.5625	0.7059	0.9344
AdaBoost	0.8919	0.8750	0.4375	0.5833	0.8710
XGBoost	0.9135	0.9444	0.5312	0.6800	0.8956
CatBoost	0.8973	0.8667	0.4333	0.5778	0.9290

Table 23 Comparative analysis of supervised machine learning algorithms- SMOTE and Grid Search

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC AUC Score
Logistic Regression	0.8757	0.6400	0.5333	0.5818	0.8970
Random Forest	0.9135	0.8500	0.5667	0.6800	0.9424
SVM	0.9135	0.7917	0.6333	0.7037	0.9406
KNN	0.9081	0.7600	0.6333	0.6909	0.8361
Naïve Bayes	0.8432	0.7143	0.1562	0.2564	0.7000
Gradient Boost	0.9189	0.9048	0.5938	0.7170	0.9173
AdaBoost	0.8757	0.6552	0.5938	0.6230	0.8735
XGBoost	0.9243	0.9091	0.6250	0.7407	0.9154
CatBoost	0.9027	0.7500	0.6000	0.6667	0.9308

Table 23 presents the performance metrics of the baseline supervised models for Case 2. After applying SMOTE with Grid Search optimization, a slight improvement in performance was observed, as shown in Table 24. Among the nine classifiers, XGBoost demonstrated the most balanced performance, achieving an F1 score of 0.7407 and an accuracy of 0.9243. The highest precision and recall achieved were 0.9091 and 0.6333, respectively, while the ROC–AUC scores ranged from 0.70 to 0.9424.

Table 24 Comparative analysis of ANN and Vanilla AE

Case	Accuracy	Precision	Recall	F1 Score	ROC AUC Score
ANN	0.9135	0.8182	0.6000	0.6923	0.9148
Vanilla AE	0.7405	0.3393	0.6333	0.4419	0.7617

The performance metrics of ANN and Vanilla autoencoder for case 2 are summarized in Table 25. ANN achieved F1 score of 0.6923 and accuracy of 0.9135. But, Vanilla autoencoder could attain F1 score of only 0.4413, accuracy of 0.7405 and a poor precision of 0.3303, indicating a limited ability to effectively distinguish handover events in the absence of contrastive learning.

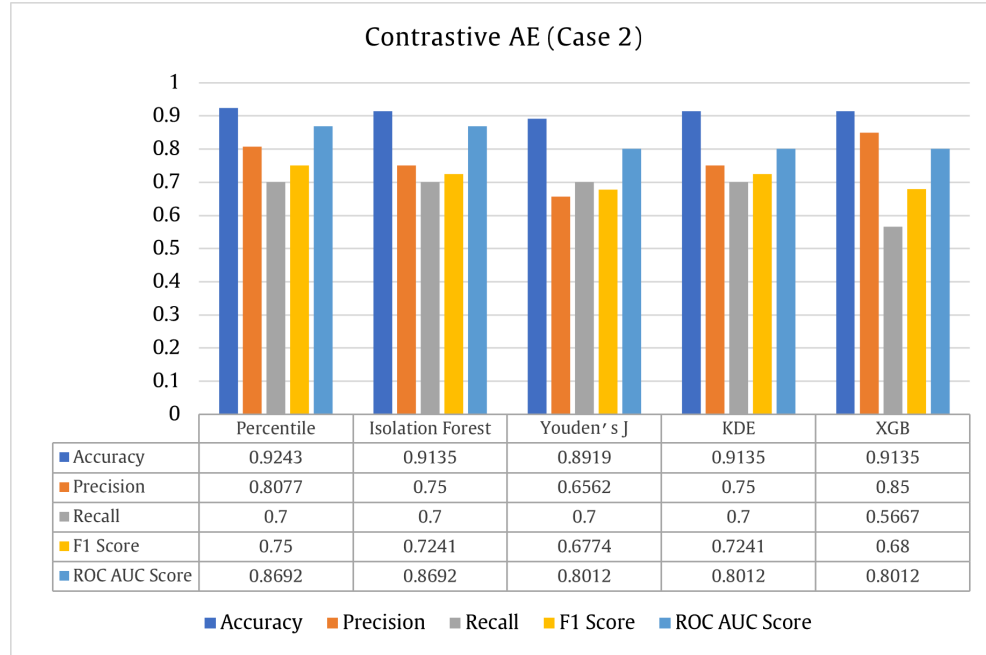
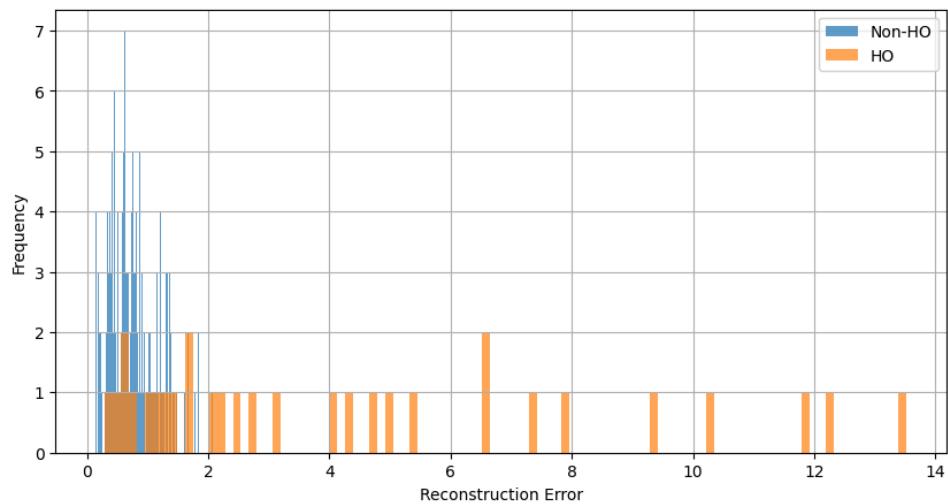
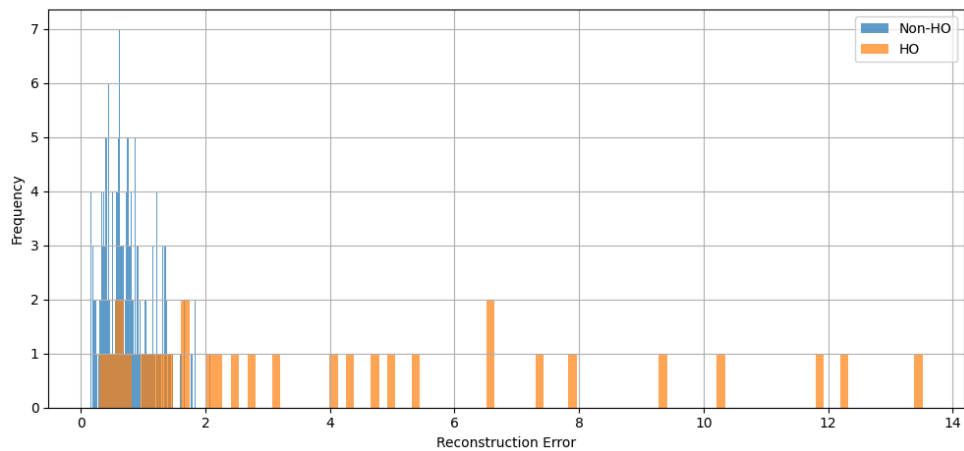


Figure. 36. Evaluation Metrics of CAE for different thresholding techniques.

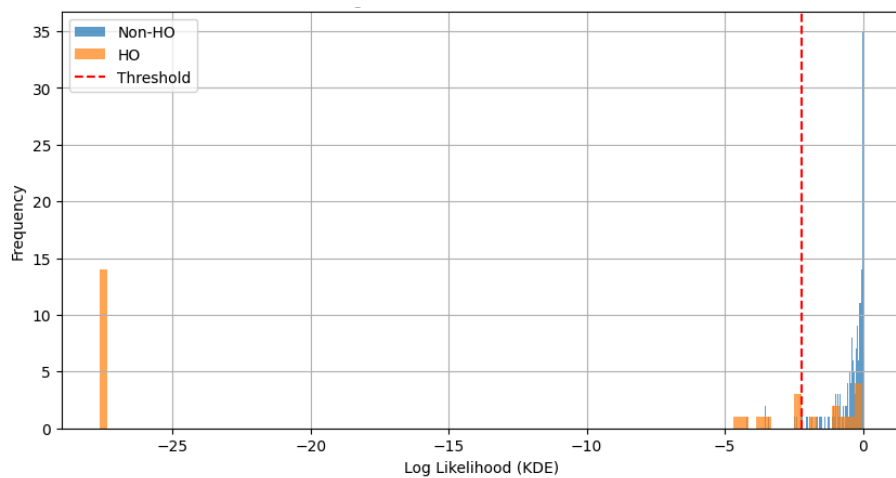
For case 2, Figure 36 presents the F1 score, accuracy, precision, recall, and ROC–AUC score of the Contrastive Autoencoder framework using five thresholding mechanisms – Percentile, Isolation Forest, Youden’s J threshold, KDE, and XGBoost. The model achieved the highest F1 score of 0.75 with the Percentile threshold, demonstrating a balanced prediction performance. Isolation Forest also yielded an F1 score of 0.7241, reflecting stable performance across varying thresholding approaches. The highest accuracy of 0.9243 was also attained with the Percentile threshold, indicating reliable overall prediction capability. Precision peaked at 0.8500 with the XGB threshold, showing that most predicted handovers were accurate, while recall values went up to 0.70, confirming the framework’s ability to capture a substantial portion of actual handover events. The ROC–AUC scores, ranging between 0.8012 and 0.8692, further demonstrate effective latent-space separation of handover and non-handover instances, even under challenging conditions.



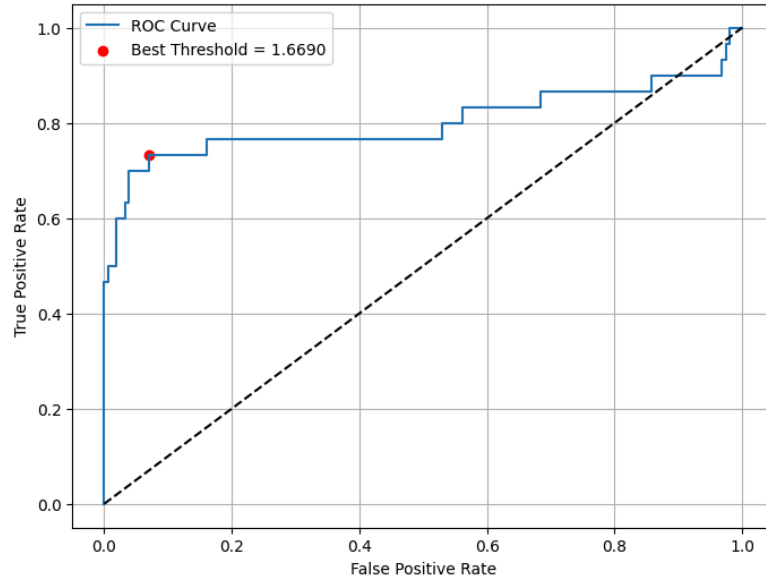
(a)



(b)



(c)



(d)

Figure 37. Thresholding of Reconstruction Error Based on (a) Isolation Forest, (b) Percentile, (c) KDE, and (d) Youden's J Threshold.

Figure 37. shows that, in Case 2, the inclusion of user speed as an auxiliary input significantly influences the learned latent representation, thereby altering the reconstruction error distribution. Consequently, the adaptive thresholding methods (Isolation Forest, KDE, and Youden's J) dynamically adjust their anomaly boundaries, capturing speed-induced variations in mobility behavior that were absent in Case 1.

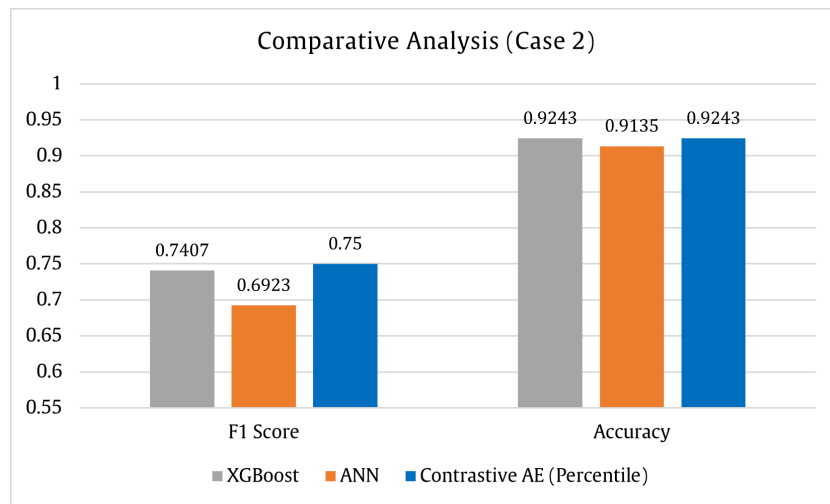


Figure 38. Comparative Analysis of Best-Performing Models

Figure 38 presents a comparative analysis of the best-performing supervised model, ANN and the proposed CAE with Isolation Forest thresholding for Case 2. The CAE framework achieved the highest F1 score of 0.75 and accuracy of 0.9243, outperforming both the supervised model

and ANN. The consistent improvement in both metrics demonstrates that the proposed contrastive learning framework reliably captures discriminative patterns and enhances handover prediction performance. The results from both Case 1 and Case 2 show that the CAE framework performs better than the state-of-the-art methods, confirming its effectiveness and robustness across different scenarios while providing accurate, balanced, and practical handover predictions that can significantly improve network QoS and mobility performance.

8.2 Evaluation of Handover Optimization Methods:

8.2.1 Q-Learning and Contextual Bandit Modeling

8.2.1.1 Evaluation Phase

To assess true generalization in a dynamic radio environment, the trained agent has to be evaluated on an unseen dataset. Testing on the same training traces risks policy memorization and gives a misleading estimate of performance in operational conditions; instead, the agent must apply its learned mapping from instantaneous radio contexts to actions when confronted with new, time-evolving measurement sequences. Accordingly, evaluation proceeds sequentially over the test trace so that results reflect the agent's behaviour in a temporally ordered, realistic scenario.

Applying a tabular action-value function to new contexts requires matching each test-state to an entry in the learned Q-table. Exact matches are unlikely because the Q-table keys are continuous, normalized feature vectors observed during training. We therefore employ a nearest-neighbour lookup in the normalized feature space: for each test vector the Euclidean distance to every stored training-state is computed and the Q-values of the nearest training-state are used to select the action. This procedure projects dynamic test contexts onto the closest learned contexts and permits direct reuse of the tabulated policy.

Two training-testing configurations were evaluated. In Case 1, the agent was trained on the Day 1 dataset and tested on the Day 2 dataset, while in Case 2, training was performed on the Day 3 dataset with testing again on the Day 2 dataset. Figures 39 and 40 present the PCA-based projections of the Q-table states from the training phase alongside the corresponding states from the test data for each case.

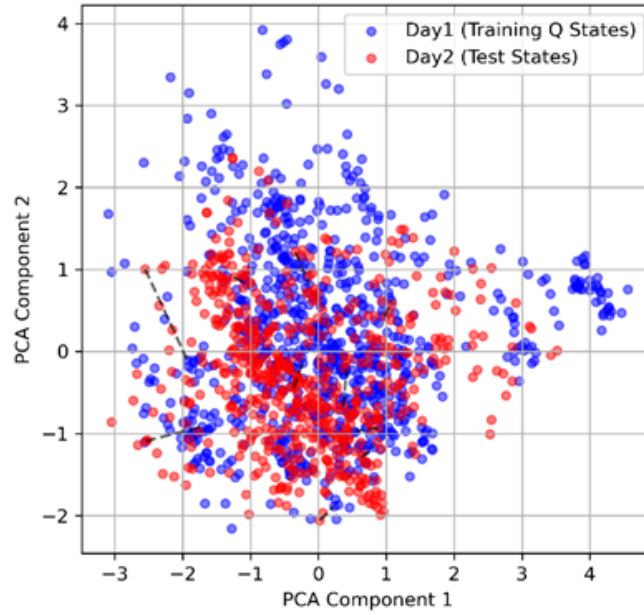


Figure 39. PCA projection of training Q-table states and test states-Case 1.

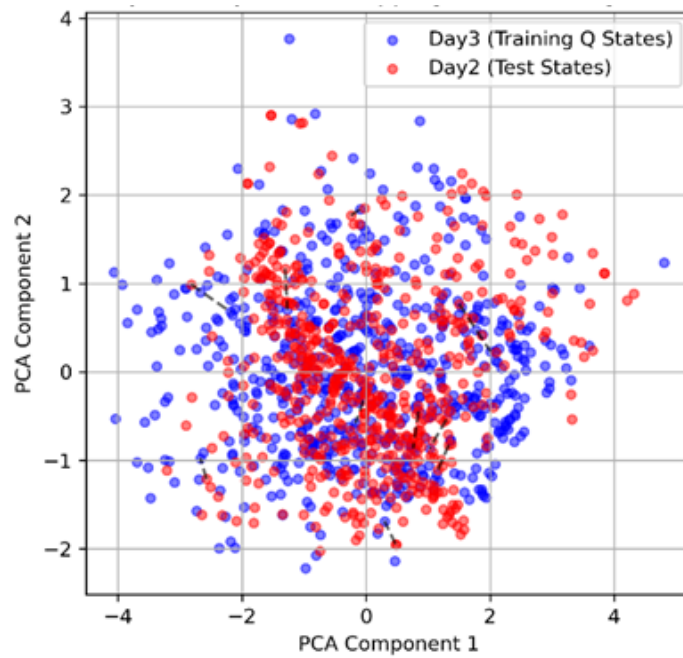


Figure 40. PCA projection of training Q-table states and test states-Case 2.

A limitation of this approach is the loss of temporal context, as nearest-neighbour matching operates on individual snapshots without considering sequential dependencies. To address this,

an external Time-to-Trigger (TTT) filter is incorporated during evaluation. This post-decision constraint ensures that handover actions are only accepted if the neighbour cell advantage persists for the required TTT duration, thereby preserving the temporal stability expected in operational handover control.

8.2.1.2 Result Analysis

This subsection presents the performance evaluation of the trained RL agent under two configurations, employing a HOM of 3 dB and TTT values of 0.7 s and 1.0 s for the both of the cases. As shown in Table 26, the RL agent successfully reduces unnecessary handovers in both cases, with the optimized handover count closely aligned to the reward function's objectives.

Table 25 RL agent's handover reduction performance across TTT settings

Case	TTT (second)	Original HO	Agent Action	TTT Blocked	Optimized HO
1	1	99	87	20	67
	0.7	99	93	21	72
2	1	99	84	18	66
	0.7	99	95	23	72

The original HO counts represent baseline handovers, while agent actions show the RL-decided handovers. The TTT blocked column indicates handovers suppressed by the agent per configured TTT, and the optimized HO reflects the effective handover count post RL filtering. The RL agent effectively reduces unnecessary handovers in both cases, achieving optimized HO aligned with the reward function for network stability and performance. Notably, TTT = 1 second results in fewer optimized handovers than TTT = 0.7 seconds due to stricter HO blocking. The consistent use of identical TTT values in training and testing phases validates the agent's adaptability and effectiveness across varying TTT configurations.

Figure 41–44 illustrate the temporal distribution of handover events for two cases under different TTT configurations, highlighting the RL agent's decision-making behavior in comparison to original handover instances and the TTT-based blocking mechanism.

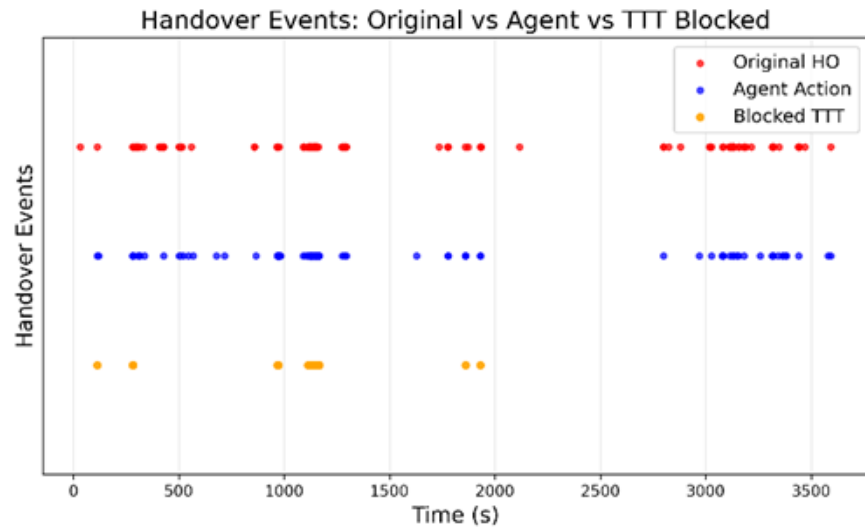


Figure 41. Temporal distribution of handover events: Case 1, TTT=1s

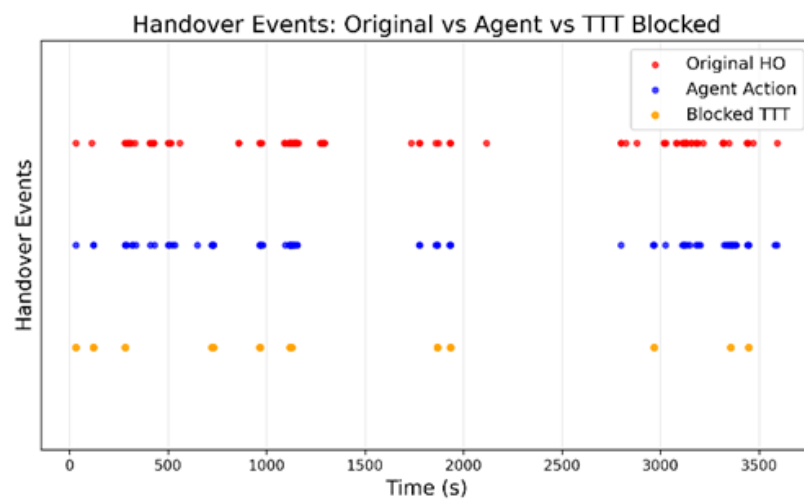


Figure 42. Temporal distribution of handover events: Case 1, TTT=0.7s

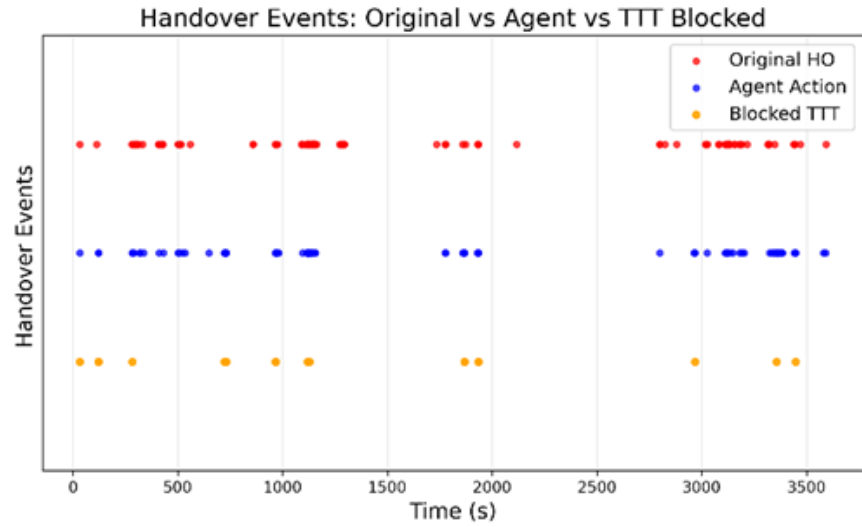


Figure 43. Temporal distribution of handover events: Case 2, TTT=1s

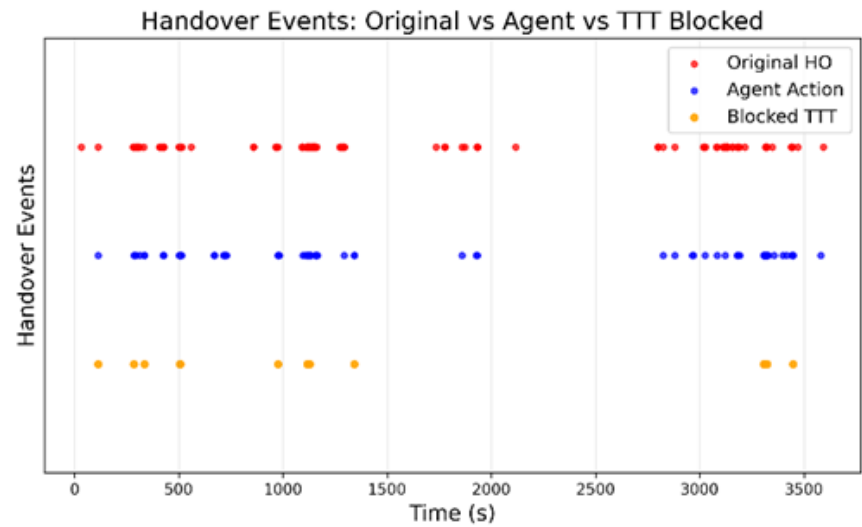


Figure 44. Temporal distribution of handover events: Case 2, TTT=0.7s

Each plot shows three event categories: original handovers, handovers accepted by the RL agent and handovers blocked by the TTT filter. The visualization shows the agent behaviour with respect to the actual handover instances and confirms that the RL agent consistently suppresses unnecessary handovers. This alignment verifies the agent's ability to incorporate external filtering (TTT) into its policy, minimizing ping-pong handovers while maintaining mobility performance. The consistent behavior across both cases and TTT settings validates the robustness and adaptability of the RL-based handover optimization framework.

8.2.2 Q-Learning via Subtractive Clustering

This section presents the performance evaluation of the proposed Q-learning-based handover (HO) optimization framework using the real-world LTE dataset collected through drive tests. The analysis focuses on two core objectives:

- (1) minimizing the overall number of handovers by selecting an optimal configuration of HOM, TTT, and clustering weights
- (2) eliminating or significantly reducing ping-pong handovers while enabling ultra-fast decision-making compared to conventional heuristic-based methods, thereby improving both efficiency and responsiveness.

8.2.2.1 Handover Count Reduction

In the baseline scenario, based on the XCAL-M logs and existing operator-defined handover configurations, 92 handovers were recorded over the Day 2 route. After applying the learned policy from the Q-learning agent, this number was successfully reduced to 80 handovers, representing a 13% decrease.

Figure 45 illustrates a bar comparison of the handover counts: the Traditional Heuristic Model recorded 92 handovers, while the Proposed Model achieved 80 handovers. This visual representation highlights the reduction in handover frequency, with the Q-learning based approach demonstrating a more efficient distribution of handover events, leading to smoother transition behavior.

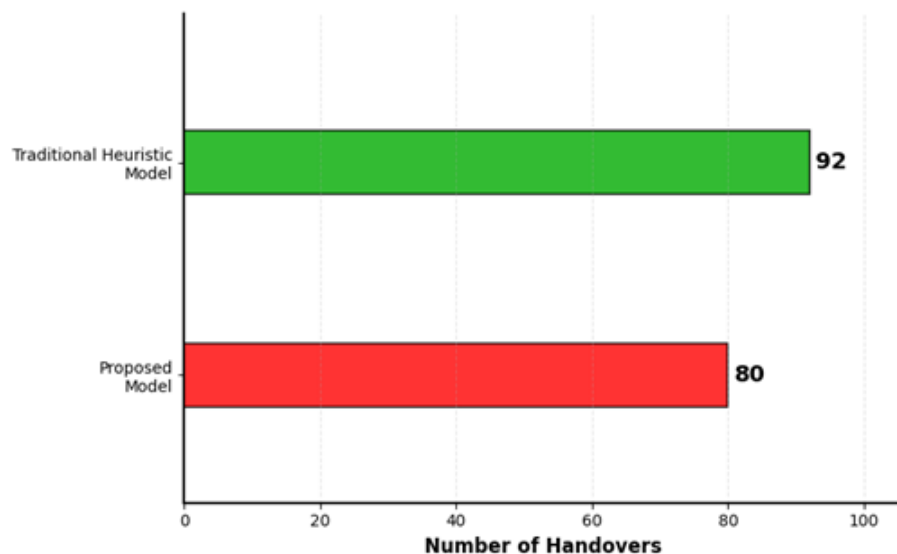


Figure 45: Handover Comparison Between Traditional and Proposed Models

8.2.2.2 Optimization Loop Summary and Best Parameter Configuration

To achieve the result above, the system evaluated 50,000 unique parameter combinations, covering varied handover margins, time-to-trigger values, and five feature weights for clustering. The full process took approximately 56 hours 37 minutes of simulation runtime. Each combination was trained on the Day 1 dataset and tested on Day 2, with handover count used as the evaluation metric.

The best-performing configuration, which yielded 80 handovers, is summarized in Table 27.

Table 26 Optimized Parameter Configuration for Minimum Handover Count

Parameter	Value
HOM Range (in dB)	{1: 0.05, 2: 0.1, 3: 0.24, 4: 0.3, 5: 0.4, 6: 0.3, 7: 0.35, 8: 0.4, 9: 0.8, 10: 0.94, 11: 0.55, 12: 0.6, 13: 0.65, 14: 1.36, 15: 0.75, 16: 0.8, 17: 0.85, 18: 0.9, 19: 0.95, 20: 1.0}
TTT Range (in ms)	{1: 40, 2: 78, 3: 146, 4: 164, 5: 222, 6: 230, 7: 268, 8: 306, 9: 364, 10: 412, 11: 420, 12: 458, 13: 496, 14: 544, 15: 572, 16: 610, 17: 648, 18: 686, 19: 724, 20: 762}
Clustering Weights	[-3.0, -3.0, -0.1, 5.5, 6.0]

8.2.2.3 Ping-Pong Handover Mitigation

A critical enhancement of the proposed framework lies in the substantial reduction of ping-pong handovers, defined as unnecessary, back-and-forth handovers between neighboring cells within a short duration. The handover comparison in Figure 46 shows the Traditional Heuristic-based model (green-dotted lines) with 92 total handovers, where ping-pong locations—marked by small squares—still exhibit clusters of tightly spaced handover events. In contrast, the Proposed Model (red-dotted lines) with 80 total handovers demonstrates a noticeable decrease in such clustered, bidirectional events.

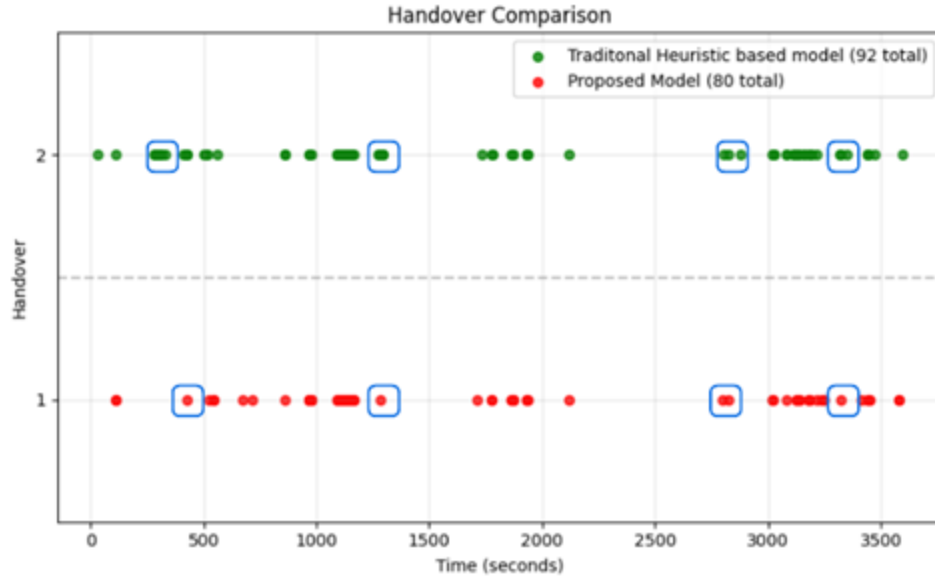


Figure 46: Ping-Pong Event Mitigation

Using equation (1), the proposed model achieves a 31% reduction, resulting in a lower ratio (15.00%) than the baseline (21.73%), as summarized in Table 28.

$$\text{Ping - pong Ratio} = \frac{\text{Ping-pong Handover}}{\text{Total number of Handover}} \times 100 \% \quad (25)$$

To achieve this improvement, the reward function in our Q-learning model incorporated a penalty for consecutive handovers occurring within the defined Time-to-Trigger (TTT) window. This temporal filtering mechanism played a pivotal role in promoting more deliberate handover decisions, thereby improving connection stability and reducing signaling overhead.

Table 27. Ping-pong Handover Summary

Model Name	Number of Total HO	Ping-Pong HO	Ping-Pong Ratio
Traditional Heuristic Model	92	20	21.73%
Proposed Model	80	12	15%

8.2.2.4 Latency Comparison with Heuristics

Conventional heuristic-based handover algorithms used by telecom operators typically require 5 to 20 milliseconds to evaluate signal conditions and determine whether a handover should be triggered [18]. In contrast, our Q-learning-based model benefits from pre-trained knowledge and pre-abstracted state definitions. When deployed, the decision process involves a simple table lookup using the current state's parameters. This enables ultra-fast inference, with a decision latency of approximately 0.016 milliseconds, representing a significant reduction in handover decision time. This rapid responsiveness is especially valuable in high-mobility scenarios where timely HO decisions are critical to maintaining service continuity.

8.2.3 Fuzzy Logic–Based Adaptive Handover Control & Deep Q-Network (DQN)–Based Reinforcement Learning

The evaluation and discussion jointly examine the effectiveness of the proposed handover optimization strategies under real-world LTE conditions. Using datasets collected from multiple drive-test sessions, the operator-defined baseline, fuzzy logic model, and adaptive DQN were compared in terms of total handover count, missed handovers, and unnecessary handovers. The results are first presented for each dataset to illustrate dataset-specific behavior, followed by a cross-day comparative analysis. The discussion then interprets these findings, explaining why the models perform differently and highlighting the advantages and limitations of each approach in dynamic mobility scenarios.

8.2.3.1 Experimental Setup

The performance of the proposed models was evaluated using real-world LTE mobility datasets collected through drive testing in Dhaka, Bangladesh. Three datasets were analyzed: Day 1 (923 samples), Day 2 (593 samples), and Day 3 (641 samples). Each dataset contains timestamped measurements of RSRP, RSRQ, CINR, and handover events.

Handover performance was assessed using three key metrics:

- **Total Handover (HO) count:** number of handover events triggered.
- **Missed HO:** instances where handover was necessary but not executed.
- **Unnecessary HO:** handovers triggered despite adequate serving cell conditions, often causing ping-pong effects.

The detailed results are summarized in Table 29, 30 & 31.

Table 28: Day1 Dataset Evaluation

Key Metrics	Operator Defined model	Fuzzy Logic model	DQN (Adaptive TTT & HOM)
Total HO count	150	137	126
Missed HO	12	5	9

Unnecessary HO	26	12	3
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Table 29: Day1 Dataset Evaluation

Key Metrics	Operator Defined model	Fuzzy Logic model	DQN (Adaptive TTT & HOM)
Total HO count	99	95	84
Missed HO	5	2	5
Unnecessary HO	17	7	1

Table 30: Day3 Dataset Evaluation

Key Metrics	Operator Defined model	Fuzzy Logic model	DQN (Adaptive TTT & HOM)
Total HO count	118	110	98
Missed HO	7	3	7
Unnecessary HO	21	10	2

8.2.3.2 Results on Day1 Dataset

On Day 1, the fuzzy logic model reduced unnecessary HOs by approximately 54% and missed HOs by nearly 58% compared to the operator-defined baseline, while slightly lowering the total HO count. The adaptive DQN further optimized handover performance, achieving a 16% reduction in total HOs and an 88% reduction in unnecessary HOs relative to the baseline. Missed HOs under adaptive DQN remained low, demonstrating a superior balance between network reliability and efficiency.

8.2.3.3 Results on Day2 Dataset

For Day 2, the fuzzy logic model achieved a reduction of 59% in unnecessary HOs and 60% in missed HOs, with minimal impact on the total HO count. The adaptive DQN outperformed fuzzy logic by reducing total HOs by 15% and nearly eliminating unnecessary handovers, while maintaining missed HOs comparable to the baseline. This indicates robust generalization of the adaptive learning strategy under varying mobility patterns.

8.2.3.4 Results on Day3 Dataset

On Day 3, the fuzzy logic model continued to provide balanced improvements, lowering unnecessary HOs by over 50% and missed HOs by approximately 57%. The adaptive DQN again demonstrated the most favorable trade-off: total HOs decreased by 17%, unnecessary HOs were minimized by over 90%, and missed HOs were maintained at the baseline level.

These results underscore the ability of adaptive reinforcement learning to maintain network stability while optimizing handover decisions.

8.2.3.4 Cross-Day Comparative Analysis

The results across the three datasets demonstrate consistent improvements from fuzzy logic and adaptive DQN over operator-defined handover parameters. To evaluate these improvements comprehensively, the models were compared across total handover counts, missed handovers, and unnecessary handovers, as summarized in Figures 47,48 & 49.

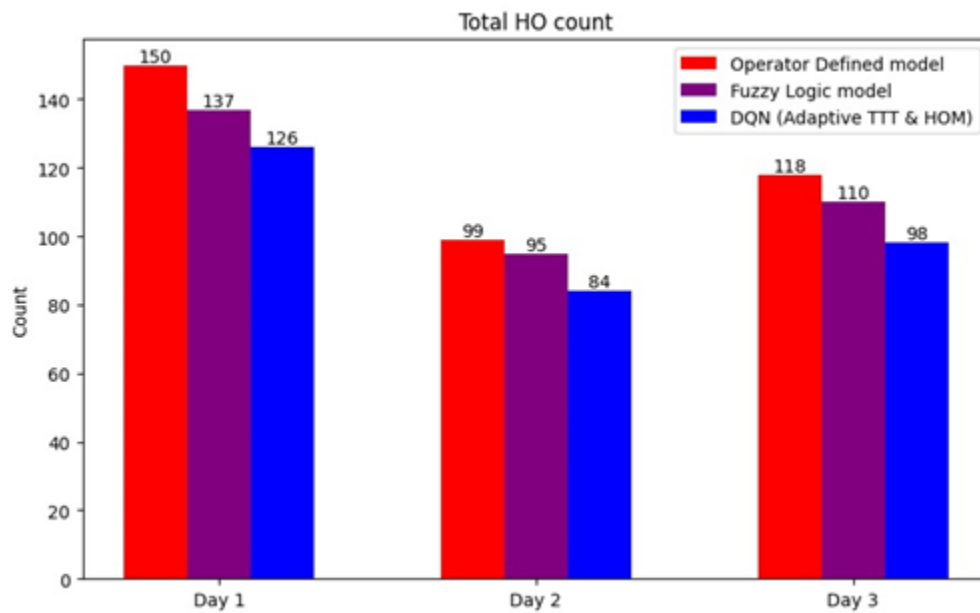


Figure 47. Total Handover Counts Across Models

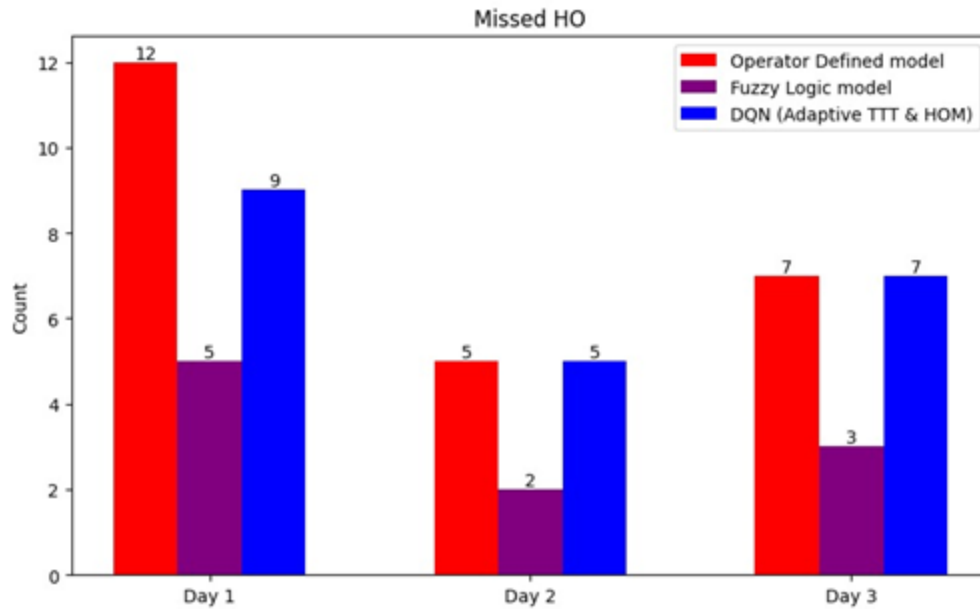


Figure 48. Missed Handover Counts Across Models

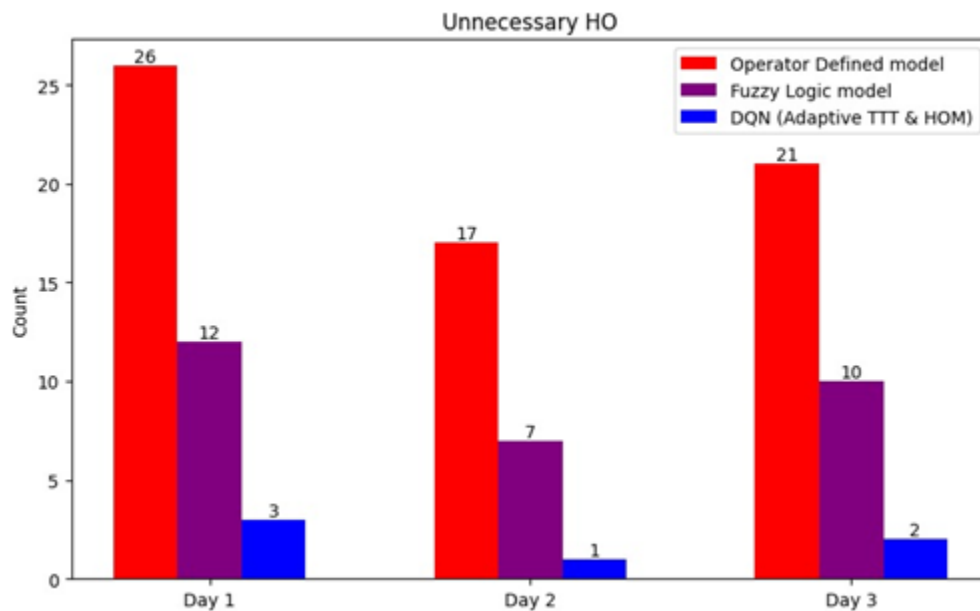


Figure 49. Unnecessary Handover Counts Across Models

Across all three datasets, several trends are evident. The operator-defined model exhibited consistently high unnecessary handover (HO) rates, highlighting the limitations of static handover thresholds in dynamic urban environments. In contrast, fuzzy logic control provided substantial reductions in both unnecessary (52–59%) and missed HOs (57–60%), while keeping total HOs close to the baseline, reflecting moderate adaptability to varying conditions. The

adaptive Deep Q-Network (DQN) consistently achieved the most favorable outcomes, with total HOs decreasing by 15–17%, unnecessary HOs reduced by 88–95% and missed HOs either maintained at or slightly below baseline levels.

Chapter Nine

Demonstration of Outcome Based Education (OBE)

9.1 Course Outcomes (COs) Addressed

The following table shows the COs addressed in EEE 4700 for Project and Thesis.

COs	CO Statement	POs	Put Tick (√)
			EEE 4700
CO1	Identify a contemporary real life problem related to electrical and electronic engineering by reviewing and analyzing existing research works.	PO2	√
CO2	Determine functional requirements of the problem considering feasibility and efficiency through analysis and synthesis of information.	PO4	√
CO3	Select a suitable solution and determine its method considering professional ethics, codes and standards.	PO8	√
CO4	Adopt modern engineering resources and tools for the solution of the problem.	PO5	√
CO5	Prepare management plan and budgetary implications for the solution of the problem.	PO11	√
CO6	Analyze the impact of the proposed solution on health, safety, culture and society.	PO6	√
CO7	Analyze the impact of the proposed solution on environment and sustainability.	PO7	√
CO8	Develop a viable solution considering health, safety, cultural, societal and environmental aspects.	PO3	
CO9	Work effectively as an individual and as a team member for the accomplishment of the solution.	PO9	√
CO10	Prepare various technical reports, design documentation, and deliver effective presentations for demonstration of the solution.	PO10	√
CO11	Recognize the need for continuing education and participation in professional societies and meetings.	PO12	

9.2 Aspects of Program Outcomes (POs) Addressed

The following table shows the aspects addressed for certain Program Outcomes (POs) addressed in EEE 4700 for Project and Thesis.

	Statement	Different Aspects	Put Tick (√)
PO3	Design/development of solutions: Design solutions for complex electrical and electronic engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for	Public health	
		Safety	

	public health and safety, cultural, societal, and environmental considerations.	Cultural	
		Societal	√
		Environmental	
PO4	Investigation: Conduct investigations of complex electrical and electronic engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Design of experiments	√
		Analysis and interpretation of data	√
		Synthesis of information	√
PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex electrical and electronic engineering problems.	Societal	√
		Health	
		Safety	
		Legal	√
		Cultural	
PO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex electrical and electronic engineering problems in societal and environmental contexts.	Societal	√
		Environmental	
PO8	Ethics: Apply ethical principles embedded with religious values, professional ethics and responsibilities, and norms of electrical and electronic engineering practice.	Religious values	
		Professional ethics and responsibilities	√
		Norms	√
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Comprehend and write effective reports	√
		Design documentation	√
		Make effective presentations	√
		Give and receive clear instructions	√
PO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	Engineering management principles	√
		Economic decision-making	√
		Manage projects	√
		Multidisciplinary environments	

9.3 Knowledge Profiles (K3 – K8) Addressed

The following table shows the Knowledge Profiles (K3 – K8) addressed in EEE 4700 for Project and Thesis.

K	Knowledge Profile (Attribute)	Put Tick (√)
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K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline	√
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline	√
K5	Knowledge that supports engineering design in a practice area	√
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline	√
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability	
K8	Engagement with selected knowledge in the research literature of the discipline	√

The following table explains or justifies how the Knowledge Profiles (K3 – K8) have been addressed in EEE 4700/4800 (Project and Thesis).

K	Explanation/Justification
K3	The systematic, theory-based formulation of engineering fundamentals essential for the engineering discipline directly supports our thesis on handover optimization by providing a solid foundation in telecommunications principles, signal processing, and network design. Understanding these core theories enables the development of advanced algorithms and models that accurately predict and optimize handover processes, ensuring improved network performance and enhanced user experience. By applying these engineering fundamentals, the project aims to address real-world challenges in mobile communications effectively, contributing to the overall advancement of the telecommunications field.
K4	The thesis on handover optimization leverages specialized knowledge in telecommunications engineering, particularly in areas like radio frequency (RF) management, mobility protocols, and machine learning. These theoretical frameworks form the foundation for developing innovative models that enhance handover decision-making processes. By using cutting-edge techniques, the research remains at the forefront of the discipline, advancing best practices in optimizing network performance and reliability in mobile communication systems
K5	In the context of handover optimization, the knowledge supporting engineering design includes understanding signal strength metrics (RSRP, RSSI, RSRQ), mobility management, and machine learning techniques. This foundational knowledge guides the design of predictive models that enhance handover decisions in real-time, ensuring efficient network performance. By applying this knowledge, the design of the optimization process reduces unnecessary handovers and improves user experience, aligning with best practices in mobile network engineering.
K6	In handover optimization, engineering practice knowledge involves the application of technologies such as XCAL for data collection during drive tests and machine learning for predictive analysis. This knowledge includes practical skills in configuring and managing network parameters (e.g., signal strength, TTT, and cell handover events), and leveraging tools for real-time network monitoring and optimization. It ensures the effective implementation of handover solutions in live networks, aligning with industry standards and practices in telecommunications engineering.
K7	

K8	In the context of handover optimization using machine learning, engagement with the research literature involves critically analyzing and applying relevant studies on signal strength metrics, mobility management, and predictive models for handover decisions. By exploring existing frameworks and methodologies, such as machine learning-based optimization techniques, the research leverages state-of-the-art findings to improve network efficiency and user experience. This engagement also includes evaluating advancements in mobility management, network performance, and the use of data-driven approaches to enhance handover timing and reduce unnecessary transitions in mobile networks.
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Use of Complex Engineering Problems

Handover optimization in mobile communication networks presents a complex engineering challenge due to the dynamic nature of network conditions, user mobility, and varying signal quality. To address this complexity, a systematic approach involving data-driven methodologies is employed. By collecting real-time data during drive tests using XCAL, we gather comprehensive information on signal strength metrics, user equipment speed, and historical handover events. Analyzing this data through advanced analytics allows us to identify patterns and correlations, which serve as the foundation for predictive modeling. This enables us to anticipate the optimal timing and conditions for handovers, minimizing unnecessary transitions that could disrupt user experience.

Moreover, incorporating machine learning techniques provides a robust framework for decision-making in handover management. Through the training of various machine learning models, we can evaluate their performance in predicting when handovers are necessary. By comparing models based on their accuracy, precision, and recall, we identify the best fit for our specific scenario. This iterative process of model selection and refinement allows us to enhance the efficiency of handover decisions, ultimately resulting in a reduction in failed handovers and improved overall network performance.

Lastly, the importance of stakeholder engagement cannot be overlooked in the optimization process. Collaboration with network operators and engineers ensures that the developed solutions align with practical deployment scenarios and operational constraints. Continuous feedback loops from real-world implementations facilitate the iterative improvement of our models and strategies, allowing us to adapt to changing network conditions and user behaviors. This integrated approach not only addresses the technical challenges of handover optimization but also ensures that solutions are viable and accepted within the operational framework of mobile networks.

Socio-Cultural, Environmental, And Ethical Impact

Socio-Cultural Impact

The handover optimization project has the potential to significantly enhance the user experience in mobile communications, which is integral to modern social interactions and cultural engagement. By reducing call drop rates and improving overall connectivity, the project supports seamless communication for individuals and communities, fostering stronger social ties and greater access to information. Enhanced mobile services can empower users in remote areas, bridging the digital divide and facilitating social inclusion. Moreover, improved connectivity can bolster local economies by supporting businesses that rely on mobile communication, enabling them to reach a broader audience and engage in e-commerce.

Environmental Impact

From an environmental perspective, optimizing handover processes can lead to reduced energy consumption in mobile networks. Fewer unnecessary handovers mean that user equipment (UE) spends less time engaging in signaling processes, leading to lower power usage. This not only benefits individual users through longer battery life but also contributes to the overall energy efficiency of network infrastructure. By minimizing the signaling load on base stations, the project can potentially reduce carbon emissions associated with energy consumption in mobile networks. Ultimately, a more efficient network can support sustainability initiatives by promoting eco-friendly practices in the telecommunications sector.

Ethical Impact

The ethical considerations of the handover optimization project revolve around data privacy and algorithmic fairness. As the project involves the collection of real-time user data through drive tests, it is essential to ensure that data handling practices adhere to privacy regulations and ethical standards. Transparency in data usage and obtaining informed consent from users will be prioritized to maintain trust. Additionally, machine learning models used for predicting handovers must be designed to avoid bias, ensuring that all users receive equitable service regardless of their location or usage patterns. Addressing these ethical concerns will not only enhance the project's credibility but also foster responsible innovation in the field of telecommunications.

9.4 Attributes of Ranges of Complex Engineering Problem Solving (P1 – P7) Addressed

The following table shows the attributes of ranges of Complex Engineering Problem Solving (P1 – P7) addressed in EEE 4700 for Project and Thesis.

P	Range of Complex Engineering Problem Solving	Put Tick (✓)
Attribute	Complex Engineering Problems have characteristic P1 and some or all of P2 to P7:	
Depth of knowledge required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach	
Range of conflicting requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues	✓
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	✓
Familiarity of issues	P4: Involve infrequently encountered issues	✓
Extent of applicable codes	P5: Are outside problems encompassed by standards and codes of practice for professional engineering	
Extent of stakeholder involvement and conflicting requirements	P6: Involve diverse groups of stakeholders with widely varying needs	
Interdependence	P7: Are high level problems including many component parts or sub-problems	✓

The following table explains or justifies how the attributes of ranges of Complex Engineering Problem Solving (P1 – P7) have been addressed in EEE 4700/4800 (Project and Thesis).

P	Explanation/Justification
P1	Handover optimization in mobile networks requires in-depth engineering knowledge, particularly in wireless communication principles (K3), signal processing (K4), and predictive modeling (K6). The challenge involves applying first principles and fundamentals, such as radio signal propagation and machine learning algorithms, to resolve complex, data-driven decisions for optimal handover timing.
P2	Conflicting requirements arise in balancing user experience, network performance, and energy efficiency. For example, reducing handovers can conserve resources but might degrade service quality if poorly timed. The optimization needs to address both technical performance and user satisfaction, making it a multifaceted issue.
P3	The problem lacks an obvious solution due to the dynamic nature of mobile networks, which requires abstract thinking and modeling. Creating predictive models for handover decisions demands originality, as real-time factors like user mobility, signal fluctuation, and network congestion need to be accounted for.
P4	Handover optimization encounters infrequent issues, especially with the emergence of new technologies like LTE, 5G. Problems such as ping-pong handovers or rare network outages require tailored solutions, often necessitating innovative analysis beyond standard procedures.
P5	
P6	
P7	Handover optimization is a high-level engineering challenge that consists of multiple sub-problems, including signal strength assessment, mobility tracking, and real-time decision-making. Each sub-problem is interdependent, requiring holistic consideration of various factors to ensure network efficiency and user satisfaction.

9.5 Attributes of Ranges of Complex Engineering Activities (A1 – A5) Addressed

The following table shows the attributes of ranges of Complex Engineering Activities (A1 – A5) addressed in EEE 4700 for Project and Thesis.

A	Range of Complex Engineering Activities	Put Tick (√)
Attribute	Complex activities means (engineering) activities or projects that have some or all of the following characteristics:	
Range of resources	A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)	√
Level of interaction	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues	√
Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways	√
Consequences for society and the environment	A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation	

Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches	√
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The following table explains or justifies how the attributes of ranges of Complex Engineering Activities (A1 – A5) have been addressed in EEE 4700/4800 (Project and Thesis).

A	Explanation/Justification
A1	The thesis project on handover optimization utilizes diverse resources such as data collection tools (XCAL), machine learning technologies, financial budgeting for drive tests, and collaboration with network engineers. It also involves managing large-scale datasets and computational resources for model training and evaluation.
A2	The project addresses significant problems, such as conflicting requirements between minimizing handovers and maintaining service quality. These technical challenges require balancing network performance, user experience, and operational efficiency, making it a multi-dimensional task with intricate interactions.
A3	The project applies machine learning in novel ways to optimize handover processes, which is a creative use of engineering principles. By integrating real-time network data and predictive algorithms, the research develops innovative solutions beyond conventional handover strategies used in mobile networks.
A4	
A5	This project extends beyond previous experiences by applying principles of signal processing and machine learning to the evolving challenges of mobile networks. The methods used build upon established concepts but adapt them to new, real-world challenges in 4G and beyond.

Chapter Ten

Conclusion

This thesis introduced an intelligent, data-driven framework for mobility management in LTE networks, combining predictive modeling and optimization techniques grounded in real-world measurements. Starting from extensive drive test data collection and stratification, the study ensured the reliability and contextual relevance of input data. The contrastive autoencoder-based handover prediction model demonstrated the potential of deep representation learning to anticipate mobility events accurately using QoS indicators. This predictive capability forms the foundation for proactive network decision-making.

On the optimization front, four complementary strategies—Contextual Bandit with Q-Learning, Subtractive Clustering with Q-Learning, Fuzzy Logic, and Deep Q-Network (DQN) Learning—were investigated to fine-tune handover control parameters. Each method offered unique advantages in balancing learning efficiency, adaptability, and interpretability. The hybridization of clustering, reinforcement learning, and fuzzy inference not only reduced handover failures and ping-pong effects but also enhanced network stability and throughput.

Collectively, the proposed framework demonstrates that incorporating intelligent learning mechanisms into traditional mobility management can significantly improve network reliability and user experience. The methodologies developed in this work can be extended to 5G and 6G architectures, where heterogeneous networks, ultra-dense deployments, and dynamic user behaviors demand context-aware, autonomous decision-making. Future research may explore federated learning, graph-based mobility models, or multi-agent reinforcement learning to further elevate the intelligence and scalability of handover management systems.

References

- [1] S. A. Garba, F. U. Ambursa, Y. S. Baguda, M. A. Shehu, "Exploring wireless connectivity and network performance: A dataset of 4G LTE user equipment measurements," *Sci. World J.*, vol. 18, pp. 558—566, 2024. doi: 10.4314/swj.v18i4.4.
- [2] K. H. Bilal, A- Mustafa, A. M. Jaheldeen, A. A. Babiker, N. Mustafa, "Handover Drive Test," *Int. J. Eng, Appl Sci. Paradigms*, vol. 22, 2015.
- [3] S. D. Yusuf, S. I. Isa, B. J. Kwaha, "Analysis of 4G/LTE Network Performance in North-Central Nigeria: A Comprehensive Drive Test Approach," *J. Eng. Res. Rep.*, vol. 26, pp. 105-122, 2024. doi: 10.9734/ierr/2024/v26i91267.
- [4] S. Farthofer, M. Herlich, C. Maier, S. Pochaba, J. Lackner, P. Dorfinger, "An Open Mobile Communications Drive Test Data Set and its use for Machine Learning," *IEEE Open J. Commun- Soc-*, 2022. doi: 10.1109/OJCOMS.2022.1234567.
- [5] E. Afroz, R. Subramanian, R- Heidary, K. Sandrasegaran, S. Ahmed, "SINR, RSRP, RSSI and RSRQ Measurements in Long Term Evolution Networks," *Int.J. Wire'. Mob. Netw-*, vol. 7, pp. 113-123, 2015. doi: 10.5121/iwmn.2015.7409.
- [6] E. M. Change, H. L. Wang, S. Y. Hur, S. J. Kao, "An efficient handover mechanism by adopting direction prediction and adaptive time-to-trigger in LTE networks," in *Comput. Sci. Appl — ICCSA 2013, Lect. Notes Comput. sci.*, vol. 7975, Springer, Berlin, Heidelberg, 2013, pp. 226-240.
- [7] M. M. S. Shafi, T. S. Ahona and Y. H. Saad, "A Weather-Aware Risk Stratification Framework for RLF Prediction Using Contrastive Autoencoders and Survival Analysis," in *The 10th IEEE Asia Pacific Conference on Wireless and Mobile (APWiMob 2025)*, 2025. [Online]. Available: <https://www.researchgate.net/publication/395962722>
- [8] E. Eldesouky, M. Bekhit, A. Fathalla, A. Salah and A. Ali, "A Robust UWSN Handover Prediction System Using Ensemble Learning," *Sensors*, vol. 21, no. 17, Sep. 2021. doi: 10.3390/s21175777.
- [9] K. M. Jaheldeen, K. H. Bilal and A. A. Mustafa, "Handover Drive Test," *Int. J. Eng, Appl. Manag. Sci. Paradigms*, vol. 22, Mar. 2015. [Online]. Available: [researchgate.net/publication/317032585](https://www.researchgate.net/publication/317032585)
- [10] M. K. Dahouda, S. Jin and I. Joe, "Machine Learning-Based Solutions for Handover Decisions in Non-Terrestrial Networks," *Electronics (Switzerland)*, vol. 12, no. 8, Apr. 2023. doi: 10.3390/electronics12081759.

- [11] F. M. Chang, H. L. Wang, S. Y. Hu and S. J. Kao, "An efficient handover mechanism by adopting direction prediction and adaptive time-to-trigger in LTE networks," in *Comput. Sci. Appl — ICCSA 2013*, vol. 7975, pp. 270–280, 2013. doi: 10.1007/978-3-642-39640-3_20.
- [12] C. L. Vielhaus et al., "Handover predictions as an enabler for anticipatory service adaptations in next-generation cellular networks," in *Proc. 20th ACM Int. Symp. on Mobility Management and Wireless Access*, 2022, pp. 19–27. doi: 10.1145/3551660.3560913.
- [13] M. Dzaferagic et al., "ML-Based Handover Prediction Over a Real O-RAN Deployment Using RAN Intelligent Controller," *IEEE Trans Netw. Service Manag.*, vol. 22, no. 1, pp. 635–647, 2025. doi: 10.1109/TNSM.2024.3468910.
- [14] K. M. Istiaque, S. Sowad, M. M. S. Shafi and M. T. Kawser, "Reducing Handover and Ping-Pong Events in LTE Networks Via Q-Learning and Subtractive Clustering," in *Proc. 2025 8th Int. Seminar on Research of Information Technology and Intelligent Systems (ISRITI)*, Oct. 2025. [Online]. Available: researchgate.net/publication/396135987.
- [15] K. M. Istiaque, M. M. S. Shafi, S. S. Sowad and M. T. Kawser, "Adaptive Handover Optimization in LTE: Comparative Evaluation of Fuzzy Logic and DQN," in *Proc. 10th IEEE Asia Pacific Conf. on Wireless and Mobile 2025 (APWiMob 2025)*, Oct. 2025. [Online]. Available: researchgate.net/publication/396134205.
- [16] M. M. S. Shafi, K. M. Istiaque, S. S. Sowad and M. T. Kawser, "Handover Optimization in LTE Networks Using Contextual Bandit Reinforcement Learning and Real-World Data," in *Proc. Int. Conf. on Computer Engineering, Network and Intelligent Multimedia (CENIM)*, Oct. 2025. [Online]. Available: researchgate.net/publication/396132842.
- [17] X. Dong, K. Zhang, C. Sun, J. Zhang, A. Zhang, and L. Wang, "Towards 250-m Gigabits-Per-Second Underwater Wireless Optical Communication Using a Low-Complexity ANN Equalizer," *Opt Express*, vol. 33, no. 2, p. 2321, Jan. 2025. doi: 10.1364/oe.549337.
- [18] A. Narwaria and A. P. Mazumdar, "SD-MoMa: ANN-Based Mobility Prediction and Adaptive Handover Management in SDWSN," *Cluster Comput*, vol. 28, no. 6, Oct. 2025. doi: 10.1007/s10586-024-05006-0.
- [19] A. I. Bello, A. Y. Musa and H. S. Abdullahi, "Comparative Analysis of ANN and ANFIS in Forecasting GSM Handover Success Rate," in *Proc. National Conf. on Eng. &*

Tech. for Econ. Transform., vol. 1, Sep. 2014. [Online]. Available: [researchgate.net/publication/283503306](https://www.researchgate.net/publication/283503306).

[20] M. U. A. Khan et al., "Optimizing Wireless Connectivity: A Deep Neural Network-Based Handover Approach for Hybrid LiFi and WiFi Networks," *Sensors*, vol. 24, no. 7, Apr. 2024. doi: 10.3390/s24072021.

[21] K. Sree, P. Chakrabarti, M. Margala, and S. N. Usha Devi, "Auto encoders with Cellular Automata for Anomaly Detection," 2024.

[22] P. Wang, J. Zhang, Y. Li, Y. Guo, and P. Li, "Breast Histopathological Image Classification Based on Auto-Encoder Reconstructed Domain Adaptation," *Applied Sciences (Switzerland)*, vol. 14, no. 24, Dec. 2024. doi: 10.3390/app142411802.

[23] S. J. Badashah et al., "Enhanced deep auto encoder technique for brain tumor classification and detection," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 38, no. 3, p. 2031, Jun. 2025. doi: 10.11591/ijeecs.v38.i3.pp2031-2040.

[24] P. V. Dinh, D. N. Nguyen, D. T. Hoang, Q. U. Nguyen, and E. Dutkiewicz, "Multiple-Input Variational Auto-Encoder for Anomaly Detection in Heterogeneous Data," Jan. 2025. [Online]. Available: <http://arxiv.org/abs/2501.08149>

[25] H. Powers and K. Edoh, "Outlier detection variational autoencoder," *Neural Comput Appl*, vol. 37, no. 21, pp. 16871–16882, Jul. 2025. doi: 10.1007/s00521-025-11357-5.

[26] Z. Lu, Z. Chu, M. Zhu, and X. Dong, "Unsupervised feature learning using locality-preserved auto-encoder with complexity-invariant distance for intelligent fault diagnosis of machinery," *Applied Intelligence*, vol. 55, no. 6, Apr. 2025. doi: 10.1007/s10489-025-06278-8.

[27] A. A. Tafere, T. S. Hailemariam, and T. T. Debella, "Deep Learning-Powered Equalization with Autoencoders for Improved 5G Communication," in *2025 IEEE Int. Conf. on Interdisciplinary Approaches in Technology and Management for Social Innovation, IATMSI 2025*, 2025. doi: 10.1109/IATMSI64286.2025.10984954.

[28] X. Tian, "A Deep Convolutional Autoencoder-Enabled Channel Estimation Method in Intelligent Wireless Communication Systems," *Int. J. Intell. Syst.*, vol. 2024, no. 1, 2024. doi: 10.1155/2024/9343734.

[29] P. Samarathunga et al., "An Autoencoder-Based Task-Oriented Semantic Communication System for M2M Communication," *Algorithms*, vol. 17, no. 11, Nov. 2024. doi: 10.3390/a17110492.

- [30] N. A. Letizia and A. M. Tonello, "Capacity-Driven Autoencoders for Communications," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 1366–1378, 2021. doi: 10.1109/OJCOMS.2021.3087815.
- [31] B. Zhang, G. Zheng, and N. V. Huynh, "A Hybrid Quantum-Classical Autoencoder Framework for End-to-End Communication Systems," Dec. 2024. [Online]. Available: <http://arxiv.org/abs/2412.20241>
- [32] Z. I. Tabi et al., "Quantum-Classical Autoencoder Architectures for End-to-End Radio Communication." doi: 10.1109/ACCESS.2021.DOI.
- [33] X. Sheng, Z. Shen, and G. Xiao, "Contrastive Predictive Autoencoders for Dynamic Point Cloud Self-Supervised Learning," 2023. [Online]. Available: www.aaai.org
- [34] B. Xue et al., "A contrastive autoencoder with multi-resolution segment-consistency discrimination for multivariate time series anomaly detection," *Applied Intelligence*, vol. 53, no. 23, pp. 28655–28674, Dec. 2023. doi: 10.1007/s10489-023-04985-8.
- [35] D. Luo, H. Zhou, J. Bae, and B. Yun, "Combining Contrastive Learning with Auto-Encoder for Out-of-Distribution Detection," *Applied Sciences (Switzerland)*, vol. 13, no. 23, Dec. 2023. doi: 10.3390/app132312930.
- [36] C. Wu et al., "Attack detection model for BCoT based on contrastive variational autoencoder and metric learning," *J. Cloud Comput.*, vol. 13, no. 1, Dec. 2024. doi: 10.1186/s13677-024-00678-w.
- [37] E. Nikougofar, "Outlier Detection in IoT using Trained Autoencoder and Contrastive Loss," Aug. 19, 2024. doi: 10.21203/rs.3.rs-4657376/v1.
- [38] X. Zheng, Z. Wu, H. Meng, and L. Cai, "Contrastive auto-encoder for phoneme recognition," in *ICASSP, IEEE Int. Conf. on Acoust. Speech Signal Process. - Proc.*, 2014, pp. 2529–2533. doi: 10.1109/ICASSP.2014.6854056.
- [39] N. Liu et al., "Unsupervised Contrastive Hashing with Autoencoder Semantic Similarity for Cross-Modal Retrieval in Remote Sensing," *IEEE J Sel Top Appl Earth Obs Remote Sens*, vol. 18, pp. 6047–6059, 2025. doi: 10.1109/JSTARS.2025.3538701.
- [40] H. M. Pham, A. Saeed, and D. Ma, "C-MELT: Contrastive Enhanced Masked Auto-Encoders For ECG-Language Pre-Training," May 2025. [Online]. Available: <http://arxiv.org/abs/2410.02131>
- [41] Z. Shakir, A. Y. Mjhoor, A. Al-Thaedan, A. Al-Sabbagh, and R. Alsabah, "Key performance indicators analysis for 4 G-LTE cellular networks based on real

measurements,” *Int. J. Inf. Technol. (Singapore)*, vol. 15, no. 3, pp. 1347–1355, Mar. 2023. doi: 10.1007/s41870-023-01210-0.

[42] S. M. Kala et al., “Architecture, Performance, and Usability of Mobile Cellular Network Monitoring Applications for Data Driven Analysis,” *IEEE Access*, vol. 12, pp. 88426–88444, 2024. doi: 10.1109/ACCESS.2024.3412752.

[43] H. Zhu and Y. Peng, “Research on adaptive handover scheme based on improved genetic algorithm,” in *Procedia Computer Science*, 2020, pp. 557–562. doi: 10.1016/j.procs.2020.02.022.

[44] W. K. Saad, I. Shayea, A. Alhammadi, M. M. Sheikh, and A. A. El-Saleh, “Handover and load balancing self-optimization models in 5G mobile networks,” *Eng. Sci. Technol., an Int. J.*, vol. 42, Jun. 2023. doi: 10.1016/j.jestch.2023.101418.

[45] M. B. Patil and L. Math, “A novel approach for optimization of handover mechanism using metaheuristics algorithms,” *Measurement: Sensors*, vol. 24, Dec. 2022. doi: 10.1016/j.measen.2022.100467.

[46] W. S. Hwang, T. Y. Cheng, Y. J. Wu, and M. H. Cheng, “Adaptive Handover Decision Using Fuzzy Logic for 5G Ultra-Dense Networks,” *Electronics (Switzerland)*, vol. 11, no. 20, Oct. 2022. doi: 10.3390/electronics11203278.

[47] N. Zohar, “Beyond 5G: Reducing the Handover Rate for High Mobility Communications,” *J. Commun. Netw.*, vol. 24, no. 2, pp. 154–165, Apr. 2022. doi: 10.23919/JCN.2022.000001.

[48] M. Alnabhan et al., “Efficient handover approach in 5G mobile networks,” *Int J Adv Sci Eng Inf Technol*, vol. 10, no. 4, pp. 1417–1422, 2020. doi: 10.18517/ijaseit.10.4.11988.

[49] I. Zubeiri, Y. El Morabit, and F. Mrabti, “Genetic algorithm for vertical handover (GAfVH) in a heterogeneous networks,” *Int. J. Electr. Comput. Eng.*, vol. 9, no. 4, pp. 2534–2540, Aug. 2019. doi: 10.11591/ijece.v9i4.pp2534-2540.

[50] I. Shayea et al., “Individualistic Dynamic Handover Parameter Self-Optimization Algorithm for 5G Networks Based on Automatic Weight Function,” *IEEE Access*, vol. 8, pp. 214392–214412, 2020. doi: 10.1109/ACCESS.2020.3037048.

[51] S. Alraih et al., “Robust Handover Optimization Technique with Fuzzy Logic Controller for Beyond 5G Mobile Networks †,” *Sensors*, vol. 22, no. 16, Aug. 2022. doi: 10.3390/s22166199.

- [52] A. I. Mbulwa, H. T. Yew, A. Chekima, and J. A. Dargham, "Self-Optimization of Handover Control Parameters for 5G Wireless Networks and Beyond," *IEEE Access*, vol. 12, pp. 6117–6135, 2024. doi: 10.1109/ACCESS.2023.3346039.
- [53] W. K. Saad, I. Shayea, B. J. Hamza, H. Mohamad, Y. I. Daradkeh, and W. A. Jabbar, "Handover parameters optimisation techniques in 5g networks," *Sensors*, vol. 21, no. 15, Aug. 2021. doi: 10.3390/s21155202.
- [54] K. Jha et al., "CSVAG: Optimizing Vertical Handoff Using Hybrid Cuckoo Search and Genetic Algorithm-Based Approaches," *Sustainability (Switzerland)*, vol. 14, no. 14, Jul. 2022. doi: 10.3390/su14148547.
- [55] B. H. Prananto, Iskandar, and A. Kurniawan, "A New Method to Improve Frequent-Handover Problem in High-Mobility Communications Using RIC and Machine Learning," *IEEE Access*, vol. 11, pp. 72281–72294, 2023. doi: 10.1109/ACCESS.2023.3294990.
- [56] S. Hatipoğlu et al., "Machine Learning-Based Handover Performance Prediction for Beyond 5G Communication Networks," 2024. [Online]. Available: <https://www.researchgate.net/publication/382248189>
- [57] V. O. Nyangaresi, A. J. Rodrigues, and S. O. Abeka, "ANN-FL Secure Handover Protocol for 5G and Beyond Networks," in *Lect. Notes Inst. Comput. Sci. Soc. Informatics Telecommun. Eng., LNICST*, 2021, pp. 99–118. doi: 10.1007/978-3-030-70572-5_7.
- [58] J. Wang et al., "Deep Reinforcement Learning-based Satellite Handover Scheme for Satellite Communications," in *13th Int. Conf. on Wireless Commun. Signal Process., WCSP 2021*, 2021. doi: 10.1109/WCSP52459.2021.9613411.
- [59] S. He, T. Wang, and S. Wang, "Load-Aware Satellite Handover Strategy Based on Multi-Agent Reinforcement Learning," in *Proc. - IEEE Global Commun. Conf., GLOBECOM*, 2020.
- [60] A. Abdelmohsen et al., "LTE handover parameters optimization using Q-learning technique," in *Midwest Symp. on Circuits and Systems*, Jul. 2018, pp. 194–197. doi: 10.1109/MWSCAS.2018.8623826.
- [61] Q. Liu et al., "Reinforcement learning-based joint self-optimisation method for the fuzzy logic handover algorithm in 5G HetNets," *Neural Comput Appl*, vol. 35, no. 10, pp. 7297–7313, Apr. 2023.

- [62] M. M. S. Shafi, K. M. Istiaque, S. S. Sowad, and M. T. Kawser, "Drive-Test-Based LTE Handover Dataset for Cellular Mobility Studies in Urban Bangladesh," *Mendeley Data*, vol. V1, 2025. doi: 10.17632/n2pvmtn2j.1.
- [63] Y. Ullah et al., "A Survey on Handover and Mobility Management in 5G HetNets: Current State, Challenges, and Future Directions," *MDPI*, Jun. 01, 2023. doi: 10.3390/s23115081.
- [64] S. K. Thillaigovindhan et al., "A Comprehensive Survey on Machine Learning Methods for Handover Optimization in 5G Networks," *Multidisciplinary Digital Publishing Institute (MDPI)*, Aug. 01, 2024. doi: 10.3390/electronics13163223.
- [65] L. P. Kaelbling, M. L. Littman, and A. W. Moore, "Reinforcement learning: A survey," *J. Artif. Intell. Res.*, vol. 4, no. 1, pp. 237–285, Jan. 1996.
- [66] D. Zidic et al., "Analyses of ping-pong handovers in real 4G telecommunication networks," *Computer Networks*, vol. 227, May 2023. doi: 10.1016/j.comnet.2023.109699.
- [67] D. Boughaci, "Solving optimization problems in the fifth generation of cellular networks by using meta-heuristics approaches," in *Procedia Computer Science*, 2021, pp. 56–62. doi: 10.1016/j.procs.2021.02.008.
- [68] V. R. Gannapathy et al., "An Adaptive TTT Handover (ATH) Mechanism for Dual Connectivity (5G mmWave—LTE Advanced) during Unpredictable Wireless Channel Behavior," *Sensors*, vol. 23, no. 9, May 2023. doi: 10.3390/s23094357.
- [69] R. Karmakar, G. Kaddoum, and S. Chattopadhyay, "Mobility Management in 5G and Beyond: A Novel Smart Handover with Adaptive Time-to-Trigger and Hysteresis Margin," *IEEE Trans Mob Comput*, vol. 22, no. 10, pp. 5995–6010, Oct. 2023. doi: 10.1109/TMC.2022.3188212.
- [70] G. Alsuhli et al., "Mobility Load Management in Cellular Networks: A Deep Reinforcement Learning Approach," *IEEE Trans Mob Comput*, vol. 22, no. 3, pp. 1581–1598, Mar. 2023. doi: 10.1109/TMC.2021.3107458.
- [71] Q. Liu, C. F. Kwong, S. Wei, L. Li, and S. Zhang, "Intelligent Handover Triggering Mechanism in 5G Ultra-Dense Networks Via Clustering-Based Reinforcement Learning," *Mobile Netw. Appl.*, vol. 26, no. 1, pp. 27–39, Feb. 2021. doi: 10.1007/s11036-020-01718-w.

- [72] R. Wang et al., "A novel handover scheme for millimeter wave network: An approach of integrating reinforcement learning and optimization," *Digit. Commun. Netw.*, vol. 10, no. 5, pp. 1493–1502, Oct. 2024. doi: 10.1016/j.dcan.2023.08.002.
- [73] M. Benzaghta et al., "Data-Driven Cellular Mobility Management via Bayesian Optimization and Reinforcement Learning," May 2025. [Online]. Available: <http://arxiv.org/abs/2505.21249>.
- [74] M. Wunder, M. Littman, and M. Babes, "Classes of multiagent Q-learning dynamics with -greedy exploration," in *Proc. 27th Int. Conf. Mach. Learn. (ICML'10)*, 2010, pp. 1167–1174. doi: 10.5555/3104322.3104470.
- [75] K. Bataineh, M. Naji, and M. Saqer, "A comparison study between various fuzzy clustering algorithms," *Jordan J. Mech. Ind. Eng.*, vol. 5, pp. 335–343, 2011.
- [76] M. N. I. Alonso and F. Arias, "The mathematics of Q-learning and the Hamilton-Jacobi-Bellman equation," *SSRN Electron. J.*, Jan. 2025. doi: 10.2139/ssrn.5083336.
- [77] 3GPP, "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification," *3GPP TS 36.331 V15.3.0*, 2018. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/36_series/36.331/.
- [78] M. Saeed, M. El-Ghoneimy, and H. A. Kamal, "Solving LTE Handover Problem Using a New Fuzzy Logic Optimization Technique," *Int. J. Adv. Sci. Tech. Res. Issue*, vol. 6.
- [79] C. G. Patil and M. T. Kolte, "An Approach for Optimization of Handoff Algorithm Using Fuzzy Logic System." [Online]. Available: <https://www.researchgate.net/publication/228452405>
- [80] A. Alhammadi et al., "Advanced handover self-optimization approach for 4G/5G hetnets using weighted fuzzy logic control," in *ConTEL 2019 - 15th Int. Conf. Telecommun., Proc.*, Jul. 2019. doi: 10.1109/ConTEL.2019.8848507.
- [81] W. Tashan et al., "Optimal Handover Optimization in Future Mobile Heterogeneous Network Using Integrated Weighted and Fuzzy Logic Models," *IEEE Access*, vol. 12, pp. 57082–57102, 2024. doi: 10.1109/ACCESS.2024.3390559.
- [82] D. Fan et al., "Joint Traffic Prediction and Handover Design for LEO Satellite Networks with LSTM and Attention-Enhanced Rainbow DQN," *Electronics (Basel)*, vol. 14, no. 15, p. 3040, Jul. 2025. doi: 10.3390/electronics14153040.

- [83] K. L. Tsai, H. Y. Liu, and Y. W. Liu, "Using fuzzy logic to reduce ping-pong handover effects in LTE networks," *Soft comput*, vol. 20, no. 5, pp. 1683–1694, May 2016. doi: 10.1007/s00500-015-1655-z.
- [84] F. B. Mismar and B. L. Evans, "Deep Q-Learning for Self-Organizing Networks Fault Management and Radio Performance Improvement," in *Conf. Rec. - Asilomar Conf. Signals, Syst. Comput.*, Jul. 2018, pp. 1457–1461. doi: 10.1109/ACSSC.2018.8645083.
- [85] C. Lee, J. Jung, and J. M. Chung, "Intelligent Dual Active Protocol Stack Handover Based on Double DQN Deep Reinforcement Learning for 5G mmWave Networks," *IEEE Trans Veh Technol*, vol. 71, no. 7, pp. 7572–7584, Jul. 2022. doi: 10.1109/TVT.2022.3170420.
- [86] J. Yang et al., "DQN-ALrM-Based Intelligent Handover Method for Satellite-Ground Integrated Network," *IEEE Trans Cogn Commun Netw*, vol. 9, no. 4, pp. 977–990, Aug. 2023. doi: 10.1109/TCCN.2023.3262239.
- [87] W. Wang et al., "Adaptive UE Handover Management with MAR-Aided Multivariate DQN in Ultra-Dense Networks," *J. Netw. Syst. Manag.*, vol. 33, no. 1, Jan. 2025. doi: 10.1007/s10922-024-09895-9.